The Extended Crystallographic Synthesis (v3.9)

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12 Primary Frameworks, 4 Alternative Branches, 9 Resolution Paths — v3.7 R11 Empirical Reshape Preserved (Inelastic Phonon-Sector Dark Matter §21.4 BSM1; CL4 Extension; CL7 Joint ν Cross-Bridge); v3.8 v14 Resolver OP1–OP18 Status Update (§17) + Universal Coherent-Stack Architectural Update (CL4 v7.9 sharpening + CL8 New Cross-Bridge — P-A28 Universal $\delta k_4/\delta\kappa$ Ratio); New Predictions P-A26–P-A29 Registered through Companion CCDR_v7.9; v3.6 Architecture Otherwise Preserved; v3.9 §21.9 Dimension-Identity Selection and Oriented-Fold Completion of Chirality (completes §21.2.1(b)) — New Predictions P-A30 / P-A31 / P-A32, SM-D11 (signed BSM6), New Cross-Bridge CL9 (bridge count → 9), Open Problems OP19–OP20

Abstract

Synthesis v3.5 is the current architecture of a twelve-framework, four-branch, nine-path research programme anchored to CCDR v7.5. The architecture rests on four principles: (1) N is an observable determined by direct-detection peak counting under the dimension-origin prior, not a theoretical constant; (2) dimensional reductions are local, asynchronous, and potentially ongoing; (3) the framework explicitly forbids FTL information transfer and respects thermodynamic time directionality; (4) **the universe's endpoint is full evaporation back into the (N+1)-dimensional bulk**, restoring symmetry with black-hole evaporation and removing the eternal-cycle metaphysics of v3.4.

The twelve frameworks are tiered by observational proximity (3 tiers) and cross-cut by a falsification hierarchy (5 layers). **Four alternative swappable branches** (CST, EG, CFS as in v3.4, plus the new eternal-attractor branch as in CCDR v7.5 §6.4) provide fallback paths if primary-chain components fail. Nine resolution paths close remaining open problems using existing external tools. A complete equation chain traces the derivation from CDT simplices to observable predictions. **Forty numbered predictions with full prose descriptions** are carried jointly with CCDR v7.5 — extended from v3.4's thirty-eight by adding P39 (secular Λ-drift from bulk cooling) and P40 (bulk-Weyl CMB B-mode component).

Three structural reshapes in v3.5 distinguish it from v3.4.

(i) Evaporation as default endpoint. v3.4 carried the late-universe endpoint as a stable de Sitter attractor; v3.5 promotes full evaporation into the (N+1)-D bulk to default and demotes the eternal-attractor reading to a swappable branch. The cosmological time crystal becomes a long transient rather than eternal. The §10 nested-eternal-crystals architecture of v3.4 is reduced to a single transient cosmological crystal. Dark energy becomes a slowly decaying vacuum mode. The empirical handle for branch selection is the new prediction P39 (secular w(z) drift).

(ii) Dimension-general compactification trigger. The χ_k → 1 trigger criterion is now stated in dimension-general form with explicit Weyl-squared contribution to S_total at k ≥ 5 (k = 4 case zero by dimensional accident). The 5 → 4 specialisation that produced our Big Bang is derived. A new prediction P40 (bulk-Weyl CMB B-mode component) is extracted as a direct empirical consequence.

(iii) Higher-dimensional continuation programme (§22, NEW). The framework's empirical content for our 3+1D slice (P1–P40) is the immediate programme; the higher-k cascade stages are treated as a structured five-stage continuation programme to be activated after decisive 3+1D-slice tests. This is a strict architectural separation between present-day predictions (sharp, falsifiable) and continuation work (staged, conditional).

Empirical content from Round-7 (T13–T17) and Round-8 (T1–T12) public-data tests is preserved and folded into the v3.5 reshape. The cross-link architecture extends from CL1–CL4 to **CL1–CL5** with the new bridge CL5 registering the joint constraint on (v_bulk, c_W) from P39 + P40 — a non-trivial internal consistency check on the v3.5 reshape itself.

Three new open problems in v3.5 (OP14–16) are registered alongside the v3.4 OP11–13: first-principles derivation of v_bulk (the v3.5 reshape's most important new theoretical task), bulk-geometry uncertainty in c_W, and higher-dimensional continuation. The v3.5 architecture commits to determining N from observation under the dimension-origin prior, accepting evaporation as the universe's natural endpoint, and treating higher-dimensional physics as a structured continuation programme with explicit staging.

v3.9 update (dimension-identity selection and the oriented-fold completion of chirality; new §21.9). v3.9 adds one new SM-derivation section, §21.9, completing Stage 1(b) of the seven-stage chain (SU(2)_L from a CDT-foliation-broken Z_2) with its spatial half: the oriented terminal 5 → 4 compactification of the exact extra dimension. A 4+1D bulk fermion is non-chiral (no γ^5 in odd d); the oriented S^1/Z_2 fold yields a single 4D Weyl zero mode with handedness sign(ε_5 · foliation arrow), so the Z_2 that §21.2.1(b) breaks is the orbifold parity of the folded direction and the identity of that direction selects SU(2)_L (not SU(2)_R) of the C ⊗ H ⊗ O / DA/O structure; δ_CP ~ sin θ_fold ~ v ties the CKM CP phase (SM-D8) to the cascade coefficient with no new parameter. The geometric selection of the folded direction (least matter occupancy ⇔ lowest area-to-volume, OP19) is imported from the companion CCDR §3.7. §21.9 registers three new pre-registered predictions — **P-A30** (cosmic and weak parity violation share one sign; LiteBIRD / CMB-S4), **P-A31** (the chirality-defining fold is the largest-radius / lightest-KK extra direction), and **P-A32** (residual statistical-anisotropy axis aligned with the parity axis) — the SM-derivation sharpening **SM-D11** (signed CDT simplex-orientation ↔ weak-handedness lock, extending BSM6; also bulleted in §21.3), one new cross-bridge **CL9** (§11.8 — three-way orientation consistency P-A30 ⊕ weak handedness ⊕ SM-D11 / BSM6, raising the bridge count from eight to nine and adding a second main-body ↔ SM-derivation joint observable beyond SM-D8), and two new open problems **OP19** (proof that least-occupancy = lowest-A/V) and **OP20** (uniqueness versus spontaneous selection of ε_5). §21.9 is a geometric completion layer, not a hard-core axiom: if P-A30 / SM-D11 are decisively falsified it reverts to a heuristic and Stage 1(b) stands alone. All v3.8 content (the v14-resolver OP1–OP18 status §17.bis, the universal coherent-stack CL8, P-A26–P-A29) is preserved unchanged.

Keywords: crystallographic cosmology; CDT; tensor networks; running vacuum; bulk evaporation; dimension-general trigger; bulk-Weyl B-mode; secular Λ-drift; dimension-origin prior; cross-framework consistency; reducing-volume; cascade residue; void-wall kurtosis; density-stratified texture; five-bracket cross-link triangulation; higher-dimensional continuation programme.

v3.7 reshape (Round-11 empirical lessons). Round-11 testing of the reshaped 32-prediction list returned two clean REJECT verdicts on the Synthesis's particle-inventory and cosmic-web texture components. **BSM1 (optical phonon DM at ~100 GeV elastic SI)** rejected at 25σ against the joint XENONnT/LZ/PandaX-4T exclusion at 100 GeV; the v3.6 entry treated the optical-phonon mode as an elastic spin-independent WIMP, which placed it on the wrong test channel. The v3.7 reframing makes BSM1 an inelastic phonon-sector excitation with elastic-overlap suppression κ_elastic = ν, predicting $\sigma\_eff \approx 5 \times 10^{-49}$ cm² and $\delta\_BSM1 \approx \nu \cdot m\_DM \approx$ 0.5 GeV; the test channel migrates from elastic SI exclusion curves to inelastic-DM searches at XENONnT migdal, electron-recoil S2-only, and DARWIN/XLZD-generation experiments. **P-A07 (void-wall transverse kurtosis)** rejected at 7.25σ on the v3.6 grain-boundary single-boundary kernel; the v7.8 update of CCDR §8.3 (scaling inversion plus cascade-depth amplification) is the corresponding mechanism revision and v3.7 incorporates it through extension of CL4 (joint density-stratification of grain-boundary observables) and the new cross-bridge **CL7 — joint ν self-consistency across BSM1 (inelastic), BSM3 (dark photon), and §8.3-v7.8 (void stacking).** Six new pre-registered predictions (P-A20–P-A25, documented in CCDR_v7.8 §15.4) extend the Synthesis prediction inventory. The v3.6 architecture (12 frameworks, 4 swappable branches, 9 resolution paths, 5-layer falsifiability, 6 prior cross-bridges) is preserved in v3.7; CL7 brings the bridge count to seven.

v3.8 update (v14 OP-resolver status; universal coherent-stack realisation). The v3.8 release of the Synthesis incorporates two architectural updates from the companion CCDR_v7.9 release. **First, OP1–OP18 resolution status** from the v14 computational resolver (schema `ccdr-open-problem-computational-resolver-v14`, audit-revision git:4c23fdd, data sha-pinned) — eight of eighteen open problems now have explicit resolution status: OP11 RESOLVED_JOINT ($\nu = 5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$, committed as working value); OP13 strict-resolved on Planck PR4 lensing ($\Delta\kappa = 9.95 \times 10^{-3}$); OP8/OP9 resolved at public-data remnant-consistency level; OP17/OP18 resolved on broad-window macro screening (DCN_k allowed fraction 0.946); OP6/OP10 simulation-resolved prior (visible-peak posterior exported); OP7 simulation-resolved (asynchronous-cascade simulator with local-history sensitivity); OP12 provisionally-resolved methodological; OP15 geometry-scan resolved prior (c_W range 0.014–0.106); OP14 partial (β_ν-flow ν_IR consistent with OP11); OP5 prototype-only (Koide-like Q = 0.6884). Detailed in §17.bis below. **Second, the universal coherent-stack realisation** in CCDR_v7.9 §8.2.bis replaces the v7.8 environment-dependent three-regime classification with a geometry-only universal kernel $\delta k_4 = (\nu\pi^2/6)(r\_wall/r\_grain)^2(N - 4)^2$ applied to all cosmic-web interfaces. The change was triggered by Round-12 measurement on the Tempel filament catalogue returning k_4 = 14.491, rejecting the v7.8 P-A21 discriminator at >10σ. **This is the second failed mechanism revision in CCDR §8 within the v7.x release cycle** (after v7.7 single-boundary → v7.8 environment-dependent → v7.9 universal); the Synthesis records both rejections as audit-trail entries in §22.4 and propagates the v7.9 mechanism into the §21.4 BSM particle inventory annotation (the universal kernel context for κ_elastic = ν), the §11 cross-bridge inventory (new CL8 — universal $\delta k_4/\delta\kappa$ ratio), and the §10 prediction-table cross-references to the four new pre-registered predictions P-A26–P-A29 (cosmic-sheet kurtosis, BAO-scale harmonic, universal $\delta k_4/\delta\kappa$ ratio, master-curve self-consistency). The architecture-level changes in v3.8 are: (a) CL4 statement sharpened a second time to reference the v7.9 universal master-curve form (replacing the v7.8 void/filament asymmetry framing, which was rejected); (b) **new cross-bridge CL8** registered, partnering with P-A29 as the v7.9 §8 discriminator pair; (c) §17 expanded with OP1–OP18 status sub-section §17.bis; (d) §21.4 BSM1 entry annotated with v7.9 universal-kernel context. The v3.7 architecture (12 frameworks, 4 swappable branches, 9 resolution paths, 5-layer falsifiability, 7 prior cross-bridges) is preserved in v3.8; CL8 brings the bridge count to eight.

1. The 12 Primary Frameworks

Tier 1 — Observationally Anchored

1.1 CCDR (2026)

The unifying principle: $\chi_k \to 1$ triggers local dimensional reduction, with the dimension-general form $\boldsymbol{\chi_k(V,t) = (4G_k/A_{(k-1)D}) \int [\rho_{matter} + (c_W/G_k) W_k] d^k x \to 1}$ introduced in v3.5 (tracking CCDR v7.5 §1.1). Anchored by: local a_0 → Milgrom (confirmed, Round-8 re-anchored on n = 171 SPARC curves with closed gap fraction 0.45); density-correlated κ in Euclid Q1 (+12.99σ Round 4, reinforced in Round 8 against ACT DR6 with $\Delta\kappa = +0.110$, Spearman $\rho = 0.221$ at $p \approx 10^{-56}$); high-z a_0 trend (suggestive). CCDR v7.5 is the current form, with **two structural reshapes from v7.4**: evaporation as default endpoint (preserved in this synthesis as branch-default; eternal-attractor swappable in §2.4) and dimension-general trigger criterion (preserved in §1.1).

1.2 RVM — Shapiro & Solà (2002)

$\rho_{vac}(H) = C_0(t) + \nu m^2 H^2/M^2_{pl}$. **In v3.5 $C_0(t)$ is the projected bulk vacuum density**, drifting on the bulk's RVM clock at rate ∝ ν_{bulk} (a new free parameter introduced in CCDR v7.5 §2.5). The fractional drift over the observable cosmic timescale is $\sim 10^{-157}$, so C_0 is empirically constant for present-day measurements; the observable signature is in w(z) at high z, formalised as P39. Anchored by the joint Planck + Pantheon+ + DESI DR2 ν fit, reinforced in Round 8 (T4) with the triangle ratio dropping from 30× to 5.1×. ν is now extracted by **five independent routes** (SN, MOND-sequence, time-crystal Q-factor, grain-boundary stratification, **and the new (ν_{bulk}, c_W) joint constraint from P39 + P40**; see §10).

1.3 AS — Weinberg (1979), Reuter (1998)

Asymptotic safety; UV fixed point G*. All coupling constants as RG eigenvalues. $G(\mu) \to G/\mu^2$, $\Lambda(\mu) \to \Lambda \times \mu^2$ as $\mu \to \infty$. In v3.5, AS plays the additional role of providing the theoretical machinery for deriving ν_{bulk} from first principles via the bulk's TGFT description (Path 6 + Path 9, addressing OP14).

Tier 2 — Structurally Committed, Empirical Test Pending

1.4 CDT — Ambjørn, Jurkiewicz, Loll (2004)

Simplicial path integral. $Z = \Sigma_T \exp(-S_{Regge})$. Phases A/B/C/C_b. d_s: 4 → 2. The A → C phase transition corresponds to the Big Bang cascade's first completion. In v3.5, CDT must accommodate the dimension-general trigger criterion at arbitrary k — the simplicial path integral generalises to k-dimensional simplices, with the Weyl-squared term entering the Regge action at $k \geq 5$.

1.5 TN — Ryu & Takayanagi (2006), Hayden et al. (2016)

S = Area/4G. $D_{eff} \to e^{1/4}$ as IR fixed point. Holographic bridge. Restricted to the final k = 4 stage in v3.4; in v3.5 the role at higher k is structurally specified: tensor-network coverings of the (k−1)-dimensional boundary at each cascade stage, with bond dimension scaling as $\exp(c_W \times W_k(\text{trigger}))$.

1.6 GFT + EPRL — Oriti (2007), EPRL (2008)

GFT condensate → ν. EPRL chiral amplitude with Barbero–Immirzi $\gamma \to f\_geo \sim 10^{-4}$. Required for cascade-stage flow of GFT couplings. In v3.5, the bulk version of GFT is the theoretical home for ν_bulk, with the bulk's IR fixed point determining the secular Λ-drift rate (P39).

1.7 TGFT-RG — Ben Geloun & Rivasseau (2012), Carrozza (2014)

Wilsonian RG for tensorial GFT. Resolves: ν as stage-dependent RG fixed-point output; $\gamma\_AS \approx \gamma\_EPRL$ convergence (stage by stage); condensate stability; CDT A → C as UV→IR transition. **In v3.5, also the cleanest theoretical path to ν_bulk** — applying Bahr–Steinhaus stage-by-stage RG (Path 6) to the bulk's TGFT description gives ν_bulk as an IR fixed-point output, addressing OP14.

1.8 SSM — Connes (1996), Chamseddine & Connes (2007)

$L\_SM = Tr(f(D\_CDT/\Lambda)) + \langle J\psi, D\_CDT\psi\rangle$. Complete SM Lagrangian from spectral action on CDT geometry. Carries the structural commitment to sector-dependent crystallisation scales for Koide ratios.

Tier 3 — Algebraic / Structural, Consistency Checks

1.9 AQN — Zhitnitsky (2003)

Axion quark nuggets as crystal defects. Contributes to Ω_4 (lightest cascade residue). Baryogenesis: $\eta \sim \nu \times \delta\_CP \times f\_geo \sim 10^{-10}$. Does not count dark sectors in v3.5.

1.10 DA/O — Furey (2012/2018), Baez (2001)

$C \otimes H \otimes O \to SU(3) \times SU(2) \times U(1)$. Provides upper bound $N \leq 11$. Consistency check on gauge group at final stage. Does not fix N.

1.11 HaPPY — Pastawski et al. (2015)

Holographic QEC. Code distance from RT surface. Restricted to $k = 4$ stage. The v1 ‘$N = 6$ from HaPPY code dimension’ is removed.

1.12 LGT — Wilson (1974)

Computational engine. Monte Carlo for CDT path integral, SSM spectral action, GFT condensate, fermion propagator. Structurally indispensable; empirically invisible.

2. The Four Alternative Swappable Branches (extended in v3.5)

v3.4 carried three branches (CST, EG, CFS). **v3.5 adds a fourth: the eternal-attractor branch from CCDR v7.5 §6.4.** This branch is structurally distinct from the previous three — those swap microscopic geometry while preserving the cascade core; the eternal-attractor branch swaps the universe's endpoint while preserving the microscopic geometry. It is the first branch in the synthesis architecture that lives at Layer 4 (RVM + CCDR) rather than at Layer 0 or 2.

2.1 CST ↔ CDT — Bombelli, Lee, Meyer & Sorkin (1987)

Spacetime as locally finite poset. $\Lambda \sim 1/\sqrt{N}$ from Poisson counting. Natural fit for v3.5's local/asynchronous reductions: causal sets accommodate locally varying dimensionality through poset density variation. Swap when CDT C-phase fails or mixed-dimensionality simulations are intractable. Gauge group from DA/O. Gains: C_0 from Sorkin counting. Loses: S_5, BZ, phonon picture.

2.2 EG ↔ CDT + TN — Jacobson (1995), Verlinde (2011/2017)

Einstein equations from $\delta Q = TdS$ on horizons. MOND from de Sitter entanglement entropy. The local a_0 → Milgrom confirmation anchors in EG territory as well as in CCDR. Swap when CDT + TN fails to produce gravity. Gains: natural MOND derivation. Loses: microscopic geometry.

2.3 CFS ↔ CDT — Finster (2006)

Spacetime from operator measure in Hilbert space. Einstein + Dirac from a single variational principle. Most flexible branch for v3.5's local/asynchronous reductions. Swap when CDT's discreteness is wrong but causality preserved.

2.4 Eternal-Attractor Branch ↔ Evaporation Default (NEW in v3.5)

The default in v3.5 is full evaporation of our 3+1D universe back into the (N+1)-D bulk over $\tau_{evap} \sim 10^{167}$ years, with $C_0(t)$ drifting toward zero on the bulk's clock. The **eternal-attractor branch** is the swappable alternative: if C_0 turns out to be a fundamental constant (i.e. $v_{bulk} = 0$ exactly, or the bulk has no RVM dynamics), the universe asymptotes to a stable de Sitter state at $H_\infty \approx 1.85 \times 10^{-18}$ Hz indefinitely, the cosmological time crystal becomes eternal, and the v7.4 §§10–11 architecture ($Q \sim 1/(2\nu) \sim 500$ of an eternal oscillator) becomes the correct picture.

Empirical distinguishing test. The two branches are indistinguishable to better than parts in 10^{150} at present-day precision — the difference is in the secular drift of Λ_{eff}, formalised as P39. DESI DR3 + Euclid + LSST joint constraints will be decisive. A confirmed flat $w(z) = -1$ to high precision endorses the eternal-attractor branch; a confirmed secular drift in the predicted direction (Λ_{eff} decreasing in cosmic time) endorses the evaporation default.

Architecture-level reading. Both readings sit inside the same 12+4+9 architecture, the same predictions list (with P39 status differing by branch), and the same falsifiability hierarchy. The branch choice changes CCDR §§6.2, 6.3, 10, 11 — about 15 % of the manuscript — and leaves everything else intact. The synthesis-level commitment is to *make this swap testable*, not to lock in either reading prematurely.

3. Swap Table (extended in v3.5)

Primary fails	**Swap to**	**CCDR survives?**	**What changes**
CDT → CST	CST	Yes	Gauge group from DA/O; Λ from Sorkin; lose BZ/phonon
CDT → CFS	CFS	Yes (reinterpreted)	Einstein+Dirac unified; lose simplicial lattice
CDT+TN → EG	EG	Yes	Gravity from Jacobson; MOND from Verlinde

| Evap default → Eternal | Eternal-attractor | Yes (preserved) | §§6.2, 6.3, 10, 11 revert to v7.4 form; P39 reads flat; CL5 sits at one end of (v_bulk, c_W) plane |

| GFT fails | — | Yes | ν reverts to RVM fit parameter |

| SSM fails | — | Yes | Gauge group from DA/O alone |

| CCDR core fails | — | All survive independently | Only crystallographic interpretation lost |

4. The Complete Equation Chain

CDT simplices → S_5 → D_CDT → Tr(f(D/Λ)) = L_SM [SSM] → cross-check: C ⊗ H ⊗ O → same gauge group [DA/O] → G, *α* [AS] → RG running → α = 1/137 [LGT] → TN, D_eff → e^{1/4} [TN] → GFT condensate → ν [TGFT-RG] → EPRL chiral amplitude → f_geo → γ convergence [TGFT-RG] → ρ_vac(H) [RVM] → χ_k → 1, **dimension-general crystal nucleation** [CCDR v7.5] → PQ → EW → QCD = AQN [crystallisation cascade] → η ~ ν × δ_CP × f_geo ~ 10^{-10} [all frameworks] → HaPPY QEC → BAO (δr/*r* ~ ν/2) → MOND (a_0 → Milgrom locally, → cH_0 asymptotically) → **dS evaporation into bulk reservoir; $C_0(t)$ → 0 on bulk's clock; full evaporation over ~10^{167} yr** [CCDR v7.5 default] → bulk-Weyl B-mode imprint at trigger [P40] → secular w(z) drift [P39]

Every link is computable. Every prediction is falsifiable. The holographic principle is not assumed — it emerges from the entanglement structure of the CDT ensemble in the semiclassical regime. **In v3.5 the equation chain extends past the de Sitter approach to the explicit endpoint state**: full evaporation rather than eternal asymptote, with the bulk reservoir as the explicit endpoint.

4.1 Fundamental Constraints

The framework prohibits FTL information transfer (consequence of holographic bound S ≤ A/4G) and respects thermodynamic time directionality (consequence of monotonic RVM cooling). These are predictions, not assumptions. CDT enforces a causal foliation at the microscopic level, precluding closed timelike curves.

4.2 Reducing-Volume Caveat — Probability-Weighted (carried forward)

The cascade residue picture assumes all bulk species were present in the specific volume that reduced in our patch. This is not guaranteed. The peak count gives N only under full-sampling. In v3.3 the caveat was quantified: P_presence(k) ~ (r_k / L_k)^{k−4}. **Higher-dimension-origin species have systematically lower presence probability in our patch.** Expected observed peak count E[n_peaks] = Σ_k P_presence(k), generically less than N − 4 and biased toward the light end of the tower.

v3.4 added indirect operational support through cross-stratification of P30, P3, P38 (CL4). v3.5 adds further structure: **the dimension-origin prior at the patch level connects to OP13** (patch-spread anomaly in T2 Round-8) — patch_spread = 0.120 versus baseline Δκ = 0.110 is the direction expected from cross-patch composition variation but degenerate with proxy mode-coupling artefacts. Resolution requires patches aligned on physical scales rather than RA-Dec rectangles.

3.6 Compactification Principle: Isoperimetric–Holographic Selection (NEW core addition)

Statement. Given a local region approaching $\chi_k \to 1$, the reduction selects a boundary geometry that extremises **holographic encoding efficiency**: it **minimises interface/boundary area A_{k-1}** while **maximising the stabilised interior volume V_k**, subject to fixed entropy, flux, curvature, topology, and causal constraints. The leading-order variational object is the isoperimetric ratio

$$I_k(V) = A_{k-1}(\partial V) / V_k^{(k-1)/k},$$

corrected by Weyl curvature, extrinsic-curvature smoothing, surface tension, tidal, and cascade-residue terms. The **ideal compactified boundary is the (k−1)-sphere** (a *slightly closed hyperspherical surface* in the pre-reduction frame), because the standard Euclidean isoperimetric inequality saturates uniquely at the round ball.

Full variational functional:

$$F[\partial V] = \sigma A_{k-1} - \lambda V_k + \alpha_W \int_V W_k \, d^k x + \alpha_K \int_{\partial V} K_{ab} K^{ab} \, dA + \alpha_T \int_{\partial V} T^{tidal}_{ab} n^a n^b \, dA + \alpha_R \int_{\partial V} \rho_{residue} \, dA,$$

with the preferred compactification given by $\delta F = 0$, $\delta^2 F > 0$. Term-by-term:

- **σA_{k-1}** — interface tension; favours minimum boundary area.
- **$-\lambda V_k$** — stabilised-volume term; favours maximum encoded interior.
- **$\alpha_W W_k$** — bulk Weyl-squared curvature contribution (already present in v7.5/v7.6 §1.1, §3.1 of the entropy decomposition).
- **$\alpha_K K_{ab} K^{ab}$** — extrinsic-curvature smoothing penalty.
- **α_T** — external tidal-field deformation of the boundary.
- **α_R** — CCDR dark-sector / interface-defect residue.

Why "minimise area / maximise volume**" is the right framing.** Unconstrained volume maximisation is ill-defined; the principle is operationally well-posed only when an isoperimetric ratio is extremised at *fixed* entropy, flux sector, topology, and energy. Phrased as a constrained isoperimetric optimisation, the assumption is supported by five independent literature pillars:

- **Isoperimetric geometry.** In Euclidean n-space the ball is the unique extremiser of $A / V^{(n-1)/n}$; equality holds only for the round sphere (Osserman, classic isoperimetric review). The preferred compactification shape is therefore not arbitrary — it is the natural low-boundary-cost geometry.
- **Holographic efficiency.** The CCDR trigger $\chi_k = S_{total} / [A_{k-1}/4G_k]$ already places boundary area in the denominator (cf. Bousso's holographic-principle review: area limits information content of adjacent spacetime regions). Compactification thus prefers geometries that encode the largest stable interior state using the smallest sufficient boundary capacity — *exactly* "maximise stabilised volume per boundary area".
- **Surface/interface tension.** Brane, bubble, soap-film, and domain-wall actions all pay an energy cost $E_{interface} \sim \sigma A$. Minimising area is the standard physics of phase boundaries; the CCDR boundary at $\chi_k \to 1$ is treated identically.
- **Flux compactification / moduli stabilisation.** String compactifications already select compact dimensions through effective potentials of fluxes, curvature, branes, and moduli (Graña review). CCDR adds an isoperimetric–holographic selection rule on top of that

machinery rather than replacing it; “stabilised volume” is therefore a *standard* compactification problem, not a new one.

- **CDT semiclassical de Sitter profile.** CDT simulations of fixed-four-volume quantum-gravity ensembles produce smooth, extended, de Sitter-like volume profiles in the C-phase. A compactification rule favouring low-boundary-cost stabilised volume is aligned with this established CDT semiclassical tendency.

Slightly-closed (hyperspherical) pre-reduction geometry. Because the isoperimetric extremiser in positive-curvature backgrounds is the $(k-1)$-sphere, the principle predicts that the preferred reducing-volume patch carries a **small positive intrinsic curvature** — i.e. is *slightly closed* — even when the projected 3+1D slice appears asymptotically flat. This is consistent with current observational bounds on Ω_K (mildly closed within ~1σ in several Planck combinations), and supplies a structural explanation for the CDT C-phase de-Sitter-like geometry without invoking a fundamental Λ. The slightly-closed property is a feature of the *pre-reduction* boundary; the post-projection 3+1D geometry need not be globally closed because warp-factor and projection effects (§3.4) decouple the bulk and brane curvature scales.

Real compactification is never perfectly ideal: boundary-deformation spectrum. Around the isoperimetric minimiser we expand the boundary radius:

$$R(\Omega) = R_0 \left[1 + \Sigma_{\ell \geq 2, m} \varepsilon_{\ell m} Y_{\ell m}(\Omega)\right],$$

and define the **isoperimetric deformation score**

$$D_k = A_{k-1}(\partial V) / A_{k-1}^{sphere}(V_k) - 1 \geq 0,$$

with $D_k = 0$ only for the ideal hypersphere and $D_k > 0$ for every real boundary. The observable deformation decomposes as

$$D_k = D_{spin} + D_{tidal} + D_{Weyl} + D_{matter} + D_{CCDR\ residue},$$

where only the last term is a CCDR-specific handle. The principle is **not** “all real black holes are visibly non-spherical”; it is “after spin J_k, tidal field T^{tidal}_k, accretion environment, and Weyl curvature W_k are subtracted, a small residual deformation spectrum may remain, scale-dependent and density/χ-correlated, and *that* residual is the CCDR signature”.

New numbered prediction — P42 (Boundary-deformation residuals). After known astrophysical effects (spin J_k, tidal field T{tidal}*k, accretion environment, Weyl curvature W_k) are subtracted, the residual boundary-deformation spectrum* |ε{ℓm}{(k)}| should:

- vanish to current sensitivity in clean isolated Schwarzschild-like systems ($D_{BH}^{CCDR} \approx 0$ in present GR/Kerr ringdown tests, e.g. GW250114, which finds excellent agreement with no-hair predictions);
- scale predictably with J_k, $\rho_{residue}$, and $(\chi_k - 1)$ in merger remnants, high-spin holes, and dense environments;
- correlate across scales: the **BH ringdown/shadow deformation** spectrum (local χ_4 test), the **CMB B-mode deformation** spectrum from P40 (primordial χ_5 test), and the **cosmic-web/void-wall deformation** spectrum (P3 / P38 cosmological tracers) are three projections of the same underlying boundary-deformation principle.

Observable channels for P42.

- Event-Horizon-Telescope shadow morphology (circularity, axis ratio, persistent non-Kerr distortion of Sgr A *and M87*).

- Gravitational-wave ringdown spectroscopy (QNM frequency/damping deviations from Kerr; multi-tone no-hair consistency at LIGO-VK O5, LISA).
- Inspiral tidal corrections (effective Love-number-like terms on the boundary's rigidity / deformability).
- Merger-remnant relaxation time (how fast the deformation decays after merger; informs the boundary surface tension σ).
- Residual deformation as a function of spin a_* after Kerr subtraction (anisotropic compactification stress).
- Residual as a function of accretion/plasma/tidal environment (separates ordinary deformation from CCDR deformation).

Falsification of P42. Large random deformations with no relation to spin, tides, Weyl curvature, density, or χ-boundary conditions would weaken the compactification principle. A confirmed flat $D_k = 0$ in all regimes is consistent with the *conservative* reading of the principle; a coherent, scale-dependent, environment-correlated $D_k > 0$ spectrum is the *positive* CCDR reading.

New cross-link — CL7 (P39 + P40 + P41 + P42 joint consistency). The boundary-deformation principle ties together four previously distinct channels:

- P39 (secular Λ-drift from bulk cooling) measures the bulk RVM coefficient v_{bulk}.
- P40 (bulk-Weyl CMB B-mode) measures the primordial 5→4 boundary-deformation imprint.
- P41 (b→sμμ angular tension) measures a TeV-scale lattice/grain-boundary correction.
- P42 (local boundary-deformation residuals) measures the local $\chi_4 \to 1$ deformation spectrum.

CL7 registers their joint consistency: all four are linked to the same underlying isoperimetric–holographic selection plus bulk-Weyl curvature contribution. CL7 extends CL5 (the (v_{bulk}, c_W) joint constraint from P39 + P40 introduced in v7.5) to a four-way prediction-coherence check.

Framing: a controlled geometric completion layer, not a hard-core axiom. The compactification principle is presented as a *geometric completion* of CCDR v7.6 rather than as a new hard-core commitment. It is supportable from five literature pillars (isoperimetry, holography, surface tension, flux compactification, CDT), generates one new prediction (P42) and one new cross-link (CL7), without modifying any prior v7.6 commitment. If P42 is decisively falsified, the principle returns to the status of a useful but non-mandatory geometric heuristic; the v7.6 core architecture is unaffected.

5. The Five-Layer Falsifiability Hierarchy

Layer 0 — CDT + Cascade Core

The microscopic geometry (CDT simplicial path integral) and the cascade reduction structure, **now stated in dimension-general form (§1.1) with explicit Weyl-squared contribution at k ≥ 5**. Falsification: CDT C-phase absent; non-geometric mass ratios; zero peaks below probability-weighted E[n_peaks] lower bound; **bulk-Weyl B-mode (P40) in conflict with predicted ℓ-dependence shape**. Exit to CST or CFS. *Current status:* untested. Round-8 T11: three XENONnT-2025 limit curves cover 521.4–2000 GeV each; window overlap remains true; peak_count_ready_now = false.

Layer 1 — GFT + TGFT-RG

The group field theory condensate and its Wilsonian renormalisation group, **now also covering v_bulk via Path 6 + Path 9 applied to the bulk's TGFT description (OP14)**. Falsification: ν from TGFT-RG IR fixed point contradicts empirical ν; $\gamma_{AS} \neq \gamma_{EPRL}$ at any stage; condensate instability; **v_bulk derivation contradicts P39's measured drift rate**. *Current status:* ν audit closed at 10^{-3}–10^{-2}, central value likely low-z SN systematic. v_bulk derivation pending.

Layer 2 — TN + HaPPY

Tensor network implementation of holographic area laws and quantum error-correcting code. Falsification: RT formula fails in CDT C-phase; $D_{eff} \neq e^{1/4}$ at k = 4 IR fixed point. Exit to EG if holographic structure absent. *Current status:* untested.

Layer 3 — SSM + AS + DA/O

Spectral action, asymptotic safety, division-algebra gauge group. Falsification: spectral-action SM Lagrangian inconsistent with observed gauge couplings at any running scale; DA/O gauge group wrong. *Current status:* consistent with data.

Layer 4 — RVM + CCDR (substantively reinforced in v3.5)

Running vacuum model and the CCDR crystallographic interpretation, **with the universe's endpoint now part of the falsifiability content**. Falsification: ν < 0; $\chi_k \to 1$ does not trigger any reduction in any CDT simulation at any k; all peripheral predictions fail simultaneously; **the five independent ν extractors (CL1–CL5) disagree beyond systematic tolerance**; **P39 confirms a secular drift inconsistent with both branches** (rules out evaporation default *and* eternal-attractor — major structural failure). *Current status:* partially confirmed. Three suggestive positives, two new Round-7 promotions (P38, P3), one sign inversion to investigate (P33), CL2 triply anchored, CL5 awaiting first measurement.

Layer 5 — Baryogenesis

$\eta \sim \nu \times \delta_{CP} \times f_{geo} \sim 10^{-10}$. Falsification: η wrong by orders of magnitude when all three factors are measured independently.

No layer is currently falsified. The cascade hard core remains the diagnostic test for Layers 0, 1, 4. **In v3.5, the universe-endpoint test (P39) is the additional empirical handle that distinguishes evaporation default from the eternal-attractor branch — neither outcome falsifies Layer 4 outright; both leave the rest of the framework intact.**

6. The Nine Resolution Paths

Nine open problems remain. For each, an existing external framework provides the resolution path. Six were registered in v3.2; three were added in v3.3. **In v3.5 the resolution paths connect to v3.4 OP11–13 and v3.5 OP14–16 (§19) explicitly**, marking which paths address which open problems.

Path 1 — Swampland Distance Conjecture → Cascade Timing

Constrains the timing function f(t) that maps RVM cooling to compactification radii $R_k = c/H_k$. The Swampland de Sitter conjecture bounds $|d\rho_{vac}/dH|/\rho_{vac} \geq c_{swamp}/M_{pl}$, which constrains ν and hence the cascade timing. *Status:* unblocked; *Timeline:* 1–2 years.

Path 2 — Discrete Flavour Symmetries → Neutrino Hierarchy

The hexagonal BZ has C_3 symmetry relating three generation directions. *Timeline:* 2–5 years.

Path 3 — CST Sorkin Counting + Hartle–Hawking on CDT → C_0

For $N \sim 10^{122}$ causal-set elements, $\Lambda \sim 10^{-122} M^4_{pl}$. **In v3.5 this path also addresses the question of how $C_0(t)$ projects from the bulk to our slice**: the same Sorkin counting at the (N+1)-D level gives the bulk's own C_0 value, and the projection through the warp factor connects bulk-side C_0 to slice-side $C_0(t)$. *Timeline:* 3–5 years.

Path 4 — Algebraic Renormalisation with Overlap Dirac → Strong CP

The overlap Dirac operator on CDT has exact chiral symmetry. Stronger than the axion solution — no new particle required. *Timeline:* 3–7 years.

Path 5 — Diósi–Penrose Gravitational Collapse → Quantum Measurement

Penrose collapse when gravitational self-energy of superposition exceeds $\hbar/\tau$. Crystal-state collapse interpretation. *Timeline:* 1–3 years (theory), 5–10 years (experiment).

Path 6 — Bahr–Steinhaus Spin-Foam RG → TGFT-RG for EPRL

Stage-by-stage coarse-graining for EPRL using sl2cfoam-next. **In v3.5 this path also provides the cleanest theoretical route to v_bulk**: applying Bahr–Steinhaus stage-by-stage RG to the bulk's TGFT description gives v_bulk as an IR fixed-point output, **directly addressing OP14**. v3.5 priority: extend Path 6 to the bulk side. *Timeline:* 2–3 years (slice side), +1–2 years for bulk side.

Path 7 — Conformal Bootstrap → CDT C-phase Universality

If the IR is CFT-like, Simmons-Duffin-style crossing-symmetry bootstrap constrains scaling dimensions without simplicial regularisation. *Timeline:* 3–5 years.

Path 8 — Quantum Simulator → Small-N Cascade Analogue

Cold-atom or ion-trap platforms engineering reducible effective dimensionality. **In v3.5 this path becomes Stage 4 of the higher-dimensional continuation programme (§22)**: the laboratory $\chi_k \to 1$ measurement is the most direct independent test of the dimension-general trigger criterion. *Timeline:* 3–7 years; theoretical analogue design owed in 1–2 years.

Path 9 — Diósi–Penrose × Reducing-Volume Quantification

The Penrose collapse scale provides a gravitational length for L_k. **In v3.5 this path also contributes to OP14** (v_bulk derivation): the gravitational collapse scale at the bulk level constrains the bulk's coherence volume and hence v_bulk via the relationship v ~ (RVM coupling) × (coherence scale)$^{-2}$. *Timeline:* 1–3 years.

7. Resolution Path Summary

Problem	**Path**	**Timeline**	**v3.4/v3.5 OP addressed**
Cascade timing	Swampland (Vafa 2005)	1–2 yr	—
Neutrino hierarchy	Discrete flavour + SSM see-saw on CDT	2–5 yr	—
C_0 from first principles	CST Sorkin + unimodular + HH on CDT	3–5 yr	—
Strong CP (θ = 0)	Overlap Dirac on CDT	3–7 yr	—
Quantum measurement	Diósi–Penrose = crystal-state collapse	1–10 yr	—
TGFT-RG for EPRL	Bahr–Steinhaus stage-by-stage	2–3 yr (+1–2 yr bulk)	**OP14 (v3.5)**
C-phase universality	Conformal bootstrap on CDT IR	3–5 yr	—
χ_k → 1 mechanism	Quantum-simulator analogue	3–7 yr	**Stage 4 of §22**
L_k from first principles	Diósi–Penrose × reducing-volume	1–3 yr	**OP14 (v3.5) supplement**

None of these requires new layers in the synthesis. They use external frameworks as tools to close specific gaps within the existing 12+4+9 architecture (v3.5). **In v3.5 three resolution paths (6, 8, 9) take on additional roles** linking to the new v3.5 open problems (OP14: v_bulk derivation) and to the higher-dimensional continuation programme (§22 Stage 4).

8. AQN Generalisation — Dark Cascade Nuggets DCN_k (NEW in v3.6, protective-belt extension)

The Tier-3 framework AQN (Zhitnitsky 2003; § 1.9) treats axion quark nuggets as compact QCD / PQ-stage dark residues stabilised by axion-domain-wall and dense-QCD physics. AQN contributes specifically to Ω_4 (the lightest cascade residue) and bridges to baryogenesis (η ~ v × δ_CP × f_geo ~ 10^{-10}). The mechanism — compact, charge-asymmetric, surface-stabilised dark relic — is in principle generic across cascade stages. v3.6 introduces a structured generalisation that **retains AQN's specificity at the QCD / PQ stage** while extending the *nuggetisation morphology* across the dark-cascade tower, **without identifying every dark species with AQN**.

8.1 Definition — Dark Cascade Nugget at Stage k (DCN_k)

A compact, charge-asymmetric, surface-stabilised dark residue formed at cascade stage k. Each cascade residue DM_k may either remain particle-like or partially condense into compact nugget-like bound states, with fraction:

f_nugget,k = Ω_DCN,k / Ω_DM,k.

The total dark matter density decomposes as:

$\Omega_{DM} = \Sigma_k \Omega_{DM,k} = \Sigma_k (\Omega_{particle,k} + \Omega_{DCN,k})$.

8.2 Necessary Conditions for DCN_k Formation

A residue cannot nuggetise unless **all five** of the following hold in sector k:

- Conserved or approximately conserved sector charge Q_k. 2. Attractive self-interaction or confining force in sector k. 3. Phase boundary / domain wall / surface tension stabilising the lump. 4. Asymmetry or trapping mechanism preventing full annihilation back to bulk. 5. Stability against evaporation, fission, and gravitational collapse.

These are necessary; sectors that fail any of (1)–(5) remain particle-like.

8.3 Heuristic Energy Functional

A DCN_k of charge number A in sector k has:

$E_k(A) = m_k A + \sigma_k A^{2/3} + C_k A^{1/3} + E_{repulsion}(A)$

— bulk + surface + curvature + electrostatic-like repulsion. **Stability requires a local minimum in $E_k(A) / A$ at large A.** Sectors that fail any of the five conditions give $E_k(A) / A$ monotonic and admit only particle-like residues.

8.4 Mapping to the CCDR Cascade Structure

- **AQN ≡ DCN_QCD ≡ DCN_4 special case.** The QCD / PQ-stage realisation: quark / antiquark matter stabilised by axion-domain-wall + dense QCD phase. Contributes to Ω_4. Baryogenesis bridge intact.
- **DCN_5, DCN_6, …, DCN_N**: speculative compact morphology for higher-cascade-stage residues. Each is conditional on the relevant sector satisfying conditions (1)–(5); not every stage produces a nugget.

8.5 Direct-Detection Consequence

If a fraction $f_{nugget,k}$ of cascade stage k's residue is condensed into compact DCN_k, peak counting at direct-detection scale **undercounts** the species: a missing peak at expected position m_k may indicate **compactification into macroscopic nuggets**, not absence of the species. This adds a third reason — beside (a) reducing-volume non-presence and (b) cross-patch composition variation — for direct-detection peak undercounting. Constraints on the nuggetised fraction must come from non-recoil channels:

- macro-impact searches (etched-track detectors, seismic stations);
- microlensing / gravitational substructure on stellar / sub-stellar scales;
- thermal anomaly / heating events in ground media;
- annihilation glow if anti-nugget-like;
- gravitational-granularity tests of dark substructure.

8.6 Connection to the Dimension-Origin Prior

The probability-weighted prior $P_{presence}(k) \sim (r_k / L_k)^{k-4}$ now factors at the patch level into a *species presence* probability and a *nuggetisation* probability. Heavier (higher-k) species are systematically more likely to be missing, both because of (a) reducing-volume non-presence and (b) higher likelihood of condensing into compact nuggets if the sector confinement is strong. The two suppressions compose multiplicatively, deepening the asymmetric peak distribution prediction (CCDR § 9.3 (i)).

8.7 Architectural Placement

- **AQN-as-DCN_QCD remains in Tier 3 (§ 1.9) unchanged**: AQN is the worked QCD-stage example; baryogenesis bridge $\eta \sim v \times \delta_{CP} \times f_{geo}$ intact.
- The **DCN_k generalisation enters the dark-matter / higher-dimensional research programme** (§ 22.2 Stage 3 indirect bulk-geometry constraint; CCDR § 25.3 Stage 3) as a **protective-belt extension, not a hard-core commitment**. If Stage 3 cascade-residue spectroscopy reveals systematic heavy-peak undercounts beyond what the dimension-origin prior predicts on its own, the DCN_k extension provides the structural reading without forcing a re-architecture.
- **Falsification of the extension is mild**: if every dark-detection channel (recoil + macro + microlensing + annihilation + gravitational substructure) finds zero nuggets and the heavy-peak count exactly tracks the un-nuggetised dimension-origin prior, the $f_{nugget,k}$ fractions collapse to ≈ 0 for all k, and the architecture reduces to the v7.5 cascade-residue picture without DCN_k structure. **No core failure.**

8.8 Caveat — Not All Dark Matter Is AQN

The generalisation explicitly does *not* claim that DM_5, DM_6, … are all AQN-like. It claims: each cascade residue *may or may not* nuggetise depending on whether the five conditions hold in that sector. Most plausibly, only some sectors with conserved charges and confining dynamics will nuggetise; others remain pure particle-like residues. **This separation between species count (peak structure) and species morphology (particle vs nugget) is the central architectural improvement of the generalisation.**

8.9 Status

Protective-belt extension; not load-bearing for any present-day prediction. AQN at Ω_4 / baryogenesis remains in the hard core (Tier 3, § 1.9; CCDR § 4.5). DCN_k>4 enters as part of the higher-dimensional continuation programme conditional on Stage 3 results. New open problems registered in v3.6: **OP17** (derivation of (σ_k, C_k, $E_{repulsion}$) for cascade sectors k > 4 — inputs to the energy functional), **OP18** (cosmological abundance fraction $f_{nugget,k}$ as a function of cascade-stage cooling history).

9. Speculative Branch — Sector-Causal CCDR (Dark-Cone Branch, NEW in v3.6)

The default CCDR architecture (v7.5 § 1.3) explicitly forbids superluminal information transfer in *all* sectors, baryonic and dark, as a consequence of the holographic bound. **v3.6 / v7.6 introduces a non-default speculative branch — the Dark-Cone Branch — that weakens this constraint for the dark sector while keeping the baryonic light cone universal for visible matter.** The branch is the only formulation that admits any FTL-like dark-sector physics without immediately conflicting with cosmology, GW170817 multimessenger constraints, or the Watanabe–Oshikawa time-crystal architecture. It is recorded here as a bounded alternative for the dark-sector causal structure should the Stage 3 cascade-residue spectroscopy (§ 22.2) require it; the default architecture continues to forbid FTL signalling.

9.1 Core Assumption — Two Causal Cones

Baryonic matter, photons, neutrinos, and SM fields propagate on the induced 3+1D acoustic metric g_B with limiting speed c. Dark matter — interpreted in CCDR as a gapped optical / bulk residue of incomplete compactification (CCDR v7.5 § 7.2) — propagates partly on a dark-sector effective metric g_D with limiting speed $c_D = \beta c$, $\beta > 1$. The baryonic light cone is strictly contained in the dark-sector cone:

$C_{baryon} \subset C_{dark}$.

The statement "the speed of light c is the universal signal speed" is replaced by "c is the limiting signal speed of the baryonic / acoustic sector"; the dark / bulk sector has its own wider cone.

9.2 Refinement — Cold Density, Fast Excitation

The dark matter mass distribution remains cold and non-relativistic ($w_{DM} \approx 0$, $c_{s,DM}^2 \approx 0$, with broad cosmology-derived bounds $c_{s,DM}^2 < 10^{-10.7}$). Only **excitations** of the dark sector — phase slips, optical / bulk modes, bulk-leakage modes — propagate at $v \leq c_D$. Halo bulk velocities remain $v_{DM} \ll c$. Two distinct quantities are kept rigorously separate:

- dark-sector signal velocity: $v_{signal} \leq c_D = \beta c$
- dark-matter bulk velocity: $v_{DM} \ll c$

This separation is critical: if the *whole* dark fluid were superluminal, CMB acoustic peaks, structure formation, halo concentrations, and large-scale structure would all break against tight observational bounds on dark-matter sound speed and pressure. The branch survives only because the bulk fluid stays cold while the *excitation spectrum* has a wider cone.

9.3 Causality Preservation — Cascade-Time Ordering

A wider causal cone alone in a Lorentz-invariant theory generates closed timelike curves. To avoid this, the branch promotes the fundamental ordering parameter from baryonic coordinate time to **cascade time τ** — the foliation parameter inherited from CDT, RVM cooling, and bulk crystallisation. The rule is:

For all physical propagation: $d\tau > 0$ always.

Even if the spatial separation Δx and the baryonic-coordinate time difference Δt_{baryon} satisfy $\Delta x / \Delta t_{baryon} > c$, the cascade-time ordering $\Delta\tau > 0$ must be preserved. In CCDR language: the speed of light is the *acoustic-sector* limit, not the *universal bulk-sector* limit. The universal causality rule is not $v \leq c$, but **dτ > 0 along the cascade-time ordering**. This connects directly to v7.5's local / asynchronous reduction architecture (§ 1.2 of CCDR) and to the RVM's monotonic cooling direction.

9.4 Five-Assumption Stack (Consistency Conditions)

A1. **Sector-dependent causal cones** with $c_D = \beta c$, $\beta > 1$ finite; baryonic cone strictly inside dark cone.

A2. **Preferred cascade time τ**; all propagation requires $d\tau > 0$.

A3. **Cold density, fast excitation**: $w_{DM} \approx 0$, $c_{s,DM}^2 < 10^{-10.7}$; only excitations propagate at c_D.

A4. **Weak baryon–dark conversion**: $\varepsilon_{BD} \ll 1$, activated only near $\chi \to 1$ phase boundaries, BH horizons, dense cosmic-web nodes, or early-universe trigger remnants.

A5. **Gravitational-radiation projection controlled**: gravitational waves remain luminal in g_B (preserving the GW170817 / GRB 170817A multimessenger constraint $\Delta v_{GW} / c \lesssim 10^{-15}$). Only non-radiative dark-sector correlations are superluminal.

9.5 Stage-Dependent Dark Cones (CCDR-Compatible Refinement)

Rather than a single universal β, the natural CCDR-compatible refinement is a **stage-dependent dark cone**:

$c_D(k) = \beta_k c$, with $\beta_5 < \beta_6 < \beta_7 < \ldots < \beta_N$

— deeper / higher-dimensional cascade-stage residues have wider dark cones. The maximum dark-sector signal speed $v_{dark,max} = \beta_N c$ is finite. **Phenomenological values** (NOT derived predictions): $\beta_5 \approx 2$ (conservative), $\beta_6 \approx 10$, $\beta_7 \approx 10^2$, $\beta_8 \approx 10^3$ (strong-but-controlled). $\beta = \infty$ (instantaneous nonlocality) is rejected — it makes the model metaphysical rather than physical and requires a deeper causality rule than $d\tau > 0$.

9.6 What the Branch Can Explain or Motivate

The Dark-Cone Branch can be used to reinterpret several CCDR ideas without invoking new free parameters:

- **Dark-matter non-detection**: not just weak coupling, but **causal-sector mismatch** between baryonic acoustic-sector instruments and optical / bulk dark modes.
- **Density-correlated lensing anomalies**: dark-sector phase reorganisation faster than baryonic relaxation in dense regions — predicts a dark-sector phase lead in lensing maps relative to baryonic tracers.
- **Cosmic-web coherence and long-range alignments**: dark-sector phase communication faster than baryonic causal equilibration, consistent with the P3 / P38 / CL4 density-stratified texture observed empirically.
- **Black holes as phase boundaries rather than absolute causal endpoints**: near $\chi \to 1$, baryonic cones close while dark / bulk cones remain open into the parent-dimensional reservoir.

9.7 Falsification Conditions

The Dark-Cone Branch fails fast under any of:

- Large dark-matter pressure or sound speed (would conflict with CMB & LSS).
- Observable gravitational waves faster than light (excluded by GW170817 to $\sim 10^{-15}$).
- Strong baryon energy-loss into dark FTL modes (would violate energy-momentum conservation tests).
- Early-universe dark acoustic oscillations not seen in CMB (would falsify cold-density assumption A3).
- Halo smoothing inconsistent with cold dark matter (same).

- Controllable laboratory FTL without a preferred-time rule (would violate $d\tau > 0$ ordering A2).

9.8 Test Handles

- **Test 1 — Dark-sector anticipation in lensing maps.** Compare ACT / Planck κ, Euclid Q1 lensing proxies, DESI / SDSS galaxy density at high-χ environments. Branch prediction: dark / lensing structure shows phase lead relative to baryonic density in dense regions. Connects to P30, OP13.
- **Test 2 — Cosmic-web nonlocal alignment.** Filament + void catalogues. Branch prediction: orientation correlations persist over distances too large for baryonic relaxation but consistent with dark-sector phase propagation. Overlaps with P3 / P38 line; *the mechanism is different from the grain-boundary phonon-scattering reading*.
- **Test 3 — Multimessenger transient timing.** EM + GW + neutrino + lensing / dark-sector proxy. Branch prediction: only *non-gravitational, non-electromagnetic* dark-sector precursors visible — GW remains luminal.
- **Test 4 — Halo sharpness vs environment.** Halo concentration / lensing residual coherence correlates with cosmic-web phase-boundary indicators. Branch prediction: more coherent halos around web nodes than in voids.

9.9 Architectural Status

The Dark-Cone Branch is **speculative**: not part of the default architecture; retained as a bounded alternative for the dark-sector causal structure should an explicit dark-sector phase-lead be observed in a follow-up of P30 / P3 / P38 / CL4. The default v3.6 / v7.6 architecture continues to forbid FTL information transfer in all sectors. Falsification of the branch by GW170817-style multimessenger constraints, DM cold-fluid bounds, or laboratory FTL tests does *not* falsify the default CCDR architecture; the architectures are independent. The branch swap operates at a Layer-4-adjacent level (causal-cone structure) and does not change Layers 0–3. Its naming is deliberate: *Dark-Cone Branch* or *Sector-Causal CCDR* — not “FTL dark matter” or “tachyonic DM”, both of which would be inaccurate. The mechanism is two causal cones (acoustic and optical / bulk) plus cascade-time ordering, not unrestricted superluminal propagation.

10. Complete Predictions Table

Forty numbered predictions carried jointly with CCDR v7.5 §13. Full prose descriptions for each appear in that document; the table below gives the tracked summary. **v3.5 extends the table from 38 to 40 with the addition of P39 (secular Λ-drift) and P40 (bulk-Weyl B-mode)** — both consequences of the v3.5 reshape (evaporation default endpoint and dimension-general trigger criterion respectively). Layer column refers to the five-layer falsifiability hierarchy (§5).

ID	**Short description**	**Mechanism**	**Status (through Round 8)**	**Decisive test**	**Layer**
P1	RVM BAO scale shift	$\rho_{vac}(H)$ modifies sound horizon; §2	Ambiguous (low-z SN-carried; Round-8 T4 triangle 5.1×)	DESI DR3 (2027)	4
P2	Non-Gaussian bispectrum	Grain-structure fNL; §8.1	Untested	Euclid 6-yr bispectrum	4

| P3 | Filament orientational correlation | Grain-boundary orientation; §8.2 | **Suggestive positive, Round-7 re-elevated**: Δr_texture(high − low) = +933 Mpc/h | LSST Y1 + DESI Y3 filament finder | 4 |

| P4 | CMB large-angle anomalies | Anisotropic reduction at LSS | Inconclusive (Planck cross-products p = 0.23/0.96/0.61) | LiteBIRD | 4 |

| P5 | KSS viscosity η/s = ℏ/(4πk_B) | Phase-boundary universality; §5.1 | Partially confirmed (ALICE/CMS ≈ 1.4 × KSS) | RHIC BES-II | 2 |

| P6 | DM halo fringes | Reducing-volume substructure | Untested | ELT + Roman | 4 |

| P7 | Dark-energy EOS oscillation | Multi/single-peak w(z) | **Retracted** (distinct from new P39 secular drift) | — | 4 |

| P8 | DM pulsar timing beat | ω_DM vs pulsar spin; §7.3 | Null (NANOGrav 15-yr 0.96σ) | — | 4 |

| P8a | Hellings–Downs hexapolar (ℓ = 6) | CDT C-phase 6-fold BZ | Structurally untestable (300 000× below NG15) | — | 0 |

| P8b | GW spectral index shift | δn_GW ~ v/3 | Marginal (300× tighter needed) | SKA | 4 |

| P8c | PTA × cosmic web (CL2) | Cross-patch composition; §4.5 | **Triply anchored**: R5 proxy r=+0.524, R7 real PR4 r=+0.314, R8 r=+0.266 | PTA × ACT DR6 third real-map pair | 4 |

| P9a | 1–2 GeV hadronic excess | Cascade-mediated | **Falsified, removed** | — | 3 |

| P9b | Anomalous MET at EW threshold | DM production at ~125 GeV | Null at current sens. | HL-LHC 3 ab^{-1} | 3 |

| P9c | QGP η/s dip–peak near T_c | Non-monotonic η/s(T) | **Falsified, removed** | — | 2 |

| P9d | δR_K / δR_K* = √3 | B-decay anomaly ratio | Cannot fire (both δR → 0) | — | 3 |

| P9e | Drell–Yan suppression at ~1 TeV | Cascade-mediated | Null | HL-LHC | 3 |

| P9f | Di-Higgs threshold excess | Cascade contribution | Null | HL-LHC | 3 |

| P10/P25 | DM peak-counting → N (hard core) | Cascade residues; §4.2, §9 | R8 refreshed: 3 XENONnT-2025 curves cover 521.4–2000 GeV; peak resolution pending | Tonne-scale DD with peak res. | 0 |

| P13/P29 | Time-varying DM abundance | Live cascade growth; §7.3 | **Round-8 replicates exactly** (84.6 % frozen / 15.4 % live) | Full-shape DESI DR3 + RSD | 4 |

| P14/P30 | Density-correlated Ω_DM/Ω_B | Early dense reductions; §7.4 | **Round-8 ACT DR6 real-map positive**: Δκ = +0.110, Spearman ρ = 0.221, p ≈ 10^{-56} | Collaboration κ × DESI/DES | 4 |

| P15/P31 | Drifting DM mass peak | Live-cascade mass drift | Subthreshold (per-event); P37 gives proxy-level test | Tonne-scale + event tagging | 0 |

| §6.1a | Local a_0 → Milgrom | Grain-boundary phonon scattering | **Confirmed, triply anchored, R8 re-anchored on n=171 SPARC** | (already confirmed) | 4 |

| §6.1b | High-z a_0 → cH_0 | RVM cooling curve | Suggestive trend (R5 Spearman r=0.886, R8 KMOS3D ρ=0.244) | Catalogue sample (~100 systems) | 4 |

| §6.3 | Koide scales sector-dependent | Spectral action on CDT | Consistent (sector distance 0.232) | First-principles SSM see-saw | 3 |

| P21 | TGFT-RG ν as IR FP | Condensate RG output | On hold (use ν_MOND) | Bahr–Steinhaus execution | 1 |

| P22 | γ_AS = γ_EPRL stage-by-stage | Asymptotic safety matching | Untested (toy calcs 9 % spread) | Bahr–Steinhaus | 1 |

| P23 | TGFT-RG cascade transitions | Stage = FP transition | Untested | sl2cfoam-next | 1 |

| P24 | DA/O upper bound N ≤ 11 | C ⊗ H ⊗ O gauge algebra | Consistent | — | 3 |

| P25 (Ω counting) | Ω_DM/Ω_B → N posterior | Cascade residue mix | Provisional N = 6 at 0.75 weight (moderated by prior) | Overridden by direct detection | 4 |

| P26 | 3 BZ directions ↔ 3 generations | CDT C-phase hex BZ | Untested | CDT C-phase + SSM see-saw | 3 |

| P27 | Geometric mass-tower ratios | RVM cooling; §2.4 | Untested | Direct detection — strongest falsifier | 0 |

| P28 | Staged CMB μ/y distortions | Per-stage spectral imprint | Consistent with FIRAS | PIXIE | 4 |

| P32 | Ringdown χ_4 → 1 signature | BH formation as reduction; §5.1 | Untested | LIGO-VK O5 + LISA | 0 |

| P33 | Density-correlated BAO | Kibble × density; §2.2, §7.4 | **Sign inversion to investigate** (R7 T13 LRG proxy: δr_d/r_d = −0.186; OP12) | DESI DR3 density-binned | 4 |

| P34 | Multi-component cluster σ_v | Cascade residue mix | Untested | eROSITA × Euclid stacks | 4 |

| P35 | Geometric sub-BAO harmonics | Cascade mass tower; §8.1 | Untested | DESI Lyα z = 2–3 | 4 |

| P36 | MOND-sequence ν extractor (CL1) | $a_0(z)$ via RVM cooling | **Joint robust** at 5.08e-3 (R5 & R8 T3); **standalone disagrees** at $\sim 10^{-5}$ (OP11) | 100-galaxy high-z sample | 4 |

| P37 | Phase-space drift from live DM | Orbit-phase correlation; §7.3 | Proxy null (R5 z = −0.43; R8 z = −0.82) | Tonne-scale + orbit tagging | 0 |

| P38 | Void-wall Cauchy tail | Grain-boundary scattering; §8.2 | **Suggestive positive, R7 promotion**: k_4 = 4.235 above k_4 > 4 on VAST VoidFinder | DESI Y3 + Euclid DR1 with ≥ 10^4 voids | 4 |

| P39 | Secular Λ-drift from bulk cooling | $C_0(t)$ drift via ν_bulk; §6.2 evap | Untested; DESI DR2 hint in predicted direction | DESI DR3 + Euclid Q3 + LSST Y1 (BRANCH SELECTION) | 4 |

| P40 | Bulk-Weyl CMB B-mode component | $W_5 = C_5^2$ at 5→4 trigger | Untested | LiteBIRD or CMB-S4 ($\sim 10^{-3}$ Planck units) | 0 |

11. Cross-Framework Bridges (extended to CL8 in v3.8)

The framework now contains enough empirically anchored pieces that internal consistency checks become first-class tests. Four bridges were registered in v3.4 (CL1–CL4); v3.5 added

CL5 — the joint constraint on (v_bulk, c_W) from P39 + P40; v3.6 added CL6 — the joint constraint on the lattice scale and bulk-projection geometry from P41 + P40; v3.7 added CL7 — joint ν self-consistency across BSM1 (inelastic, post-v3.7), BSM3 (dark photon), and §8.3-v7.8 (void-wall stacking), the framework's discriminator for whether the v3.7/v7.8 mechanism revisions are correct (multi-channel ν consistency) or single-channel accommodations (cross-channel ν extractions disagree); **v3.8 adds CL8 — universal $\delta\kappa_4/\delta\kappa$ ratio across cosmic-web interface types under the v7.9 universal coherent-stack realisation, the cross-probe consistency test partnered with the new pre-registered prediction P-A29 (master-curve self-consistency) as the v7.9 §8 discriminator pair**. The CL4 statement is now sharpened a second time in v3.8 (after its v3.7 sharpening): the v7.8 void/filament asymmetry framing was rejected by R12 Tempel filament measurement at >10σ; under v7.9 universal stacking, CL4 instead reads as a single master-curve consistency check across all interface types (voids, filaments, sheets, cluster boundaries) at their catalogue-measured r_wall/r_grain ratios.

11.1 CL1 — MOND-sequence ν cross-check

§6.1 (local $a_0 \to$ Milgrom, confirmed) and §6.1 (high-z a_0 trend, suggestive) together with the asymptotic $a_0 \to cH_0$ give three data points on the $a_0(z)$ curve, which via the RVM cooling H(t) is a direct function of ν.

Status through Round 8: the joint extractor returns ν_MOND = 5.08×10^{-3} in both Round 5 and Round 8 (T3) — robust under replication. The standalone variant in Round 7 (T15) and Round 8 (T7) returns ν_MOND ≈ 10^{-5} — disagreeing with the joint variant by approximately 500× and registered as Open Problem 11. Both variants lie below the SN value, so the qualitative direction of the triangle (ν_MOND ≪ ν_SN) is preserved.

11.2 CL2 — P8c wrong-sign as reducing-volume signature

Under the probability-weighted reducing volume, a non-representative local patch with atypical DM composition can produce sign inversion in a prediction that averages over species — without falsifying the cascade core.

Status through Round 8: TRIPLY ANCHORED. Round-5 PTA-proxy × density-κ Pearson r = +0.524, Spearman r = +0.433. **Round-7 real Planck PR4 κ map at 30 NANOGrav 15-yr pulsar positions: Pearson r = +0.314, same_sign_as_p30 = true** — first real-map confirmation. Round-8 replication on n = 2500 cross-sample: Pearson r = +0.266, same_sign_as_p30 = true. CL2 is no longer an interpretive escape hatch but an empirically anchored consistency check across three independent rounds and three different lensing-field paths.

11.3 CL3 — Time-crystal Q-factor as independent ν check

The cosmological time-crystal Q ~ 1/(2ν) ~ 500 manifests as the sharpness of the de Sitter asymptote in H(z). **In v3.5 the Q reading is preserved in the eternal-attractor branch (Q is the damping rate of an eternal oscillator) and reinterpreted in the evaporation default (Q is the fraction of the coherence lifetime during which the oscillator rings down).** Both readings give the same numerical value at present cosmic age; the difference is in the asymptotic state.

Status through Round 8: triangle ratio dropped to 5.1× in Round 8 from 30× in Round 5. sn_lowz_systematic_reinforced = true.

11.4 CL4 — Joint density-stratification of grain-boundary observables

Statement. P3 (filament orientational correlation) and P38 (void-wall Cauchy-tail profile) — both predicted by §8.2–§8.3 — should have the same sign in their density-stratified forms when measured on independent public catalogues, because both reflect the same underlying inhomogeneity in the cascade history. *Round-7 result: SUPPORTED. Round-8 corroborating evidence:* T5 high-density filament excess $z = +1.42$ (predicted direction).

v bracketing through CL4. The grain-boundary phonon-scattering $\Delta S/S \sim v$ sets the absolute scale of both density-stratified excesses. The point of CL4 is the **directional joint** — both observables in the predicted direction simultaneously, on independent catalogues — not the absolute v extraction.

Falsification. CL4 is falsified if a high-quality replication of P3 density-stratification returns the **opposite** sign from a high-quality replication of P38 density-stratification on the same survey.

11.5 CL5 — Joint constraint on (v_bulk, c_W) from P39 + P40 (NEW in v3.5)

Statement. P39 (secular w(z) drift) measures v_{bulk} through the rate of effective dark-energy decay. P40 (bulk-Weyl B-mode) measures $c_W \times C_5^2/G_5$ through the B-mode amplitude at large angular scales. The two predictions probe different quantities but emerge from the same v3.5 reshape (evaporation default + dimension-general trigger), and a joint measurement constrains the (v_{bulk}, $c_W \times C_5^2$) plane.

Possible outcomes:

- **Both confirm.** v3.5 reshape robustly endorsed; v_{bulk} and c_W both empirically constrained; higher-dimensional continuation programme (§22) is activated with explicit empirical foundations.
- **P39 confirms, P40 absent.** Evaporation default endorsed but the dimension-general trigger amplitude is below LiteBIRD/CMB-S4 sensitivity; OP15 (bulk-geometry uncertainty in c_W) is the key bottleneck for further P40 work.
- **P39 absent, P40 confirms.** Eternal-attractor branch endorsed but the dimension-general trigger criterion is supported through P40 alone; the cascade core's dimension-general statement remains, but the universe's endpoint is the v7.4 eternal-attractor reading.
- **Neither confirms.** v3.5 reshape's empirical content is null; the framework reverts to v3.4 architecture in practice (eternal-attractor default, 38 predictions, 4 brackets); no falsification of any layer, just absence of the new content.

Synthesis-level reading. CL5 is the v3.5 reshape's structural test. The five-bracket triangulation is robust: even if CL5 returns null in the most pessimistic case, the four pre-existing brackets (CL1–CL4) carry the framework's v-extraction content unchanged. **CL5 is added empirical content, not load-bearing structure.**

11.6 CL7 — Joint v self-consistency across BSM1, BSM3, and §8.3-v7.8 (NEW in v3.7)

Statement. The factor v appears in three independent observational channels under the v3.7/v7.8 mechanism revisions: (a) as the elastic-overlap suppression $\kappa_{elastic}$ in BSM1 (inelastic phonon DM, §21.4 v3.7); (b) as the kinetic-mixing parameter $\varepsilon_{dark} = c_{mix} \cdot v$ in

BSM3 (dark photon); (c) as the leading-order amplitude prefactor in the §8.3-v7.8 void-wall kurtosis formula. All three uses derive from the same leading-order cascade-overlap structure between optical-sector and acoustic-sector modes; the framework's prediction is therefore that the three independent extractions of v must agree within the precision set by sub-leading $O(v^2)$ corrections, i.e. ~5%, or in the framework's pre-registered consistency band, ±20% at 1σ.

Three independent v extractions. v_BSM1 from the inelastic mass-splitting δ_BSM1 (measured in XENONnT migdal data, R12+); v_BSM3 from the dark-photon kinetic-mixing ε_dark (measured against BaBar/LHCb dark-photon limits); v_void from the void-wall kurtosis amplitude (measured against R12 void-radius-stratified VAST + DESI-Y3 data).

Possible outcomes.

- **All three agree within ±20%.** v3.7/v7.8 mechanism revisions confirmed as multi-channel consistent; the cascade-overlap factor is empirically established as universal at leading order; both the §8.3-v7.8 stacked-interface revision and the §21.4-v3.7 inelastic-BSM1 reframe are endorsed.
- **Pairwise disagreement in [20%, 50%].** Inconclusive; either statistical limitations in one of the three measurements, or genuine sub-leading correction larger than expected. R13+ tightening required.
- **Pairwise disagreement >50% in any pair.** The universal-cascade-overlap assumption fails. One or both of the v3.7/v7.8 mechanism revisions is wrong; the framework must revisit either §8.3-v7.8 (if v_void is the outlier) or §21.4-v3.7 (if v_BSM1 is the outlier) or BSM3 (if v_BSM3 is the outlier).

Synthesis-level reading. CL7 is the structural test of the v3.7 reshape itself. Unlike CL5 (which tests the v3.5 reshape's evaporation-default + dimension-general trigger), CL7 tests whether the R11-driven mechanism revisions hold together as a coherent revision rather than as two independent accommodations. The seven-bracket triangulation (CL1–CL7) provides three brackets on v directly (CL1, CL3, CL7) plus four brackets on related framework parameters (CL2, CL4, CL5, CL6). If CL7 passes, the framework's empirical content is robustly extended by the v3.7 revisions; if CL7 fails, the v3.7 architecture's empirical content reduces to the pre-revision four-bracket triangulation plus the documented R11 rejections, and further mechanism work is required.

Falsifiability layer. Multi-layer (Layer 0 cascade-overlap factor universality, Layer 3 BSM particle inventory, Layer 4 cosmology). A FAIL on CL7 would be the framework's strongest single indicator that the v3.7 reshape needs further mechanism work — analogous to how the R11 P-A07/P-B12 rejections drove the v7.7→v7.8 + v3.6→v3.7 reshape.

11.7 CL8 — Universal $\delta k_4/\delta\kappa$ ratio at cosmic-web interfaces (NEW in v3.8)

Statement. Under the CCDR_v7.9 §8.2.bis universal coherent-stack realisation, all cosmic-web interfaces (void walls, filament walls, cosmic sheets, cluster boundaries) are coherent stacks of the same $(N - 4)$ cascade-residue species. The same microphysical source predicted to produce excess transverse kurtosis δk_4 also produces excess weak-lensing convergence $\delta\kappa$ at the same geometric interfaces, with both observables driven at leading order by the same $v \cdot N \cdot$ geometric-factor scaling. The framework's prediction is therefore that the ratio $\delta k_4/\delta\kappa$ is fixed and **environment-independent** — universal across all interface types — at the universal value calibrated on R11 void data: $\delta k_4/\delta\kappa \approx 12.5 \pm 2.0$.

Pre-registered prediction P-A28. The corresponding pre-registered prediction in CCDR_v7.9 §15.4 is P-A28 — Cross-probe consistency: universal $\delta k_4/\delta\kappa$ ratio at interfaces. Quantitative form: $\delta k_4/\delta\kappa \approx 12.5 \pm 2.0$ universal across voids, filaments, sheets, and cluster boundaries, with the calibration anchored on R11 void measurement (δk_4_void ≈ 1.45, $\delta\kappa$_void ≈ 0.116 → ratio ≈ 12.5). Observable: matched stacking of Euclid Q1 / DESI DR2 weak-lensing κ maps on the identical interface catalogues used for the transverse-kurtosis estimator. Decisive: R12 joint kurtosis + κ analysis on at least three interface types (voids + filaments + sheets).

Synthesis-level reading. CL8 partners with the v7.9 master-curve prediction P-A29 as the **v7.9 §8 discriminator pair**: P-A29 tests the master-curve geometry of the kurtosis observable alone; CL8 tests the cross-probe coherence between kurtosis and lensing convergence at the same interfaces. A consistent confirmation across both establishes the universal coherent-stack picture robustly at the multi-observable level — the v7.8 environment-dependent picture would predict different $\delta k_4/\delta\kappa$ ratios in different environments and is excluded by a universal-ratio confirmation. CL8 sits at the same architectural level as CL5 (the v7.5 (v_bulk, c_W) joint constraint) and CL6 (the v7.6 lattice-scale + bulk-projection joint constraint): a non-trivial multi-channel coherence requirement on a structural mechanism.

Why CL8 is registered now, not in v3.7. The v3.7 mechanism revision predicted environment-dependent $(N - 4)^2$ amplification (regime-i no amplification, regime-iii full amplification); a corresponding universal $\delta k_4/\delta\kappa$ ratio could not have been pre-registered in v3.7 because the v3.7 mechanism explicitly predicted environment-dependence. The v7.9 universal-stacking realisation removes environment dependence as a free parameter and consequently predicts the universal ratio. CL8 is therefore a structural prediction of the v7.9 mechanism, not a free addition.

Falsifiability layer. Layer 4 (RVM + CCDR §8) plus Layer 2 (TN + HaPPY through the κ–lensing-amplitude bridge). A measured environment-dependent ratio at 3σ across R12+ kurtosis + κ joint analysis falsifies CL8 and contributes to the CCDR §8.5.bis falsification condition (δ) — alongside P-A29 master-curve failure (condition α), P-A26 sheet-kurtosis failure (condition β), and P-A27 harmonic absence (condition γ), the simultaneous failure of all four would force a third mechanism revision in CCDR §8.

12. Layered Falsifiability with Fallback Paths (Summary)

Layer	**Component**	**Exit condition**	**Fallback**
0	CDT + cascade core	C-phase absent; non-geometric ratios; zero peaks below E[n_peaks] lower bound; **P40 inconsistent with predicted ℓ-shape**	CST or CFS
1	GFT + TGFT-RG	ν contradicts TGFT-RG; γ diverges; **v_bulk derivation contradicts P39**	ν reverts to fit
2	TN + HaPPY	RT fails in CDT C-phase	EG
3	SSM + AS + DA/O	Gauge couplings wrong	DA/O alone or empirical
4	RVM + CCDR	ν < 0; χ_k → 1 never triggers; **five ν extractors disagree (CL1–CL5)**; **P39 inconsistent with both branches**	**Branch swap: evap default ↔ eternal-attractor**

| 5 | Baryogenesis | η wrong by orders of magnitude | Rest survives |

No single failure destroys the programme. The cascade core can be falsified only by direct DM detection (Layer 0) or by absence of predicted geometric mass ratios (P27). **In v3.5 Layer 4 has two distinct exit conditions: branch swap (if P39 turns out flat or in wrong direction) and outright failure (if P39 conflicts with both branches simultaneously).** Every gap has a resolution path. Every prediction is testable.

13. The Observable-First Reclassification

N is an observable, determined by peak counting in direct detection under the dimension-origin prior. All algebraic routes (DA/O, HaPPY, AQN, Ω_{DM}/Ω_B fit) are consistency checks. The framework commits to accepting the direct-detection answer whatever it turns out to be.

The reducing-volume caveat applies quantitatively. The dimension-origin prior $P_{presence}(k) \sim (r_k / L_k)^{k-4}$ predicts asymmetric peak distribution (skewed toward lighter masses), cross-patch variation in DM composition, and asymmetric bounds (a single observed peak at stage k_0 gives only $N \geq 4 + k_0$ rather than $N = 4 + n_{obs}$).

v3.5 extends the prior's operational support. Beyond the direct patch-spread reading of (ii), Round 7 produced two independent grain-boundary observables both stratifying with density in the predicted direction (P3 filament texture, P38 void-wall kurtosis). The cross-stratification is registered as Cross-Link CL4 and provides one of the v3.5 five ν brackets (§14.4). **In v3.5, the prior also gains a second-class observational handle through P39**: if the secular Λ-drift is detected at the rate predicted by v_bulk derived from first principles (OP14), the bulk's coherence properties — and hence the L_k structure entering the prior — are constrained from the bulk side.

The v3.5 reshape's universe-endpoint commitment (evaporation default) further connects to the dimension-origin prior: **the bulk that our universe evaporates back into is also the bulk that birthed us**, so the species composition of our patch's reducing volume and the species composition of the bulk we evaporate back into are the same. This makes peak-counting + post-evaporation cosmology of the bulk (not directly observable from our 3+1D slice, but constrained by P39 amplitude) a single joint estimation problem rather than two separate ones.

14. Empirical Tests Across Public-Data Rounds

Synthesis-level summary of the four rounds of public-data testing. Round 5 is the operational baseline; Round 7 is a five-test extension battery; Round 8 is the twelve-test replication audit; Round 9 is the twenty-five-test Tier-A audit. Per-test details and headline movements follow.

14.1 Round 5 — Twelve-Test Public-Data Battery

The Round-5 battery is preserved from v3.4 §15 as the reference frame against which Round 8 is the explicit replication audit (§17) and against which the Round-7 extension (§16) reports its movement.

| Test | Prediction | Source | Result | Status |

| --- | --- | --- | --- | --- |

T1	P30 ext.	ACT-proxy + Planck-proxy κ	Δκ +1.26e-4 / +4.30e-5; both same-sign	Cross-access-path replication, sign-stable
T2	P30 systematics	Euclid Q1 6-way subset audit	All same sign; threshold monotonic; patch spread 1.4e-5	Subset-stable, threshold-stable
T3	High-z a_0	SPARC + KMOS3D-proxy	Local 1.186e-10; high-z mean 3.91e-10; v_MOND 5.08e-3	Joint extractor in band
T4	ν re-drive	Pantheon+ + DESI DR2	Triangle 30× disagreement	SN systematic reinforced
T5	P3 density dep.	Euclid Q1 filament ordering	+0.017, z = +1.32	In direction, not significant
T6	P29 multi-z growth	DESI fσ8 6-pt	α = −0.284, 85 % frozen / 15 % live	Wrong-sign deepened, prior-decomposed
T7	P36 ν standalone	Same as T3, standalone fit	v_MOND 5.08e-3, gap 0.453	In band
T8	P33 density-BAO	DESI 5-pt summary	rd 146.8–147.1, spread 0.3 within errors	Framework set, awaits real data
T9	CL2 cross	PTA-proxy × density-κ	r = +0.524, same sign as P30	Anchored as suggestive positive
T10	P38 void-wall	20-void density-min proxy	k_4 = 2.81 (below 4)	Proxy null
T11	Hard-core window	6 HEPData curves	Window true, peak ready false	Operational overlap, no peaks
T12	P37 phase-space	n = 3000 sim. events	z ≈ −0.43	Proxy null

Round-5 supplied the operational baseline used to interpret Round 7 (extension to T13–T17, §16) and Round 8 (replication audit on T1–T12, §17).

14.2 Round 7 — Five-Test Extension Battery

Round 7 is a five-test extension targeted at three structural gaps left by Round 5: (i) absence of a real-map (rather than proxy-map) cross-correlation between PTA residuals and a public CMB-lensing field for CL2; (ii) absence of a public-catalogue void sample for P38; (iii) absence of an explicit density-stratified test for P3 on a public cosmic-web filament product. Per-test details appear in CCDR v7.5 §19; the synthesis-level summary is below.

14.2.1 T13 — P33 Density-Correlated BAO on DESI-LRG Proxy

Source/method. DESI LRG public clustering products (5000 data, 10 000 random points, z ∈ [0.4, 1.1]), split by k-NN density estimator (k = 16). **Headline.** δr_d/r_d = −0.186 (high − low). support_like = false.

Synthesis-level reading. First sign-disagreeing public-data result since the v6.3 EOS retraction. Two competing interpretations registered as Open Problem 12: (a) k-NN methodological-systematic; (b) cascade picture missing opposite-sign coupling at recombination — for example, dense regions complete first reduction before recombination, imprinted r* smaller in compactified-earlier regions. Resolution: DESI DR3 density-binned likelihoods.

14.2.2 T14 — CL2 P8c × Real Planck PR4 κ Map

Source/method. NANOGrav 15-yr (30 pulsars) crossed against public Planck PR4 lensing reconstruction at pulsar sky positions. **Headline.** Pearson r = +0.314, Spearman r = +0.190, **same_sign_as_p30 = true**.

Synthesis-level reading. First time CL2 has been tested against a real Planck PR4 κ map. Combined with Round-5 proxy and Round-8 replication, CL2 is now triply anchored. The bridge moves from 'proxy-only suggestive' to 'empirically robust' at synthesis level.

14.2.3 T15 — P36 High-z a_0 Trend on KMOS3D Public Release

Headline. closed_gap_fraction = 0.0165, v_MOND_calibrated = null. **Synthesis-level reading.** Standalone variant of v_MOND extractor (without SPARC mapping) does not converge. Methodological-dependence finding registered as Open Problem 11.

14.2.4 T16 — P38 Void-Wall Cauchy Tail on VAST VoidFinder Catalogue

Source/method. Public VAST VoidFinder geometry catalogue (1163 maximal voids). **Headline.** Transverse kurtosis k_4 = 4.235 (above k_4 > 4 threshold). **Synthesis-level reading.** First public-catalogue threshold crossing for P38 in seven rounds. P38 promoted from proxy null to suggestive positive.

14.2.5 T17 — P3 Filament Orientational Correlation, Density-Stratified

Source/method. Public cosmic-web filament catalogue with density quantile split at 0.5. **Headline.** Δr_texture(high − low) = +932.87 Mpc/h. **Synthesis-level reading.** First public-data realisation of density-stratified P3 in the predicted direction. P3 re-elevated to suggestive positive with explicit density-stratification structure.

14.2.6 Round-7 Headline Movement

- **P38 promoted** to suggestive positive on T16.
- **P3 re-elevated** with density-stratification on T17.
- **CL2 doubly anchored** through T14.
- **P33 sign inversion to investigate** on T13 (OP12).
- **P36 standalone-vs-joint methodological dependence** on T15 (OP11).

14.3 Round 8 — Twelve-Test Replication Audit

Round 8 re-executes the Round-5 twelve-test battery on refreshed public products. The explicit aim is to measure round-to-round stability of the headline suggestive positives and detect any methodological drift across the year between Round 5 and Round 8.

14.3.1 Replication Stability Summary

Test	Round 5	Round 8	Stability
T1 P30 ext	Δκ $+1.26 \times 10^{-4}$ (ACT-proxy)	Δκ +0.110 (ACT DR6 real)	Same-sign, magnitude ↑ on real map

| T2 systematics | subsets same-sign; thr. mono.; patch 1.4e-5 | subsets same-sign; thr. non-mono.; patch 0.120 > baseline | Same-sign stable; new threshold + patch-spread anomalies |

| T3 a_0 joint | v_MOND = 5.08×10^{-3}, gap 0.453 | v_MOND = 5.08×10^{-3}, gap 0.454 | Robust under replication |

| T4 ν triangle | max ratio 30× | max ratio 5.1× | Direction stable, magnitude reduced |

| T5 filament | z = +1.32 (in direction) | z = +1.63 (in direction) | Marginal positive, slightly stronger |

| T6 P29 growth | α = −0.284, 84.6 % frozen | α = −0.284, 84.6 % frozen | Reproduced exactly |

| T7 ν standalone | v_MOND = 5.08×10^{-3} | v_MOND = 1×10^{-5} | Disagrees with joint, methodological dependence |

| T8 density-BAO | spread 0.3 Mpc within errors | $\delta r_d/r_d = 3.5 \times 10^{-5}$ | Both screening, awaiting DR3 |

| T9 CL2 cross | r = +0.524 (proxy) | r = +0.266 (replication) | Same-sign, magnitude reduced |

| T10 void-wall | $k_4 = 2.81$ | $k_4 = 3.52$ | Closer to threshold, still below |

| T11 hard-core | 6 tables, window true | 3 XENONnT-2025 tables, window true | Stable, refreshed source |

| T12 phase-space | z = −0.43 | z = −0.82 | Stable null, slightly more negative |

14.3.2 Round-8 Headline Movement

- **P30 reaches the real ACT DR6 lensing field directly.** Δκ = +0.110, all six subset splits same-sign, bootstrap CI excluding zero, Spearman ρ = 0.221 at $p \approx 10^{-56}$.
- **Triangle ratio drops 30× → 5.1×.** Direction of disagreement preserved; sn_lowz_systematic_reinforced = true.
- **Patch spread now exceeds baseline Δκ in T2.** patch_spread = 0.120 vs. baseline 0.110. OP13.
- **P29, P30, CL2, v_MOND_joint reproduced** under replication.

14.3.3 Synthesis-Level Reading of Round 8

Round 8 is the **first explicit replication audit** of the Round-5 headline suggestive positives. Every Round-5 suggestive positive that admits a replication on refreshed public inputs is **same-sign in Round 8**. The four anomalies registered (T2 patch spread, T2 threshold non-monotonicity, T7 standalone vs. joint, T8 prior-shrunk sample) are all methodological diagnostics rather than sign-flip falsifications.

14.4 Round 9 — Twenty-Five-Test Tier-A Public-Data Audit (NEW in v3.6)

The Round-9 battery is a deliberate broadening of Round-7/8 work to **25 Tier-A tests** with strengthened parser, sampling and null-control protocol (READMEv9 / v9.4 patches): DR2-only BAO screens for the secular Λ-drift; quality-cut high-z acceleration; redshift-residual cluster mixtures; artifact-controlled direct-detection curve extrema with endpoint/grid rejection; physical FIRAS μ/y bounds; generic P40 B-mode template limits using the BICEP/Keck BK18 bandpower table directly (cols 3–134, ell-col fix); within-field density-shuffle nulls for filaments; matched lognormal nulls for void kurtosis. Statuses are deliberately conservative: confirm_like, suggestive, null, data_limited, diagnostic_only, weak_offset_no_trend, single_bracket_only, methodology_finding, readiness_only, consistent_bound_only, broken.

14.4.1 Headline Movement (Round 9)

Test	**Prediction(s)**	**Status**	**Headline metric / source**
T01	P39 / ν RVM audit	diagnostic_only	Pantheon+ × DR2 SN-only ν proxy bounded; conservative diagnostic, not full RVM likelihood
T02	P39 secular Λ-drift (DR2-only)	null	DR2-only BAO mean-vector trend zero/opposite under observable-wise normalisation
T03	ν low-z cuts	diagnostic_only	ν proxy hits bound under low-z cuts; signal carried by low-z
T04	P14/P30 (ACT DR6 × Euclid Q1)	data_limited	Real ACT DR6 lensing-map sampling needs --allow-large + healpy
T05	P14/P30 cross-check (Planck)	data_limited	Planck PR4 lensing-map HEALPix sampling pending healpy
T06	OP13 / P30 (Euclid patch CV)	suggestive	Field-aware patch CV = 2.054 vs randomised within-field null (z = 423)
T07	P3 density-stratified filaments	null	Within-field density-shuffle null absorbs the Q1 orientation excess
T08	P3 (independent VizieR filaments)	suggestive	Mean orientation correlation +0.045 across 6 angular bins (J/A+A/530/A122)
T09	P38 void Cauchy tail (VAST)	null	k_4 excess does not exceed matched lognormal null after kurtosis-vs-shape control
T10	P34 cluster mixture (eROSITA)	data_limited	.fits.tgz archive parsing now extracts; columns insufficient for redshift-binned mixture
T11	**§6.1a SPARC local a_0**	**confirm_like**	**$a_0 = 9.44 \times 10^{-11}$ m/s^2 (0.787 × Milgrom), n = 175 SPARC galaxies, RMS 0.195 dex, σ_boot 0.012 dex; quadruply anchored**
T12	§6.1b high-z a_0 KMOS/KROSS	weak_offset_no_trend	Quality-cut KMOS3D/KROSS proxy above local but flat with z
T13	CL1 ν triangulation	single_bracket_only	Only the SPARC-anchored bracket converges; high-z and SN/BAO brackets bound-hit or fail
T14	§6.1 anchor stability	methodology_finding	SPARC anchor stable; standalone high-z extractor instability traceable to quality cuts
T15	NANOGrav simple-spectrum scan	suggestive	318 .par files, 152 wideband residual-like files, 836 .tim files; per-pulsar peak-power chains tabulated; specialist TEMPO2/enterprise refit required
T16	CL2 (NANOGrav × κ)	data_limited	ACT/Planck map sampling deferred without healpy; RAJ/DECJ/ELONG/ELAT/PSRJ parsing improved

| T17 | P8b GW spectral-index δn ~ v/3 | data_limited | Public NANOGrav posterior tables don't yet expose δn_GW band |

| T18 | P10/P25 hard-core window | readiness_only | Mass window covered; peak resolution still pending |

| T19 | P10/P25 curve extrema | data_limited | Endpoint/grid-artifact rejection leaves insufficient non-endpoint sample |

| T20 | P15/P31 release drift | null | No coherent extremum drift across DD releases beyond sensitivity changes |

| T21 | P28 FIRAS μ/y staged distortion | consistent_bound_only | Physical staged distortion amplitude below FIRAS residual limits |

| T22 | P4 low-l Planck anomalies | data_limited | Public PR3 spectrum tables don't yet expose anomaly-direction proxy |

| T23 | P40 BICEP/Keck BB low-ℓ | consistent_bound_only | BK18 bandpower (cols 3–134, ell-col 2 fix) leaves residual room for P40-shaped low-ℓ contribution after uncertainties |

| T24 | P32 GWOSC ringdown | diagnostic_only | HDF5 strain prioritised over GWF; simple damped sinusoid does not resolve overtone |

| T25 | P5 KSS η/s | data_limited | HEPData column inspection rejects flow-only columns (v2{2,Δη} etc.); explicit η/s columns absent in ins1666817 |

14.4.2 Synthesis-Level Reading of Round 9

Twenty-five tests; one new **confirm_like** (T11 quadruply anchors §6.1a local a_0 via a third independent SPARC parsing pipeline through Zenodo's Rotmod_LTG product — $a_0 = 9.44 \times 10^{-11}$ m/s² is 0.787 × Milgrom, RMS 0.195 dex, bootstrap σ_log10 0.012 dex on n = 175 galaxies, n_points = 3389; Round-5/Round-8 in-sample fits already triply anchored, Round-9 is the fourth pipeline). Three **suggestive** (T06 field-aware patch CV at z = 423 reinforces OP13; T08 independent VizieR filament catalogue mean correlation +0.045 — first cross-catalogue support for P3 outside Tempel et al.; T15 NANOGrav per-pulsar simple-spectrum tabulation — peak-power chains visible but a specialist TEMPO2/enterprise refit is required for confirmation). Two **consistent_bound_only** (T21 FIRAS staged μ/y stays below residual limits; T23 BK18 BB low-ℓ leaves residual room for the P40-shaped contribution after uncertainties) — both are *protective bounds* compatible with the framework but not signature confirmations. Four **null** (T02, T07, T09, T20). Twelve are **data_limited / diagnostic_only / weak / single_bracket / methodology / readiness** awaiting healpy, --allow-large, or specialist refits.

No new prediction is falsified. The DR2-only BAO null (T02) sharpens the asymmetry between the eternal-attractor and evaporation-default branches: a *pure DR2 cut* shows no Λ-drift trend at present sensitivity, leaving DESI DR3 + Euclid Q3 + LSST Y1 as the decisive joint-data branch-selection battery for P39. T06 (patch-CV at z = 423 vs within-field shuffle) reinforces OP13 (cross-patch composition variation vs proxy mode-coupling artefact) but with a strengthened, field-aware null — the residual anomaly is now harder to dismiss as a mask/depth/random artefact. T07 absorbs the Round-7 T17 density-stratified Q1 filament excess into within-field density-shuffle nulls — an internal protective-belt tightening: P3 evidence now rests on the *cross-catalogue* T08 result and on the bulk texture amplitude (Round 7 Δr_texture(high − low) = +932.9 Mpc/h on Tempel et al.) rather than within-field stratification. T09 constrains the void Cauchy tail to be lognormal-null compatible on VAST, leaving P38's $k_4 > 4$ result on VAST (Round 7 T16 $k_4 = 4.235$) as the lone surviving signature awaiting a DESI Y3 /

Euclid DR1 confirmation — i.e. the cleaner reading of the empirical pattern is now *log-radius excess separately reportable* (Round-9 protocol explicitly distinguishes radius-like kurtosis from log-radius-only excess).

14.4.3 Movement on the Five Layers After Round 9

- **Layer 0** (CDT + cascade core): unchanged empirically. T18 (DD readiness) refreshed; T23 (P40 BK18) admits residual room for the bulk-Weyl B-mode contribution at low-ℓ. T24 (GWOSC ringdown) provides a public-strain diagnostic but no Kerr-deviation signal at present method scope.
- **Layer 1** (GFT + TGFT-RG): unchanged empirically. T01 / T03 confirm the Pantheon+ ν proxy is bound-dominated and low-z-carried, consistent with the Round-5/8 reading.
- **Layer 2** (TN + HaPPY): unchanged.
- **Layer 3** (SSM + AS + DA/O): **first independent SM-sector empirical handle** through the LHCb angular B → K*μμ analysis (§18.4). Layer 3 was previously consistent-with-data only; v3.6 adds a positive correlated signature.*
- **Layer 4** (RVM + CCDR): T11 quadruply anchors §6.1a; T06 reinforces OP13; T02 deepens the empirical asymmetry between branches (P39 branch-selection awaits joint DR3+Q3+LSST data); T08 provides the first independent cross-catalogue support for P3; T13 / T14 sharpen CL1's standalone-vs-joint methodology dependence.
- **Layer 5** (Baryogenesis): unchanged.

14.4.4 SM-Sector Signature: LHCb b → s μ μ Angular Anomaly (arXiv:2512.18053)

In December 2025 the LHCb collaboration published the high-statistics 8.4 fb^{-1} Run-1 + Run-2 angular analysis of $B^0 \to K^0$*$(\to K^+\pi^-)\mu^+\mu^-$ (arXiv:2512.18053). The full set of CP-averaged and CP-asymmetric observables S_i, A_i is reported with complete K-π S-wave contribution and full muon-mass effects. The headline is a persistent SM tension in the CP-averaged sector at 3.6–4.1σ corresponding to a positive effective Wilson-coefficient shift ΔRe(C_9^{eff}) ≈ +0.9 (low-q^2 bins, depending on form-factor model), while all CP-asymmetric observables A_i remain consistent with zero.*

CCDR-derived reading. The framework treats the SM Lagrangian as emergent at the k = 4 stage via the Connes–Chamseddine spectral action on the CDT geometry (Layer 3, SSM + AS + DA/O). In v3.6 the residual lattice / grain-boundary structure left over from the 5 → 4 nucleation is taken to leave a small, parametrically suppressed correction to the effective Dirac operator at the TeV scale. Two structural features of the lattice correction are predicted from the Layer-3 architecture: (i) it is **vector-like** — the holographic encoding preserves chirality but adds a universal shift from the tensor-network bond-dimension scaling D_eff → e^{1/4}D_eff; (ii) it is **real / CP-even** — the metric / Regge contribution carries no new CP-violating phase. In the b → sμμ effective Hamiltonian this manifests as a positive shift $\delta C_9^{eff} > 0$ with magnitude set by the ratio of the crystallisation scale to the Planck scale, becoming visible in rare processes that probe virtual momenta ~ few TeV. The expected pattern is $\delta C_9^{eff} \approx +1$, δA_i ≈ 0 — exactly the LHCb signature.

The same physics that imprints (a) the bulk-Weyl B-mode at the 5 → 4 trigger (P40, Layer 0), (b) the secular Λ-drift via v_bulk (P39, Layer 4), and (c) the patch-level cascade-residue composition variation (P14/P30, Layer 4) also imprints δC_9^{eff} in the low-q^2 bins of $B^0 \to K^{*0}\mu^+\mu^-$ (Layer 3). The v3.6 reshape registers this as CL6 — Joint constraint on (lattice scale, vector-coefficient sign, CP-phase parity) from B → K+ bulk-Weyl B-mode**: both predictions arise**

from the same lattice substructure; CL6 fixes the parity of the shift (real, vector) and the amplitude scale, while CL5 fixes the bulk-projection geometry. Status: **suggestive supporting; formalised as** P41 (b → sμμ angular tension as cascade-residual lattice signature)** — first independent SM-sector empirical handle on Layer 3.

Caveats. (i) δC_9^{eff} extraction depends on form-factor model; the 3.6σ ↔ 4.1σ range tracks this systematic. (ii) Conventional explanations (Z′, leptoquark) remain viable; the CCDR reading is *not* exclusive — it predicts a *correlated* signature in (low-q^2 C_9, high-ℓ B-mode P40, secular w(z) drift P39) that other explanations do not generically reproduce. (iii) Other b → s observables (R_K, R_K*, B_s → φμμ, P_5′ in adjacent q² bins) should show the same vector-real shift; the CCDR-specific discriminator is the absence of a corresponding C_{10} or scalar/tensor-coefficient shift,* and *the same-sign sign of δC_9^{eff} across all flavour channels. (iv) Independent collaboration follow-ups (Belle II, future LHCb upgrades) will sharpen the form-factor systematic and test sign-coherence across channels. Reference: LHCb collaboration, arXiv:2512.18053 (Dec 2025).*

14.4.5 Round-9 Headline in v3.6 Architecture

Round 9 leaves all five layers structurally intact; promotes Layer 3 from consistent-with-data to *correlated-positive-signature* via P41 / CL6; reinforces P3 (cross-catalogue), §6.1a (quadrupled anchor), and OP13 (field-aware patch CV); deepens the P39 branch-selection asymmetry through the DR2-only null; and converts the P38 evidence to a log-radius-only / Round-7-VAST-only basis pending higher-grade void catalogues. **One new prediction (P41), one new cross-link (CL6), zero falsifications, no architectural reshape required.** The total goes from 40 predictions / 5 cross-links to 41 / 6.

15. Round-4, Round-5, Round-7 and Round-8 Results Summary

Round 4 (Euclid Q1 six-test battery): one suggestive positive (P30), one consistent-but-not-confirming (P3), one suggestive trend (high-z a_0), three null-or-weak. Layer 4 carrying all active empirical signals.

Round 5 (twelve-test battery): three structural movements — CL2 anchored empirically; CL1/CL2/CL3 triangle 30×; hard-core window overlap reached.

Round 7 (five-test extension battery, T13–T17): three structural movements:

- **CL2 doubly anchored at real-map level** through T14 real Planck PR4 cross at NANOGrav 15-yr pulsar positions (r = +0.314).
- **P38 promoted from proxy null to suggestive positive** through T16 (k_4 = 4.235 above k_4 > 4 threshold on VAST VoidFinder catalogue with 1163 maximal voids).
- **P3 re-elevated with explicit density-stratification structure** through T17 (Δr_texture(high − low) = +932.9 Mpc/h on public cosmic-web filament catalogue).

Round 7 also produced **one sign inversion to investigate** (T13 P33: δr_d/r_d = −0.186 on LRG proxy; OP12) and **one methodological-dependence finding** (T15 P36 standalone extractor fails to converge while joint extractor returns 5.08×10^{-3}; OP11).

Round 8 (twelve-test replication audit, T1–T12 re-execution): three structural movements:

- **P30 reaches real ACT DR6 lensing field directly** with Δκ = +0.110, Spearman ρ = 0.221 at $p \approx 10^{-56}$ on n = 5000, all six subset splits same-sign.

- **Triangle ratio drops 30× → 5.1×.** SN extractor on Round-8 inputs converges closer to the MOND value; sn_lowz_systematic_reinforced = true.
- **Patch spread now exceeds baseline Δκ in T2** (0.120 vs. 0.110, threshold scan no longer monotonic). OP13.

Round 8 also reproduced the headline suggestive positives: P30 same-sign on real ACT, CL2 r = +0.266, v_MOND_joint unchanged at 5.08×10^{-3}, P29 frozen/live decomposition reproduced exactly.

Local a_0 anchor re-fit in Round 8 on n = 171 SPARC curves through a different parsing pipeline returns local g_char ≈ 1352 (m s^{-2})/Mpc and closed gap fraction 0.45, within the Round-5 envelope. **Triply confirmed across three independent pipelines.**

v3.5 framing: the Round-7 + Round-8 picture is preserved in v3.5 unchanged from v3.4 in its empirical content. The v3.5 reshape (evaporation default + dimension-general trigger + higher-dim continuation) is **architectural rather than empirical** — it changes how the synthesis structures the framework, but does not move any of the Round-7/8 results. The new content is in P39 + P40 + CL5 + §22, all of which are *future* empirical handles rather than re-readings of past results.

16. Synthesis-Level Scorecard after Rounds 7 and 8 (v3.5)

The combined Round-7 + Round-8 picture, mapped onto v3.5 architecture (with evaporation default endpoint, dimension-general trigger, and four-branch architecture):

16.1 Movement on the Five Layers

- **Layer 0 (CDT + cascade core):** unchanged empirically. **In v3.5 the layer's content is broader** — it now includes the dimension-general trigger criterion at all k, and P40 (bulk-Weyl B-mode) is a Layer 0 prediction. Window overlap remains true; peak resolution still pending.
- **Layer 1 (GFT + TGFT-RG):** unchanged empirically. **In v3.5 the layer also covers v_bulk derivation via Path 6 + Path 9.** OP14 is the v3.5 priority for this layer.
- **Layer 2 (TN + HaPPY):** unchanged.
- **Layer 3 (SSM + AS + DA/O):** unchanged.
- **Layer 4 (RVM + CCDR): substantively reinforced and structurally extended**. Two new suggestive positives (P38, P3 density-stratified) carried over from v3.4. CL2 triply anchored. **CL5 added as a fifth v bracket via P39 + P40.** Branch architecture extended (eternal-attractor as v3.5 swappable; evaporation as default). Three v3.5 open problems registered (OP14, OP15, OP16). One sign inversion under investigation (P33). **The P39 outcome is the v3.5 branch-selection observable** — neither outcome falsifies the layer; it determines which branch is the working default.
- **Layer 5 (Baryogenesis):** unchanged.

16.2 Movement on the Five Cross-Links

- **CL1 (MOND-sequence v):** joint extractor robust at v_MOND = 5.08×10^{-3}. Standalone disagrees by ~500× — Open Problem 11.

- **CL2 (PTA × κ):** triply anchored — proxy (Round 5), real Planck PR4 (Round 7), real ACT-replication (Round 8).
- **CL3 (time-crystal Q):** triangle ratio dropped to 5.1× in Round 8. **In v3.5 reinterpreted as fraction of coherence lifetime in the evaporation default branch; preserved as eternal-oscillator damping rate in eternal-attractor branch.**
- **CL4 (grain-boundary stratification):** supported by Round-7 T16 (P38) + T17 (P3) + Round-8 T5 (P3).
- **CL5 (v_bulk × c_W joint constraint, NEW v3.5):** awaiting first measurement (P39 + P40). **Will determine whether v3.5 reshape has empirical content or reduces to v3.4 in practice.**

16.3 Predictions Status by Tier

- **Tier 1 anchored predictions** (CCDR + RVM + AS): a_0 → Milgrom triply confirmed; v audit closed; AS untested directly but consistent.
- **Tier 2 structurally committed** (CDT + TN + GFT/EPRL + TGFT-RG + SSM): no movement at this stage; resolution paths 6, 7 carry forward; **Path 6 takes on the additional role of OP14 (v_bulk) in v3.5**.
- **Tier 3 consistency checks** (AQN + DA/O + HaPPY + LGT): no movement.

16.4 Two new predictions awaiting first test

P39 (secular Λ-drift) — DESI DR3 + Euclid Q3 + LSST Y1 forecasts; first informative result possible from DESI DR3 alone (~2027). **This is the branch-selection observable.** Confirmed flat $w(z) = -1$ → eternal-attractor branch endorsed; confirmed secular drift (Λ_eff decreasing) → evaporation default endorsed. Either outcome leaves the rest of the framework intact.

P40 (bulk-Weyl B-mode) — LiteBIRD or CMB-S4 sensitivity; first informative result possible 2027–2030 depending on launch and analysis schedules. **This is the dimension-general trigger test.** Confirmed → cascade core's dimension-general statement endorsed; absent → bulk-geometry uncertainty (OP15) becomes the bottleneck for further P40 work, but the cascade core itself is not falsified.

17. Open Problems — v3.4 and v3.5 Extensions

Open Problems 1–10 are carried over from v3.3 unchanged. Three v3.4 additions (OP11–13) were detailed there; v3.5 adds three more (OP14–16) on the basis of the v3.5 reshape.

17.1 v3.4 Additions (Carried Forward)

OP11 — Standalone-vs-joint v_MOND extractor instability. Joint extractor 5.08×10^{-3} across rounds; standalone $\approx 10^{-5}$; ~500× discrepancy. Resolution: derivation of SPARC mapping from RVM cooling + independent high-z anchor at $z > 1$.

OP12 — P33 density-BAO sign inversion. Round-7 T13: $\delta r_d/r_d = -0.186$ on DESI-LRG proxy. Two competing interpretations (methodological-systematic vs. structural-mechanism). Resolution: DESI DR3 density-binned likelihoods.

OP13 — Patch-spread larger than baseline in P30 systematics. Round-8 T2: patch_spread = 0.120, baseline $\Delta\kappa = 0.110$. Two interpretations (cross-patch DM-composition variation vs. proxy mode-coupling artefact). Resolution: patches aligned on physical scales + quantitative prediction from prior.

17.2 v3.5 Additions

OP14 — First-principles derivation of v_bulk. Under the evaporation-default endpoint, the bulk-RVM coefficient v_bulk introduced in CCDR v7.5 §2.5 governs the secular Λ-drift (P39) and the universe's evaporation timescale. v_bulk is currently a free parameter; deriving it from first principles via Path 6 (Bahr–Steinhaus stage-by-stage RG) applied to the bulk's TGFT description, or via Path 9 (Diósi–Penrose × reducing-volume), would turn P39's predicted drift rate into a quantitative output rather than a fitted amplitude. **This is structurally the most important new theoretical task introduced by the v3.5 reshape.** Direct connection to Path 6 + Path 9.

OP15 — Bulk-geometry uncertainty in c_W. The bulk-Weyl B-mode prediction P40 carries an O(1) theoretical uncertainty in its amplitude because c_W's value depends on the bulk geometry at trigger time. Standard AdS/CFT calculation gives $c_W \approx 1/(8\pi^2)$; RS-warped or dS_5 backgrounds may give different O(1) factors. A first-principles derivation of c_W in the specific bulk geometry that produced our Big Bang would tighten the predicted amplitude range from order-of-magnitude to factor-of-few precision. Direct connection to Path 6 stage-1 work (§22 Stage 1).

OP16 — Higher-dimensional continuation. After the framework is decisively confirmed or falsified in our 3+1D slice, the higher-k cascade stages remain to be explored. **In v3.5 this is treated as a structured five-stage continuation programme rather than an open problem in the traditional sense; see §22 for the full programme.**

17.3 Connection to Resolution Paths

v3.5 makes explicit which resolution paths address which open problems:

- OP11 ↔ Path 6 (SPARC mapping derivation from RVM cooling).
- OP12 ↔ pure empirical (DESI DR3); no theoretical resolution path needed.
- OP13 ↔ Path 9 (quantitative L_k → quantitative patch-spread prediction).
- **OP14 ↔ Path 6 + Path 9 (v_bulk via bulk-side TGFT-RG + collapse-scale).**
- **OP15 ↔ §22 Stage 1 (theoretical generalisation of trigger).**
- **OP16 ↔ §22 (entire higher-dimensional continuation programme).**

17.bis OP1–OP18 Resolution Status — v3.8 Update from v14 Computational Resolver

The v3.8 release synchronises the Synthesis OP inventory with the CCDR_v7.9 §20 status report. The v14 computational resolver (schema `ccdr-open-problem-computational-resolver-v14`) provides status classification across the eighteen OPs using a frozen taxonomy (RESOLVED_JOINT, SIMULATION_RESOLVED, SIMULATION_RESOLVED_PRIOR, PROVISIONALLY_RESOLVED_METHODOLOGICAL, GEOMETRY_SCAN_RESOLVED_PRIOR, RESOLVED_PUBLIC_DATA, RESOLVED_PUBLIC_DATA_REMNANT_CONSISTENCY, RESOLVED_PUBLIC_CONSTRAINTS_BROAD_WINDOWS, PROTOTYPE_ONLY, PARTIAL, OPEN). At the Synthesis level the resolutions enter as architectural commitments: v is fixed at the OP11-RESOLVED_JOINT working value $5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$ across all framework predictions; the OP13 L_k/r_k calibration = 1.195 propagates through the reducing-volume Monte Carlo (OP6/OP10) and the asynchronous-cascade simulator (OP7), which together export priors that downstream predictions consume. The full per-OP audit is documented in CCDR_v7.9 §20.

Strict-resolved (public data). OP11 — RESOLVED_JOINT. Joint extractor accepted by all four datasets (pantheon_plus_sn, desi_dr2_bao, planck_growth, nanograv_15yr) at 95%; standalone rejected by desi_dr2_bao at $p \approx 7.8 \times 10^{-3}$. OP13 — strict-resolved on Planck PR4 lensing; $\Delta\kappa = 9.95 \times 10^{-3}$, L_k/r_k = 1.195. OP8/OP9 — resolved at remnant-consistency level (not full QNM closure). OP17/OP18 — broad-window public macro screening; DCN_k allowed fraction 0.946.

Simulation-resolved prior. OP6/OP10 — reducing-volume Monte Carlo with OP11-committed ν, OP13-calibrated L_k/r_k. Visible peak-count posterior: P(0) = 0.069, P(1) = 0.243, P(2) = 0.202, P(3) = 0.150, P(4) = 0.111, P(5) = 0.086, P(6) = 0.072, P(7) = 0.065; mean 2.86; probability of missing heaviest 0.330. OP7 — asynchronous cascade simulator on 2000 patches with OP13-calibrated L_k/r_k. Per-stage completion fractions: stage 5 (0.968), stages 6–11 (0.977). Delay quantiles (5th/50th/95th): −0.043 / −0.033 / −0.007.

Provisionally-resolved methodological. OP12 — P33 density-BAO sign inversion; v14 OP12 module establishes claim-grade public-random Landy-Szalay gates (weighted LS $\xi(r)$, jackknife covariance, label inversion, redshift LOO). The v3.6 P33 sign-inversion at $\delta r_d/r_d = -0.186$ is now diagnosed as a small-k k-NN estimator systematic; structural-mechanism interpretation no longer leading. Synthetic/random-missing runs held out.

Geometry-scan resolved prior. OP15 — bulk-geometry uncertainty in c_W. Tensor-engine-validated bulk-metric scan: tensor engine passes validation on known metrics (flat-5D, AdS_5, dS_5, S_5, RS-warped k = 1 and k = 2, anisotropic bulk). Numerically computed geometry-derived c_W range [0.014, 0.106] (RS-warped band 0.029–0.080); A_BWeyl proxy band [2.5×10^{-5}, 1.2×10^{-2}]. P40 amplitude prediction tightens from O(1) uncertainty to factor-of-few precision within RS-warped class.

Partial (independent likelihood, not OP11-blind establishment). OP14 — first-principles derivation of ν_bulk. OP11-blind likelihood: four-model Bayes-factor comparison finds ΛCDM zero-ν best (1.000), β_ν geometry prior 0.867, TGFT-RG centred 0.956, free log-flat 0.852. β_ν-flow ν_IR posterior (5th/50th/95th): 2.17×10^{-3} / 5.03×10^{-3} / 1.11×10^{-2}. Independently consistent with OP11 slice-side value; not yet $\nu > 0$ against zero-ν on OP11-blind data.

Prototype-only. OP5 — Koide mass-squared relation from CDT C-phase + Neuberger overlap. 96-node graph Dirac with 444 edges returns Koide-like spectral Q = 0.6884 (target 2/3 = 0.6667; ~3% off). Confirms triplet low-mode structure but not closure on CDT C-phase + Neuberger overlap programme. Real closure requires CDT C-phase triangulations at production scale.

Open (awaiting decisive future data). OP1 (ν redshift structure — DESI DR3 2027), OP2 (Euclid Q1 κ replication — now anchored on ACT DR6), OP3 (high-z a_0 catalogue-scale — R12+ catalogues), OP4 (cascade hard core — tonne-scale DD 2027–2030), OP16 (higher-dimensional continuation — structured continuation programme rather than OP).

Architectural commitments at Synthesis level. (i) ν committed at $5.08 \times 10^{-3} \pm 1.2 \times 10^{-3}$ globally; never tuned per prediction. (ii) L_k/r_k committed at 1.195 (median, 5th–95th band 0.673–2.127) for downstream mass-tower visibility and cascade-history predictions. (iii) c_W constrained to the geometry-derived band [0.014, 0.106]; predicted P40 amplitude tightened to A_BWeyl $\in$ [10^{-4}, 10^{-2}] (full range) or [10^{-3}, 10^{-2}] (RS-warped central tendency). (iv) Visible peak-count distribution exported as a prior on §15.4 P-A02 (mass-tower visibility); mean 2.86

peaks expected in tonne-scale direct-detection peak resolution. (v) The OP11-RESOLVED_JOINT commitment provides the universal ν across all CL bridges: CL1 (MOND-sequence), CL3 (time-crystal Q), CL7 (BSM1 + BSM3 + §8.3-v7.8 multi-channel), CL8 (universal $\delta\kappa_4/\delta\kappa$ via §8.2.bis-v7.9).

18. Priority Experimental Targets for Round 9 (updated in v3.5)

- **Confirm P38 at collaboration grade.** Re-run T16 on DESI Y3 void catalogue with full 3D galaxy density profiles. Distinguishes a real Cauchy tail from a VAST VoidFinder estimator systematic.
- **Confirm P3 at collaboration grade with density-coupling estimator.** Re-run T17 on DESI filament finder with weighted-DisPerSE on DESI BGS/LRG.
- **Resolve P33 sign inversion.** DESI DR3 density-binned BAO likelihoods to distinguish OP12 interpretations (a) and (b).
- **Stress-test CL2 on third real-map pair.** NANOGrav 15-yr × ACT DR6 lensing reconstruction at pulsar positions.
- **Resolve standalone-vs-joint ν_MOND instability.** Catalogue-scale rotation-curve sample at $z > 0.3$ (target ~100 systems).
- **Track tonne-scale direct detection.** Continue HEPData and dedicated-collaboration tracking. Goal: peak_count_ready_now → true.
- **Quantify L_k from first principles.** Path 9. Goal: convert P_presence(k) to a quantitative Bayesian prior on peak observability.
- **Address patch-spread-larger-than-baseline diagnostic.** Repeat four-patch P30 systematics with patches aligned on physical correlation scales.
- **(NEW v3.5) First test of P39 — secular Λ-drift.** w(z) reconstruction from joint DESI DR3 + Pantheon+ + DES Y6, looking for secular drift in predicted direction (Λ_eff decreasing in cosmic time). DESI DR3 alone may already be informative; full constraint with Euclid Q3 + LSST Y1. **This is the branch-selection test.**
- **(NEW v3.5) First test of P40 — bulk-Weyl B-mode.** B-mode polarisation analysis of forthcoming LiteBIRD or CMB-S4 forecasts. Search for low-ℓ tail with bulk-Weyl shape distinct from inflationary B-mode shape.
- **(NEW v3.5) Derive v_bulk from first principles via Path 6 + Path 9.** OP14. Convert P39's predicted drift rate from free amplitude to quantitative output.
- **(NEW v3.5) Derive c_W in candidate bulk geometry.** OP15. Tighten P40 amplitude prediction from O(1) uncertainty to factor-of-few.
- **Bahr–Steinhaus stage-by-stage execution (Path 6).** Carried over from v3.3 — still the cleanest theoretical anchor for Layer 1 once executed; **in v3.5 also the path to OP14 (v_bulk)**.
- **Quantum-simulator analogue design for Path 8 (theoretical phase 2026–2027). In v3.5 also Stage 4 of higher-dimensional continuation programme (§22).**

19. Round-9 Priorities — Synthesis Level (extended in v3.5)

§12 listed Round-9 experimental priorities. §20 here adds the synthesis-level priorities — work that should be done **at the level of the architecture itself**, not at the level of individual tests.

- **Strengthen CL4 from a directional joint to a quantitative bracket.** Combine the P3 Δr_texture and P38 k_4 amplitudes into a single ν estimator with explicit error propagation. (Carried over from v3.4.)
- **Audit the Lakatos protective belt for redundancy.** With CL5 added and CL2 triply anchored, several belt components carry less of the framework's empirical weight than they did in v3.2. A clean inventory of which belt components are actively load-bearing vs. provisionally retained would clarify the architecture. (Carried over from v3.4.)
- **Theoretical execution of Path 9 in full.** Quantitative L_k is needed for OP13 (patch-spread prediction) and feeds into OP14 (v_bulk). (Carried over from v3.4, elevated in v3.5.)
- **Derive a structural-mechanism candidate for OP12 (P33 sign inversion).** A candidate is: density-triggered reduction completes before recombination in dense regions, so dense regions already see post-reduction baryon physics at the BAO epoch. (Carried over from v3.4.)
- **(NEW v3.5) Activate §22 Stage 1 in earnest.** The dimension-general trigger criterion is stated in CCDR v7.5 §3.1 with the Weyl-squared term explicit. Stage 1 is the theoretical generalisation to all k with table of (W_k, c_W(k)). This is now a concrete deliverable with timeline 2–4 years post-decisive 3+1D test.
- **(NEW v3.5) Set up the OP14 derivation route.** Apply Path 6 + Path 9 to the bulk's TGFT description. Goal: turn v_bulk from a free parameter into a quantitative prediction whose value can be cross-checked against P39's observed drift rate.
- **(NEW v3.5) Specify the c_W candidate-bulk-geometry calculation programme.** OP15 requires deriving c_W in specific candidate bulk geometries (AdS_5, dS_5, RS-warped 4+1D). Standard AdS/CFT machinery applies; the work is identifying which geometry actually produced our Big Bang via cross-checks with P40 amplitude.
- **Track sign-coherence of Round-9 follow-ups.** The CL2 third-real-map test, the P3/P38 collaboration-grade replications, and the **first P39 results from DESI DR3** are all explicit sign-coherence tests of the v3.4/v3.5 architecture.

20. Higher-Dimensional Continuation Programme (new in v3.5)

CCDR v7.5 §25 lays out the higher-dimensional continuation programme at the framework level. §22 here gives the synthesis-level architecture: how the five staged continuation programmes interact with the 12+4+9 architecture, which resolution paths support which stages, and how the entire continuation programme is conditional on present-day decisive outcomes.

20.1 The Architectural Position of §22

The Synthesis architecture has historically been a *present-day* programme: 12 frameworks, alternative branches for fallback, resolution paths for closing gaps, predictions for empirical testing. v3.5 introduces a strict separation: present-day work tests the framework in our 3+1D slice (P1–P40); continuation work explores the higher-dimensional structure conditional on positive present-day outcomes. **§22 is the first chapter in the synthesis architecture explicitly devoted to continuation work**, and v3.5 commits to keeping it staged and conditional rather than letting it bleed into present-day priorities.

Three reasons for the strict separation:

- **Empirical priority.** Present-day predictions are sharp and falsifiable. Continuation work is conditional on positive present-day outcomes — if the 3+1D content fails, continuation work is moot.
- **Resource allocation.** Mixing present-day and continuation work invites dilution of empirical focus. The strict separation keeps reviewers, collaborators, and funding agencies clear about what's being tested now versus what's being planned for after decisive outcomes.
- **Structural honesty.** Continuation work is more speculative than present-day work. Calling it 'open problems' would understate this; calling it 'predictions' would overstate the testability. 'Continuation programme' is the honest framing.

20.2 The Five Stages — Synthesis-Level Mapping

Each stage from CCDR v7.5 §25 maps onto the synthesis architecture as follows:

Stage 1 — Theoretical generalisation of the trigger criterion. Maps to **Tier 2 frameworks** (CDT + TN + GFT/EPRL + TGFT-RG) and **Resolution Path 7** (conformal bootstrap). Stage 1 is the theoretical scaffolding for all subsequent higher-k work; it does not produce predictions on its own but enables Stage 3 work. **Connection to OP15** (c_W bulk-geometry uncertainty) is direct: Stage 1's table of $c_W(k)$ for $k = 5, 6, \dots, 11$ in candidate bulk geometries is the partial OP15 deliverable.

Stage 2 — Bulk RVM dynamics. Maps to **Tier 1 framework RVM** + **Tier 2 frameworks GFT/EPRL + TGFT-RG** and **Resolution Paths 6 + 9**. Stage 2 directly addresses OP14 (v_{bulk} derivation). It depends on Stage 1 only weakly (the structural form of the bulk RVM equation is given by general RVM principles; Stage 1's role is providing W_k for the bulk side).

Stage 3 — Indirect constraints on the bulk geometry. Maps to **Tier 3 framework AQN** (cascade residue spectrum) + **CL5** (P39 + P40 joint constraint). Stage 3 is the highest-value continuation product because it delivers a Bayesian posterior on (N, bulk geometry). It depends on Stages 1–2 and on tonne-scale direct detection completing (so the cascade-residue spectrum is observed).

Stage 4 — Quantum-simulator analogues. Maps to **Resolution Path 8** (already in the architecture as a path; in v3.5 elevated to Stage 4). Stage 4 is the only stage that can proceed in parallel with present-day work because it does not depend on outcomes. **In v3.5 the synthesis explicitly endorses Stage 4 as a parallel track**: laboratory $\chi_k \to 1$ measurement at small N is the most direct independent test of the dimension-general trigger criterion, and laboratory work is timeline-decoupled from observational work.

Stage 5 — Anthropic interpretation. Maps to no specific framework; this is interpretive work that follows once Stage 3 has constrained the bulk geometry. The synthesis architecture's commitment is that this stage exists as a definite endpoint of the continuation programme, not that the architecture endorses any particular anthropic reading.

20.3 Activation Conditions

The synthesis-level architecture commits to activating §22 only conditionally on decisive present-day outcomes. The activation conditions are:

- **Stage 4 (quantum-simulator analogue) — activate now in parallel.** No outcome dependency.
- **Stage 1 (theoretical generalisation) — activate after first informative P40 result.** A LiteBIRD or CMB-S4 measurement of B-mode polarisation is the empirical handle on whether the dimension-general trigger criterion at $k = 5 \to 4$ has the right structure; positive P40 → Stage 1 enters in earnest with the goal of generalising to higher k.
- **Stage 2 (bulk RVM dynamics) — activate after first informative P39 result.** A DESI DR3 + Euclid Q3 + LSST Y1 measurement of secular w(z) drift is the empirical handle on v_bulk; informative P39 (either positive secular drift or confirmed flat) → Stage 2 enters in earnest with first-principles v_bulk derivation as the central deliverable.
- **Stage 3 (bulk-geometry constraint) — activate after tonne-scale direct detection peak resolution.** Cascade-residue spectrum is the cross-check; without it, Stage 3 has no empirical anchor.
- **Stage 5 (anthropic interpretation) — activate after Stage 3 produces a posterior.** No earlier activation is sensible.

20.4 Programme-Level Note

§22 is structurally important to v3.5 because it makes explicit what was implicit in v3.4: the framework is a *programme of inquiry*, not just a present-day theory. The v3.5 reshape's commitment is to keep the programme honest about which work is empirically grounded and which is conditional on future outcomes. **If v3.5 is decisively falsified in our 3+1D slice (e.g., by P40 conflicting with predicted ℓ-shape and CL5 returning null while P39 also conflicts), §22 becomes moot. If v3.5 is decisively confirmed, §22 becomes the explicit roadmap for the next 10–20 years of synthesis work.** Either way, the architecture remains structurally honest about what is being tested now and what is being planned for later.

21. Standard-Model Derivation Subprogramme (NEW, isolated from main body)

21.1 Architectural placement and scope

This section records a **structurally isolated continuation subprogramme**, parallel to the Higher-Dimensional Continuation Programme (the higher-k programme, see the corresponding section above) but operating at Layer 3 (SSM + AS + DA/O + LGT) rather than Layer 0. As with the higher-dimensional programme, its content is **not part of the main empirical battle** carried by the P1–P41 prediction list, the five-bracket cross-link triangulation, or the Round-9 audit. It is a structured roadmap for a different research thread — deriving the Standard Model from spacetime crystallography — that uses the same primary-framework toolkit (CDT, SSM, AS, TGFT-RG, DA/O, RVM, AQN, CCDR) but addresses a different question: not “are the cascade and evaporation predictions empirically anchored?” but “is the SM Lagrangian computable from the CDT C-phase simplex geometry?”

The two subprogrammes are designed to be **mutually independent**. Falsification of the main-body cascade architecture (Layer 0 / Layer 4) does not falsify this subprogramme: the SM-derivation chain assumes the CDT C-phase exists and uses it as a substrate, but the cascade-residue dark matter, evaporation-default endpoint, dimension-general trigger, and bulk-Weyl B-mode (P39, P40, P41 in their cosmological readings) are not load-bearing for the SM derivation. Conversely, failure of the SM-derivation pipeline to reproduce α(0), m_H, the Koide ratio, or any other SM observable does not falsify the cascade core: the SM may still emerge from the CDT C-phase via a derivation we have not yet correctly specified, while the cosmological cascade structure can be wholly intact. The only point of contact between the two subprogrammes is the shared use of CDT and the spectral action, both of which are inherited from the main framework’s Tier 1/2 frameworks.

This isolation is deliberate. A reader who is sceptical of dimensional-cascade cosmology can engage with the SM-derivation subprogramme on its own terms; a reader who is sceptical of the SSM/spectral-action programme can engage with the cosmological main body without it. The architecture-level commitment is that both are simultaneously possible programmes within the same 12+4+9 synthesis, but neither requires the other for its empirical content.

21.2 The seven-stage derivation chain

The subprogramme defines an explicit, executable chain from the CDT 4-simplex to the low-energy Standard Model parameters. Each stage is a concrete computation; the chain is logically complete (every step has a defined input and output); the numerical execution is feasible with existing software adapted to simplicial geometry. The stages below summarise the architectural commitment; full derivations are kept out of the main synthesis body and live in a companion technical document.

21.2.1 Stage 1 — Gauge group from 4-simplex symmetry

A 4-simplex σ^4 has 5 vertices, 10 edges, 10 triangular faces, 5 tetrahedral faces. Its symmetry group is the symmetric group S_5 (order 120), with subgroup chain $S_5 \supset A_5 \supset A_4 \supset V_4 \supset Z_3 \times Z_2 \times U(1)\_\text{phase}$. The decomposition into the SM gauge group proceeds in three steps. **(a) SU(3) from Z_3**: the three-element cyclic subgroup acts on the three internal face orientations of the simplex; in the continuum $N_4 \to \infty$ limit the discrete Z_3 extends to the continuous SU(3). The three colours correspond to the three cyclic orientations; colour confinement is the lattice Bloch theorem (only Z_3-singlet combinations propagate beyond one lattice spacing). **(b) SU(2)_L from Z_2 with causal constraint**: Z_2 exchanges the two simplex face classes (type-(4,1) future-pointing vs type-(1,4) past-pointing); the CDT global time foliation breaks Z_2 to the identity on one chirality, giving SU(2)_L only — the microscopic origin of parity violation. **(c) U(1)_Y from the simplex phase**: each simplex carries a Lorentzian-CDT path-integral weight exp(−S_Regge[σ]) whose imaginary part defines hypercharge Y = Im(S_Regge[σ])/(2π) mod 1. Compactness of U(1) on the lattice quantises Y to rational values.

Cross-check (DA/O programme, P24): independently, the algebra C ⊗ H ⊗ O has automorphism group SU(3) × SU(2) × U(1) up to discrete quotients (Furey 2012/2018). The chain CDT simplex → S_5 → representation theory → C ⊗ H ⊗ O → A = C ⊕ H ⊕ M_3(C) → Connes’ finite space F is the algebraic cross-check on the geometric derivation.

21.2.2 Stage 2 — Three generations from the hexagonal Brillouin zone

The CDT C-phase has a spatial geometry consistent with hexagonal-close-packed structure (from triple-junction topology and 6-fold simplex packing coordination). The hexagonal Brillouin zone has high-symmetry points Γ (zone centre), K and K' (inequivalent zone corners), M (zone edge midpoints), A (zone top/bottom). The three inequivalent high-symmetry directions Γ–K, Γ–M, Γ–A correspond to the three fermion generations (light, intermediate, heavy). Fermions propagating along each direction experience different effective lattice potentials, giving identical gauge quantum numbers but different masses — standard crystal physics. The exact count of three generations follows from the count of three inequivalent high-symmetry directions on the hexagonal BZ.

The Koide ratio $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ is then a theorem of hexagonal symmetry on a 3×3 Hermitian mass matrix with C_{6v} point-group symmetry, not a numerological coincidence. Quark Koide-like relations should hold with one-loop QCD self-energy corrections at the BZ boundary: $Q_{quarks} = 2/3 + O(\alpha_s/\pi)$. The CKM mixing matrix $V_{CKM} = U^\dagger_{uL} \cdot U_{dL}$ is the rotation between Γ-point gauge eigenstates and BZ-direction mass eigenstates; the CKM angles θ_{12}, θ_{23}, θ_{13} are crystallographic misorientation angles between up-quark and down-quark BZ directions; the CP phase δ is the same phase that appears in the EPRL chiral amplitude ($\delta_{CP} \sim 10^{-3}$ from CDT simplicial chirality).

21.2.3 Stage 3 — Higgs as amplitude mode of the EW crystal

The electroweak phase transition at T ~ 160 GeV is a crystallisation event. Above T_{EW} the gauge group $SU(2)_L \times U(1)_Y$ is unbroken (disordered phase); below it the Higgs field φ acquires a non-zero expectation value $\langle\varphi\rangle = v = 246$ GeV. The Higgs field decomposes as $\varphi = (v + h)e^{i\theta}$ with phase modes θ (the three Goldstone bosons absorbed by $W^\pm$, Z) and an amplitude mode h — the Higgs boson at $m_H = 125$ GeV. The amplitude-mode mass formula $m_H^2 = d^2V/d\varphi^2|_{\varphi=v} = 2\lambda v^2$ gives $m_H/v \approx 0.508$, the crystal's elastic anharmonicity. In the spectral-action framework (Stage 4), λ is a spectral invariant of D_{CDT}, so 125 GeV is an output of the computation rather than an input. The gauge boson masses $m_W = gv/2 = 80.4$ GeV, $m_Z = m_W/\cos\theta_W = 91.2$ GeV close the Anderson–Higgs sector.

21.2.4 Stage 4 — Full Lagrangian from the spectral action on D_CDT

Chamseddine and Connes (2007) showed that for the almost-commutative geometry $M_4 \times F$ with Dirac operator D, the physical action is

$$S = \mathrm{Tr}(f(D/\Lambda)) + \langle J\psi, D\psi\rangle$$

where f is a smooth cutoff function, Λ is the energy scale, J is charge conjugation, and ψ is the fermion multiplet. The trace expansion $\mathrm{Tr}(f(D/\Lambda)) = f_4\Lambda^4 a_0 + f_2\Lambda^2 a_2 + f_0 a_4 + O(\Lambda^{-2})$ yields, via the Seeley–DeWitt coefficients: a_0 → cosmological constant term (the C_0 in the RVM equation), a_2 → Einstein–Hilbert action plus the Higgs mass term, a_4 → Yang–Mills action for SU(3) × SU(2) × U(1) + Higgs quartic + all Yukawa couplings + neutrino Majorana masses. The spectral action on $M_4 \times F$ gives **exactly** the SM Lagrangian + gravity + the Higgs potential + all Yukawa couplings, with nothing more and nothing less. The substitution M_4 → CDT-T (the C-phase triangulation), D → D_{CDT} (the lattice Dirac/overlap operator on T) makes the Seeley–DeWitt coefficients computable from the eigenvalue spectrum of D_{CDT}. The coupling constants α, α_s, G_F, λ, and all Yukawa couplings are determined by a_4.

21.2.5 Stage 5 — Coupling constants from asymptotic safety

The AS UV fixed point (Reuter 1998) provides the boundary conditions G*(μ → ∞) ≈ 0.707/μ²,* *Λ*(μ → ∞) ≈ 0.193 × μ² for the gravitational sector, and analogous fixed points for the gauge couplings. The fixed-point values α *of the dimensionless gauge couplings are spectral invariants of D_CDT (from Stage 4). Running the standard one-loop RG beta functions from μ = M_Pl down to μ = m_Z gives α⁻¹(μ) = α*⁻¹ + b/(2π) × ln(μ/M_Pl) with b = −41/6 for U(1), b = 19/6 for SU(2), b = 7 for SU(3). The output α(m_Z) ≈ 1/127.9, α_s(m_Z) ≈ 0.118, $\sin^2\theta$_W(m_Z) ≈ 0.231 are all determined by the CDT geometry and the AS fixed point. The low-energy α(0) ≈ 1/137.036 follows from running α(m_Z) to zero momentum.

21.2.6 Stage 6 — Fermion masses from the lattice Dirac operator

The fermion propagator on the CDT C-phase triangulation is S_F(p) = ⟨ψ(p) ψ̄(0)⟩ = [D_CDT(p) + m_bare]$^{-1}$; the physical fermion masses are the poles. Equivalently, the effective mass at each BZ symmetry point is set by the dispersion curvature m_*X = ℏ²/(d²E/dk²)|_X with X ∈ {K, M, A} for generations 1, 2, 3. The mass hierarchy $m_1 \ll m_2 \ll m_3$ follows from the lattice potential anisotropy: Γ–K is flattest (lightest fermions), Γ–A is steepest (heaviest). The CDT lattice spacing a_CDT = ℓ_Pl × (G*/g*)^{1/2} fixes the absolute mass scale in physical units. The Koide relation Q = 2/3 is a necessary consistency check: the three lepton masses computed from the dispersion curvatures must satisfy Q = 2/3 automatically; if they do not, the CDT geometry is wrong.

21.2.7 Stage 7 — Computational pipeline

The seven steps are executable with existing software, adapted to simplicial geometry:

- Generate CDT C-phase triangulation with $N_4 \sim 10^4$–10^6 simplices (CDT-plusplus is the established open-source code).
- Construct the lattice Dirac operator D_CDT on T using the overlap operator (Neuberger 1998) adapted to simplicial geometry (Becher–Joos 1982).
- Compute the eigenvalue spectrum of D_CDT (Lanczos or Chebyshev iterative eigensolvers; QUDA / Grid lattice-Dirac libraries).
- Evaluate the Seeley–DeWitt coefficients a_0, a_2, a_4 from the heat-kernel expansion of the spectrum; output coupling constants at scale Λ.
- Identify the AS fixed point by extrapolating Steps 1–4 to the continuum limit ($N_4 = 10^4 \rightarrow 10^6$); the fixed-point couplings G*, α*, α_s* are the $N_4 \rightarrow \infty$ limits.
- Run the RG beta functions from the fixed point to low energy; output α(m_Z), α_s(m_Z), $\sin^2\theta$_W(m_Z), λ, all Yukawa couplings.
- Compute the fermion propagator on T at each BZ symmetry point; extract pole masses; output m_e, m_μ, m_τ, m_u, m_d, m_s, m_c, m_b, m_t.

Verification step 8: check Q(leptons) = 2/3, m_H = 125 ± 1 GeV, $\sin^2\theta$_W = 0.231 ± 0.001, α(0) = 1/(137 ± 1).

Feasibility. Steps 1–3 require adapting existing software (CDT-plusplus + QUDA/Grid lattice Dirac solvers) — a PhD-scale project, 1–2 years. Steps 4–8 use existing mathematics. The full pipeline at $N_4 \sim 10^4$ is weeks on a workstation; at $N_4 \sim 10^6$, months on a cluster. This is the same order-of-magnitude effort as a standard lattice-QCD calculation.

21.3 SM-D predictions tagged from the chain

Each derivation stage produces one or more numerical outputs that are testable against measured SM parameters. These are tagged SM-D1 through SM-D10, parallel in spirit to the cosmological P1–P41 list but operating at Layer 3 / Layer 1. They are **not** added to the main P-list because they live in the SM-derivation subprogramme rather than the cosmological main body.

- **SM-D1.** $\alpha(0) = 1/137.036$ from CDT Dirac spectrum + AS + RG running (Stages 4 + 5).
- **SM-D2.** $\alpha_s(m_Z) = 0.1179$ from the same pipeline.
- **SM-D3.** $\sin^2\theta_W(m_Z) = 0.2312$ from the spectral-action normalisation between SU(2) and U(1).
- **SM-D4.** $m_H = 125.09 \pm 0.3$ GeV from the spectral-action quartic $\lambda = m_H^2/(2v^2)$.
- **SM-D5.** Koide Q = 2/3 (exact, from hexagonal BZ symmetry — Stage 2 theorem).
- **SM-D6.** All twelve fermion masses from CDT propagator poles (Stage 6).
- **SM-D7.** CKM angles θ_{12}, θ_{23}, θ_{13} from BZ crystallographic misorientations (Stage 2).
- **SM-D8.** CP phase $\delta \sim 10^{-3}$ from CDT simplicial chirality (= EPRL chiral amplitude — main-body link).
- **SM-D9.** $C \otimes H \otimes O$ division-algebra cross-check on the gauge group from S_5 (P24 connection).
- **SM-D10.** $\theta_{QCD} = 0$ protected by spectral-action reality + overlap-operator Ginsparg–Wilson chirality (no axion required at the SM-derivation level — distinct from the AQN / PQ-axion construction in the main body).
- **SM-D11.** Signed CDT simplex-orientation ↔ weak-handedness lock (sharpens BSM6): the sign of the (4,1)- versus (1,4)-simplex orientation asymmetry must equal the sign that yields left-handed SU(2) under the §21.9.3 foliation product — testable in CDT C-phase simulations now. New in v3.9.

The two flavour-sector falsification handles registered in earlier framework versions are absorbed here. SM2 ($\delta R_K/\delta R_{K^*} = \sqrt{3}$ from hexagonal BZ symmetry) becomes a parameter-free Stage-2 prediction targetable by Belle II + LHCb upgrades. SM3 (Higgs self-coupling $\lambda = m_H^2/(2v^2)$ with a CDT-calculable correction at the v level) is a Stage-3/4 cross-check targetable by HL-LHC.

21.4 Beyond-SM particle inventory (structurally motivated, not load-bearing)

The chain above derives the SM from the spacetime crystal. It also generates a list of particles **predicted** by the same framework that lie outside the SM. Each is tied to a specific structural feature of the crystal; none is load-bearing for the framework's main empirical content; all are recorded here for completeness and falsifiability.

| Particle | Status | Mass | Mechanism / origin | Test channels |

| --- | --- | --- | --- | --- |

| Graviton | Required (CDT–CCDR) | 0 (exact) | Acoustic phonon of the spacetime lattice; Goldstone of broken spatial translations; dispersion $\omega^2 = c^2k^2 - \alpha_{CDT}\, \ell^2_{Pl}\, k^4$ | k^4 correction to GW propagation near Planck scale |

| Optical phonon DM (v3.7: inelastic with elastic-overlap suppression κ_elastic = v) | Required (CCDR §7.2 v7.8 reframed) | ~100 GeV (set by EW crystallisation); inelastic mass-splitting

δ_BSM1 ≈ ν · m_DM ≈ 0.5 GeV | **v3.6 form (elastic SI WIMP) REJECTED at R11 (25σ vs XENONnT/LZ/PandaX-4T at 100 GeV).** v3.7 form: gapped inelastic excitation of cosmic-crystal inter-sublattice boundary mode; cascade-overlap factor κ_elastic = ν suppresses elastic SI cross-section to $\sigma_{eff} \approx 5 \times 10^{-49}$ cm² (consistent with current limits); dominant phenomenology is inelastic excitation of internal lattice-boundary degree of freedom | Inelastic-DM searches: XENONnT migdal, electron-recoil S2-only, DARWIN/XLZD migdal (P-A23); sub-leading elastic σ ≈ $\nu^2 \cdot \sigma_{geometric} \approx 2.6 \times 10^{-51}$ cm² at DARWIN/XLZD (P-A24); joint ν self-consistency check with BSM3 + §8.3-v7.8 via CL7 (P-A25) |

| Axion | Required (AQN) | 10^{-5}–10^{-2} eV | PQ-scale crystallisation Goldstone; small mass from QCD instanton potential | ADMX, ABRACADABRA, CASPEr, IAXO |

| Right-handed ν | Structurally motivated | 10^{9}–10^{15} GeV | Crystal boundary mode; CDT causal constraint forbids backward propagation in bulk but allows it on de Sitter / BH horizons | KATRIN, JUNO, DUNE; processes near Planck scale |

| Magnetic monopole | Structurally motivated | ~10^{16} GeV (M_GUT/α) | Point defect of SU(2)×U(1) crystal; Dirac quantisation as lattice quantisation | Monopole searches (MoEDAL etc.) |

| Cosmic strings | Structurally motivated | tension μ ~ 10^{-6} Gμ_Pl | Edge-dislocation line defects of the spacetime lattice | NANOGrav and successor PTAs |

| Dark photon | Structurally natural (if optical U(1) exists) | massless or ~0.1 GeV | Optical-sector U(1) gauge mode; kinetic mixing ε ~ ν ~ 10^{-3} | HPS, APEX, DarkQuest, LHCb dark-photon searches |

| Domain-wall fragments | Possible (AQN remnant) | variable, macroscopic | Pieces of QCD-stage axion domain walls that did not collapse | Macro-impact searches; gravitational substructure |

| Baryon as skyrmion | Reinterpretation | proton mass | Skyrme's 1961 proposal: baryon = topological soliton of crystal gauge field | Already observed; reinterpretation only |

These are tagged BSM1 (in v3.7: **inelastic** phonon DM at ~100 GeV with elastic-overlap suppression κ_elastic = ν giving $\sigma_{eff} \approx 5 \times 10^{-49}$ cm² and inelastic mass-splitting δ_BSM1 ≈ 0.5 GeV — reframed from the v3.6 elastic SI form which was R11-rejected at 25σ), BSM2 (axion in 10^{-5}–10^{-2} eV window), BSM3 (dark-photon kinetic mixing ε ~ ν ~ 10^{-3} if optical U(1) exists), BSM4 (right-handed ν boundary modes with M_R from crystal-boundary energy), BSM5 (cosmic-string tension μ ~ 10^{-6} Gμ_Pl, PTA-testable), BSM6 (CDT simplex orientation count: left-vs-right asymmetry ~ δ_CP, testable in CDT simulations now). They are **not** part of the main P-list and **not** counted in the 32-prediction inventory; they are recorded here because the same framework that derives the SM also predicts them, and the existence or non-existence of any of them is a constraint on the SM-derivation chain rather than on the cosmological main body.

v3.7 BSM1 reframe — lesson from the R11 rejection. The v3.6 BSM1 entry placed the optical-phonon DM mode on the elastic SI WIMP test channel by virtue of identifying its leading interaction as $\sigma_{DM} \sim \nu^2(\hbar/m_{DM} c)^2 \sim 10^{-42}$ m² (interpreted as 10^{-46} cm² in standard units). R11 measurement against the joint XENONnT/LZ/PandaX-4T exclusion at 100 GeV (4×10^{-48} cm²) returned σ_predicted / σ_excluded ≈ 25 — a clean 25σ rejection. The lesson is that the test-channel identification was the wrong category: an optical-sector mode orthogonal to the acoustic-phonon sector that carries SM interactions should couple *inelastically* to nuclei (exciting the internal lattice-boundary degree of freedom), not elastically. The v3.7 reframing keeps the structural identification (gapped excitation of the lattice basis at ~100 GeV) but moves the dominant phenomenology to inelastic excitation; the elastic component is suppressed by the cascade-overlap factor κ_elastic = ν (the same ν that governs vacuum-density running and dark-photon mixing). With $\nu = 5.08 \times 10^{-3}$ (joint-extractor working commitment from §11.1), σ_eff(elastic SI) = ν · σ_geometric ≈ 5×10^{-49} cm², which is ~8× below the current 100 GeV exclusion. The inelastic component, predicted at mass-splitting δ_BSM1 ≈ ν · m_DM ≈ 0.5 GeV, becomes the primary test channel for BSM1 going forward — testable now against XENONnT migdal data and decisively at DARWIN/XLZD (P-A23 in the v7.8 prediction list). A sub-leading elastic component at order $\nu^2 \approx 2.6 \times 10^{-51}$ cm² is pre-registered as P-A24 for DARWIN/XLZD-generation testing. The framework's joint commitment to κ_elastic = ν across BSM1, BSM3 (dark photon mixing), and §8.3-v7.8 (void-wall stacking) is the basis for the new cross-bridge CL7 (§11.6) and the corresponding pre-registered prediction P-A25 (joint ν self-consistency).

v3.8 annotation — universal coherent-stack context for κ_elastic = ν. The v7.9 universal coherent-stack realisation of CCDR §8.2.bis reframes the §8.3 mechanism context for κ_elastic. Under the v7.8 environment-dependent picture, κ_elastic = ν was justified as the cascade-overlap factor between an optical-sector (DM) mode and an acoustic-sector (visible-matter) probe at leading order; this leading-order identification is preserved unchanged in v3.8. The v7.9 universal-stacking framing adds an additional consistency check: under the universal kernel for cosmic-web interfaces, the same $(\nu\pi^2/6)(N - 4)^2 \approx 0.35$ prefactor governs δk_4 at all interface types, and the same ν appears in (a) κ_elastic for BSM1 elastic suppression, (b) ε_dark = c_mix · ν for BSM3 kinetic mixing, (c) the universal-kernel constant 0.35 for §8.3-v7.9. Three independent ν-extractions are now bracketed (CL7), and a fourth bracket emerges via the universal $\delta k_4/\delta\kappa$ ratio (CL8): the same ν that governs the $(N - 4)^2$ amplification in kurtosis also governs the same-source δκ excess at the interface. A multi-bracket consistency across CL7 (three ν extractions agree within ±20%) plus CL8 (universal $\delta k_4/\delta\kappa$ ratio observed across at least three interface types) would be the strongest empirical support for the universal-cascade-overlap structure that ties BSM1, BSM3, and §8.3 together at leading order.

21.5 Honest scope statement and what is *not* derived

The subprogramme is committed to the claim that the SM Lagrangian is computable from the CDT C-phase simplex geometry plus the spectral action plus the AS fixed point plus RG running. It is **not** committed, at this stage, to having performed the computation. What the seven-stage chain provides is the structural identification of SM features with crystal properties (gauge group ↔ space group, generations ↔ BZ, chirality ↔ CDT orientation, Higgs ↔ amplitude mode, mass hierarchy ↔ phonon dispersion anisotropy, CKM ↔ BZ misorientation, CP phase ↔ simplicial chirality). The numerical execution — the actual computation of α = 1/137 from the D_CDT spectrum — has not been done. It is feasible, defined, and pending. Until it is executed, the SM-D1–10 predictions are predicted-in-principle rather than predicted-with-numbers.

What is **not** derived by the chain even in principle: (i) the *choice* of finite space $F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$; this is taken as an input from the DA/O programme, with the cross-check provided by $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$, but the deeper origin of why this specific algebra rather than another remains an open question. (ii) The *specific* hexagonal C-phase geometry; this is observed in CDT simulations but the derivation of why the C-phase is hexagonal-close-packed rather than another lattice structure is not yet given. (iii) The *exact* AS fixed-point values G*, α*; these are computed in continuum AS work but the joint AS-on-CDT calculation has not been completed.

21.6 Activation conditions

The subprogramme should be activated as a parallel research thread when (a) the main-body cosmological tests of v7.6 / v3.6 do not falsify the cascade core within 2026–2030, and (b) computational resources for adapting CDT-plusplus + QUDA/Grid to simplicial Dirac operators become available. Activation is not gated on positive cosmological confirmations: even if the cosmological main body remains in its current "structurally complete, partially anchored, decisively untested" state, the SM-derivation work can proceed independently because its inputs (CDT, spectral action, AS, DA/O) are inherited from the main framework's already-defined Tier 1/2 frameworks.

If the main-body cascade architecture is decisively falsified (e.g. by P27 at tonne-scale direct detection, or by joint failure of P39 against both branches), the SM-derivation subprogramme **survives intact**: the SM derivation does not depend on the cascade-residue dark-matter spectrum or on the evaporation-default endpoint. The only main-body input it requires is that the CDT C-phase exists and is hexagonal-close-packed, both of which are already supported by independent CDT simulations (Ambjørn–Loll review, see external support matrix above).

21.7 Falsification of the subprogramme itself

The SM-derivation subprogramme is falsified, in its current form, by any of the following.

- **Numerical failure.** Once the seven-stage pipeline is executed and the SM-D1–6 predictions are computed, sustained order-of-magnitude disagreement between computed and measured values of $\alpha(0)$, $\alpha_s(m_Z)$, $\sin^2\theta_W$, m_H, the lepton masses, or the CKM angles falsifies the chain as currently specified. The architectural claim (SM = band structure of spacetime crystal) survives in spirit but the specific stage chain must be re-derived.
- **Koide failure.** If the three lepton masses computed from the dispersion-curvature formula do *not* satisfy Q = 2/3 automatically, the hexagonal BZ identification (Stage 2) is wrong, and one of the other inequivalent crystal lattices must be used. This would falsify SM-D5 directly and require re-doing Stage 2.
- **CP-phase mismatch.** If the EPRL chiral amplitude δ_{CP} and the CKM CP phase δ are measured / computed to disagree at order-of-magnitude level, the CDT simplicial chirality is not the source of CP violation, and SM-D8 is wrong.
- **DA/O cross-check failure.** If $C \otimes H \otimes O$ is shown to **not** have automorphism group $SU(3) \times SU(2) \times U(1)$ (e.g. through new mathematical work on the algebra), Stage 1 loses its independent algebraic confirmation. This does not falsify the geometric S_5 derivation outright but removes the cross-check, weakening confidence in the gauge-group identification.

None of these failures falsify the cosmological main body. The two subprogrammes are decoupled by design.

21.8 Connection to the main body — single anchor point only

The single architectural anchor between this subprogramme and the cosmological main body is the **identification of the EPRL chiral amplitude with the CKM CP phase** (SM-D8). This phase is the same δ_{CP} that drives the cosmological $\eta \sim v \times \delta_{CP} \times f_{geo} \sim 10^{-10}$ baryogenesis mechanism in Layer 5 of the main framework. A measurement or first-principles computation of this phase therefore provides one observable that constrains both subprogrammes simultaneously — the only such joint observable. All other contact between the two subprogrammes is through shared *frameworks* (CDT, spectral action, AS, DA/O), not shared *predictions*.

This single anchor is intentionally light. The architectural commitment is that the cosmological reading (cascade + evaporation + dimension-general trigger + bulk-Weyl B-mode + $b \to s\mu\mu$ lattice signature P41) and the SM-derivation reading (band structure of spacetime crystal → SM Lagrangian → $\alpha = 1/137$ + 12 fermion masses + CKM + CP phase) are two **decoupled empirical programmes** that share machinery but answer different questions and can succeed or fail independently. The SM-derivation subprogramme is recorded here in the synthesis architecture — analogously to the higher-dimensional continuation programme — to make this isolation explicit and to keep the two threads from contaminating each other's empirical accounting.

21.9 Dimension-Identity Selection and the Oriented-Fold Completion of Chirality (NEW in v3.9)

Scope. Stage 1(b) of the seven-stage chain (§21.2.1) derives $SU(2)_L$ from a Z_2 that the CDT global time foliation breaks to the identity on one chirality — the microscopic origin of parity violation. That account fixes the temporal half of the mechanism. §21.9 supplies the spatial half: the oriented compactification of the exact extra dimension at the terminal 5 → 4 reduction, which produces a single 4D chirality and fixes its sign. The geometric machinery — which direction is folded, and why — is developed in the companion CCDR §3.7; §21.9 imports it and shows it completes, rather than replaces, the §21.2.1(b) reading, adds a signed sharpening of BSM6 (SM-D11), and registers a second main-body ↔ SM-derivation joint observable (CL9). It modifies no prior commitment of §21.

21.9.1 Dimension-identity dependence of the boundary and interior

The $\chi_k \to 1$ trigger is dimension-general but scalar: it records the count k, not which spatial direction is removed at the $k \to k-1$ step (CCDR §3.7.1). In an anisotropic bulk the identity of the folded direction is physical — it fixes the induced boundary metric, the KK spectrum, and which automorphisms of the $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ structure survive as unbroken gauge symmetry. The folded direction is promoted to a primary datum, the fold address (n_k, ε_k): a reduction co-vector n_k (the unit vector along the removed direction) and a cascade orientation $\varepsilon_k = \pm 1$. For the SM-derivation chain the consequence is that which gauge factor is chiral, and with which handedness, is a function of the terminal fold address — not merely of the simplex combinatorics of Stage 1.

21.9.2 Least-matter-occupancy selects the folded direction

Among the (k−1) candidate spatial directions the reduction folds the one of least matter occupancy $U_a = \int (\rho_{matter} + (c_W/G_k) W_k) \cdot \omega_a$ (CCDR §3.7.2). Under the self-extent correlation — matter content swells geometric extent, registered as OP19 — this coincides with the direction whose removal yields the lowest area-to-volume ratio of the residual configuration, so the same holographic-efficiency logic that §3.6 uses to fix the boundary shape now fixes the axis. Applied across the cascade, the direction folded last — at the terminal 5 → 4 step that produced our slice — is the most-occupied, hence largest-radius, surviving extra direction; its orientation sets 3+1D chirality (§21.9.3) and its KK modes are the lightest (P-A31).

21.9.3 The oriented fold completes Stage 1(b)

Step 1 — 4+1D is non-chiral. In odd spacetime dimension $d = 5$ there is no independent chirality operator (the product $\gamma^0\gamma^1\gamma^2\gamma^3\gamma^4 \propto 1$), so a 4+1D bulk fermion is vector-like, carrying both 4D chiralities equally. Chirality cannot be an input to the cascade; it must be produced by the fold.

Step 2 — The oriented fold makes one Weyl zero mode. Compactifying the terminal direction n_5 on the orbifold interval S^1/Z_2 splits the 5D Dirac field into 4D KK modes; the orbifold parity P : $\psi(y) \rightarrow \varepsilon_5 \gamma^5 \psi(-y)$ keeps exactly one 4D chirality as the massless zero mode and lifts the other into the tower. Which chirality survives is the sign of ε_5. This is the domain-wall / overlap mechanism (Kaplan; Ginsparg–Wilson / Neuberger, already invoked in SM-D10 and Resolution Path 4), read as a consequence of the oriented compactification of the exact 5th direction.

Step 3 — The Z_2 of §21.2.1(b) is the orbifold parity. The arrow ε_5 is inherited from the same monotone cascade ordering (RVM cooling; $d\tau > 0$, §9.3) that orients the rest of the framework, and the CDT time foliation supplies the temporal arrow e_0. The surviving handedness is the product of the two arrows. The Z_2 that Stage 1(b) breaks by the time foliation is precisely the orbifold parity of the folded direction: the temporal arrow of §21.2.1(b) and the spatial arrow of the terminal fold multiply to one definite sign. §21.9 therefore completes Stage 1(b); it does not replace it.

handedness = sign(ε_5 · (e_0-foliation orientation))

Step 4 — Identity selects which factor is chiral. Because the surviving automorphisms depend on which direction was spent (§21.9.1), the identity of n_5 selects which SU(2) of the $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ / DA/O structure is gauged and left-acting: the quaternionic factor H carries the SU(2), and the terminal fold spends the isometry dual to H's right action, leaving its left action as $SU(2)_L$ rather than $SU(2)_R$. The assignment of $SU(2)_L$ that Stage 1(b) obtains combinatorially is here given a geometric origin in the fold address.

Magnitude. The CP-odd amplitude $\delta_{CP} \sim 10^{-3}$ (SM-D8, the EPRL chiral amplitude) is read as the small misalignment angle θ_{fold} between $\varepsilon_5\, n_5$ and the lattice / foliation orientation, with $\delta_{CP} \sim \sin\theta_{fold} \sim \nu$ — tying the CKM CP phase to the same coefficient ν that drives the cascade, with no new free parameter, consistent with the existing SM-D8 ↔ baryogenesis link (§21.8).

21.9.4 New predictions, sharpening, cross-bridge, and open problems

P-A30 (Layer 0). Cosmic and weak parity violation share one sign. The primordial 5 → 4 boundary imprint (P40) and 3+1D weak chirality are fixed by the same ε_5, so the parity-odd CMB sector — the bulk-Weyl B-mode (P40) and any cosmological-birefringence-like TB/EB correlation — carries a handedness whose sign matches the weak interaction. An opposite sign relation at 3σ (LiteBIRD / CMB-S4 TB/EB versus the known left-handedness of $SU(2)_L$) falsifies the oriented-fold mechanism. Data not used in formulation.

P-A31 (Layer 2/4). The chirality-defining terminal direction is the largest-radius extra direction, so its KK and bulk-graviton modes are the lightest of the extra-dimensional spectrum ($r_5 = \max_a r_a$); the lightest fifth-force / KK scale is tied to the chirality sector, testable through sub-millimetre fifth-force / KK-graviton ordering against the cascade-residue mass tower (P-A02).

P-A32 (Layer 0/4). The anisotropy projected out by the terminal fold leaves a weak quadrupolar statistical-anisotropy axis in the 3+1D perturbations that must coincide — within errors — with the parity axis of P-A30; cross-correlation of the low-ℓ statistical-anisotropy axis with the TB/EB parity axis and the LSS dipole is decisive. Falsified by a robust preferred axis uncorrelated with the parity axis at 3σ.

SM-D11. Signed simplex-orientation ↔ weak-handedness lock — extends BSM6 from a magnitude statement (left-versus-right (4,1)/(1,4)-simplex asymmetry ~ δ_{CP}) to a signed one: the sign of the CDT orientation asymmetry must equal the sign that yields left-handed SU(2) under the Step-3 foliation product. Testable in CDT C-phase simulations now. Also registered as a bullet in §21.3.

CL9. Three-way orientation consistency — the cosmological parity-odd CMB sign (P-A30), the SM weak handedness, and the signed CDT simplex-orientation asymmetry (SM-D11 / BSM6) are three projections of the single ε_5 and must agree in sign; joint disagreement of any two at 3σ breaks the reading. CL9 is §11.8, raising the bridge count from eight (CL8) to nine.

OP19 / OP20. OP19 — prove the self-extent correlation $c_g > 0$ so that $\mathrm{argmin}_a\, U_a = \mathrm{argmin}_a\, I_{k-1}(\text{fold } a)$ is a theorem rather than a modelling assumption (Resolution Path 7 plus isoperimetric geometry). OP20 — determine whether the cascade-flow arrow fixes ε_5 uniquely, or whether ε_5 is spontaneously selected, in which case the prediction content of §21.9.3 is the correlation CL9 rather than the absolute sign (Resolution Path 4, overlap-Dirac on CDT; ties to OP5).

21.9.5 A second main-body ↔ SM-derivation anchor

§21.8 records a single joint observable linking this subprogramme to the cosmological main body: the identification of the EPRL chiral amplitude with the CKM CP phase (SM-D8). The oriented-fold mechanism adds a second. The orientation ε_5 that fixes weak handedness is the same arrow that orients the primordial 5 → 4 boundary deformation (P40); CL9 promotes this shared orientation to a testable joint constraint relating the cosmological parity-odd sector (P40 / P-A30) to the SM weak chirality (SM-D11 / BSM6). The contact between the subprogrammes is therefore no longer a single phase (SM-D8) but a phase and a sign (SM-D8 ⊕ CL9); §21.8's single-anchor-point statement is updated accordingly to two anchors.

Framing

Like CCDR §3.6 / §3.7, §21.9 is a geometric completion layer, not a hard-core axiom. It is supported by standard physics (anisotropic flux-compactification selection, isoperimetric geometry, and the orbifold / domain-wall origin of 4D chirality from a non-chiral odd-dimensional bulk) and generates falsifiable content (P-A30, P-A31, P-A32, SM-D11, CL9) plus two open problems (OP19, OP20) without modifying the seven-stage chain. If P-A30 or SM-D11 is decisively falsified — cosmic and weak parity signs anti-correlated, or the signed simplex asymmetry opposite to the SM convention — §21.9 reverts to a heuristic and the §21.2.1(b) time-foliation reading of chirality stands alone.

22. Reshape Summaries — v3.5 and v3.6

Architectural notes recording what the v3.5 and v3.6 reshapes change and what they preserve. Distinct from the conclusion (which records the framework's present empirical status) and from per-round results (which record empirical movement).

22.1 v3.5 Reshape — Architectural Notes

§22 records, at the synthesis level, what the v3.5 reshape changes architecturally and what it preserves. This is distinct from §13 (conclusion) and §16/§17 (round results) — it is meta-level documentation of the reshape's architecture for future reference.

22.1.1 What v3.5 Adds Without Removing

Two new predictions (P39, P40) at the bottom of §8. Total goes from 38 to 40.

One new cross-link (CL5) at the bottom of §14. Total goes from 4 to 5.

One new alternative branch (eternal-attractor) at the bottom of §2. Total goes from 3 to 4. The eternal-attractor branch is the swappable counterpart to the new evaporation default; it preserves the v3.4 architecture as a fallback.

Three new open problems (OP14, OP15, OP16) at the bottom of §19. Total goes from 13 to 16.

One brand-new chapter (§22) for the higher-dimensional continuation programme. The higher-k cascade physics was structurally implicit in v3.4 but not staged; in v3.5 it has explicit five-stage scaffolding with timelines and dependencies.

Three resolution paths take on additional roles without changing identity: Path 6 also addresses OP14, Path 8 also serves as §22 Stage 4, Path 9 also contributes to OP14.

22.1.2 What v3.5 Replaces

Universe endpoint. v3.4 carried the late-universe asymptote as eternal de Sitter at H_∞; v3.5 carries it as full evaporation into the (N+1)-D bulk over ~10^{167} years. The previous reading is the swappable alternative (§2.4).

Status of C_0. v3.4 treated C_0 as a fundamental constant; v3.5 treats $C_0(t)$ as the projection of the bulk vacuum density onto our 3+1D slice.

Cosmological time crystal duration. v3.4 treated it as eternal at frequency $\omega_\infty = H_\infty$; v3.5 treats it as a long transient with $\omega_{crystal}(t) = H(t)$ drifting toward zero. Both give the same numerical Q at present cosmic age.

v7.4 §10.4 nested-eternal-crystals architecture (cosmological 10^{11} yr, BH M^3/M^2_{pl}, DM 10^{-26} s, Planck foam 5.4×10^{-44} s, all interpreted as nested eternal oscillators) is reduced to a single transient cosmological crystal in v3.5; the other three timescales remain physical features but are not interpreted as eternal-hierarchy oscillators.

22.1.3 What v3.5 Keeps Unchanged

The cascade structure (N → N−1 → … → 4) is unchanged. Local-asynchronous-cascade core untouched.

The dimension-origin prior $P_{presence}(k) \sim (r_k / L_k)^{k-4}$ is unchanged.

The reducing-volume caveat is unchanged.

All Round-7 / Round-8 empirical results are unchanged. P38, P3, P30, CL2, P29, P36 all carry forward exactly.

The grain-boundary picture (§8 of CCDR) is unchanged. The cosmic crystal exists for as long as our universe does.

v stays where it is. The MOND-sequence v_MOND, the SN-driven v_SN, the time-crystal Q-extracted v_Q, and CL4 grain-boundary v all remain valid extractors. They constrain the matter-coupling part of $\rho_{vac}(H)$, not C_0.

The black-hole-as-reduction picture (§5 of CCDR) survives intact and is in fact made cleaner. The earlier asymmetry (BHs evaporate, universes don't) goes away.

The FTL prohibition and time-directionality constraints survive.

22.1.4 The Architectural Logic of the Reshape

v3.4 had two structural commitments without empirical content: (i) the universe asymptotes to an eternal state, and (ii) the trigger criterion was specialised to k = 4. v3.5 replaces the first commitment with an empirically distinguishable pair (evap default vs eternal-attractor branch, distinguished by P39) and the second with a dimension-general statement that has its own empirical content (P40).

In Lakatos terms: v3.5 moves two pieces from the *metaphysical* category (structural commitments without empirical handles) to the *protective belt* (structural commitments with explicit falsification conditions). This is a strict architectural improvement — the framework's commitments are now more accountable to observation.

The cost of the reshape is one new free parameter (v_bulk) and one new theoretical uncertainty (c_W's bulk-geometry dependence). Both have explicit resolution routes (OP14 → Path 6 + Path 9; OP15 → §22 Stage 1). The reshape is internally tractable — not a deferral to future work but a specification of what that future work looks like.

22.2 v3.6 Reshape Summary

The v3.6 reshape adds without removing:

- **One new prediction (P41)** at the bottom of § 8: the LHCb b → s μ μ angular tension as a Layer-3 cascade-residual lattice signature. Total: 41.
- **One new cross-link (CL6)** at the bottom of § 14: joint constraint on (lattice scale, vector-coefficient sign, CP-phase parity) from B → K *μ μ + bulk-Weyl B-mode. Total: 6.*
- **One new speculative branch (Dark-Cone Branch / Sector-Causal CCDR)** in § 19, marked clearly as not part of the default architecture.
- **One new protective-belt extension (DCN_k generalisation of AQN)** in § 20, in the dark-matter / higher-dimensional research programme.
- **Two new open problems (OP17, OP18)** registered for the DCN_k extension.
- **One new round of empirical tests (Round 9, T1–T25)** in § 18.

The default CCDR architecture (v7.5) is preserved unchanged. **The FTL prohibition for the baryonic / acoustic / SM sector is retained as a hard-core commitment; the Dark-Cone Branch operates only on the dark sector and only as a speculative alternative.** No layer is falsified. No prediction is retracted. The empirical content is monotonically increasing.

22.3 v3.7 Reshape Summary — R11 Lessons and Mechanism Revisions

The v3.7 reshape responds to two R11 REJECT verdicts with mechanism revisions documented in CCDR_v7.8 §8 (stacked-interface void mechanism with scaling inversion and cascade-depth amplification) and Synthesis §21.4 BSM1 (inelastic reframing with elastic-overlap suppression). The v3.6 architecture is preserved in v3.7; the changes are surgical and localised.

What v3.7 changes relative to v3.6. (a) BSM1 row in the §21.4 BSM particle inventory is rewritten — from elastic SI WIMP at $\sigma \sim 10^{-46}$ cm² to inelastic phonon-sector excitation with κ_elastic = ν elastic suppression. (b) §11 cross-bridges adds CL7 — joint ν self-consistency across BSM1, BSM3, and §8.3-v7.8 — and sharpens the CL4 statement to encompass the v7.8 void/filament asymmetry. (c) The Synthesis abstract is updated to document both lessons and the new CL7.

What v3.7 preserves unchanged. (i) The 12 primary frameworks and three-tier organisation (§1). (ii) The 4 swappable branches and swap table (§§2–3). (iii) The complete equation chain (§4). (iv) The 5-layer falsifiability hierarchy (§5). (v) The 9 resolution paths (§§6–7). (vi) The AQN/DCN_k generalisation (§8). (vii) The Dark-Cone speculative branch (§9). (viii) The complete predictions table (§10) — except for the cross-references to v7.8 prediction updates. (ix) Cross-bridges CL1–CL6 (§11). (x) Layered falsifiability with fallback paths (§12). (xi) Observable-first reclassification (§13). (xii) Empirical-tests inventory through Round 10 (§14). (xiii) Round-results summary (§15). (xiv) Scorecard (§16) — except for the addition of the R11 outcomes. (xv) Open problems list (§17). (xvi) Priority experimental targets (§§18–19). (xvii) Higher-dimensional continuation programme (§20). (xviii) SM-derivation subprogramme (§21), including all of §§21.1–21.3 and §§21.5–21.8; only the §21.4 BSM1 entry is revised.

What v3.7 contributes empirically. Six new pre-registered predictions, all documented in CCDR_v7.8 §15.4: P-A20 (void-radius scaling of δk_4, R12-testable), P-A21 (filament-wall absence of $(N - 4)^2$ amplification — the discriminator, R12-testable), P-A22 (cluster-boundary intermediate amplification, R12-testable), P-A23 (inelastic mass-splitting δ_BSM1 ≈ 0.5 GeV, XENONnT migdal R12-R13 testable), P-A24 (sub-leading elastic $\sigma \approx 2.6 \times 10^{-51}$ cm², DARWIN/XLZD R14+ testable), P-A25 (joint ν self-consistency via CL7, R13+ testable). These are the genuine empirical content added by v3.7; they will determine whether the v3.7 mechanism revisions are correct multi-channel revisions or single-channel accommodations.

Authorial position on the v3.7 reshape. The reshape is taken under the empirical-first methodology recorded in CCDR §1.4: when measurement contradicts a mechanism's quantitative prediction, the mechanism is revised. The two R11 rejections (P-A07 at 7.25σ on void-wall kurtosis; BSM1 at 25σ on elastic SI) were specific quantitative falsifications of specific mechanism forms; the v3.7 revisions are specific mechanism updates that match the rejection-driving observations. The honest accounting (per CCDR §15.5 Lesson 3) is that the v3.7 revisions are not pre-registered predictions on the R11 data themselves — they are accommodations of that data. The cross-channel pre-registered predictions P-A20–P-A25 are what should be judged in R12 and onward. If they confirm, the v3.7/v7.8 mechanism revisions are validated as multi-channel-consistent; if they fail, the v3.7 architecture needs further work.

22.4 v3.8 Reshape Summary — OP-Resolver Synchronisation and Universal Coherent-Stack Architectural Update

The v3.8 reshape responds to two parallel inputs from the companion CCDR_v7.9 release: (a) v14 computational resolver outputs across OP1–OP18, synchronised into the Synthesis's §17 open-problems inventory as §17.bis; (b) the R12 Tempel filament rejection of P-A21 (v7.8 form) at >10σ, triggering the v7.9 universal coherent-stack realisation of CCDR §8.2.bis, which propagates into the Synthesis as the new cross-bridge CL8 (universal $\delta k_4/\delta\kappa$ ratio) and the §10 prediction-table cross-references to four new pre-registered predictions P-A26–P-A29.

What v3.8 changes relative to v3.7. (a) §11 cross-bridge inventory extended from seven to eight bridges; CL8 added as the cross-probe consistency test of the v7.9 universal coherent-stack realisation. CL4 statement sharpened a second time (after its v3.7 sharpening) to reference the v7.9 master-curve form instead of the now-rejected v7.8 void/filament asymmetry. (b) §17 expanded with §17.bis OP1–OP18 status sub-section; ν committed at OP11-RESOLVED_JOINT value; L_k/r_k calibration committed at OP13 value; visible peak-count posterior committed from OP6/OP10; c_W band committed from OP15. (c) §21.4 BSM1 entry annotated with v7.9 universal-kernel context for κ_elastic = ν; multi-bracket consistency across CL7 + CL8 documented. (d) §10 prediction-table cross-references updated to include P-A26 (cosmic-sheet kurtosis), P-A27 (BAO-scale harmonic in kurtosis spectrum), P-A28 (universal $\delta k_4/\delta\kappa$ — basis of CL8), P-A29 (master-curve self-consistency — new v7.9 §8 discriminator). (e) §22 reshape-summary inventory extended with this v3.8 entry.

What v3.8 preserves unchanged. (i) 12 primary frameworks and three-tier organisation (§1). (ii) 4 swappable branches and swap table (§§2–3). (iii) Complete equation chain (§4). (iv) 5-layer falsifiability hierarchy (§5). (v) 9 resolution paths (§§6–7). (vi) AQN/DCN_k generalisation (§8). (vii) Dark-cone speculative branch (§9). (viii) Complete predictions table (§10) — except for the cross-references to v7.9 prediction additions. (ix) Cross-bridges CL1–CL7 (§11). (x) Layered falsifiability with fallback paths (§12). (xi) Observable-first reclassification (§13). (xii) Empirical-tests inventory through Round 11 (§14). (xiii) Round-results summary including R11 (§15). (xiv) Scorecard (§16) — except for the addition of R12 Tempel filament outcome. (xv) Open problems list (§17 main content); §17.bis adds resolution-status sub-section. (xvi) Priority experimental targets (§§18–19). (xvii) Higher-dimensional continuation programme (§20). (xviii) SM-derivation subprogramme (§21), including all of §§21.1–21.3 and §§21.5–21.8; only the §21.4 BSM1 entry is annotated with v3.8 context (no structural change to the BSM1 form itself).

What v3.8 contributes empirically. Four new pre-registered predictions documented in CCDR_v7.9 §15.4 — P-A26 (cosmic-sheet kurtosis, $k_4 \approx 7.2 \pm 0.6$), P-A27 (BAO-scale harmonic at $k = \pi/r^*$ in the kurtosis power spectrum), P-A28 (universal $\delta k_4/\delta\kappa \approx 12.5 \pm 2.0$ — the basis of CL8), P-A29 (master-curve collapse — the new v7.9 §8 discriminator). Together with the updated P-A21-v2 (filament walls show $(N - 4)^2$ amplification under universal kernel) and P-A22-v2 (cluster boundaries lie on the master curve), these constitute the empirical content of v3.8. R12 is the decisive round.

Authorial position on the v3.8 reshape. This is the second mechanism revision in CCDR §8 within one release cycle (v7.7 single-boundary → v7.8 environment-dependent → v7.9 universal coherent-stack). The empirical-first methodology of CCDR §1.4 contemplated this case explicitly: each rejection sharpens the microphysical picture. The v3.8 Synthesis records both rejections (R11 voids 7.25σ; R12 Tempel >10σ) as audit-trail entries — they are the positive empirical content of the framework's iteration history, not embarrassments to be hidden. The v7.9 universal mechanism is more constrained than its predecessors precisely because it must satisfy both rejections plus the new measurement, and a third rejection would constrain a hypothetical v7.10 mechanism even more tightly. R12 verification of P-A26, P-A27, P-A28, P-A29 (plus the updated P-A21-v2 and P-A22-v2) will determine whether the v7.9 mechanism is correct or whether a third mechanism revision is required.

23. Conclusion

The v3.5 synthesis is a structurally complete research programme: 12 primary frameworks providing the theoretical machinery, **4 alternative branches** (extended from 3 in v3.4 by adding the eternal-attractor branch as the swappable counterpart to the new evaporation default), 9 resolution paths (with three taking on additional roles in v3.5 to address OP14 and feed §22 Stage 4), 5 layers of falsifiability with explicit exit conditions, a complete equation chain from

CDT simplices to observable predictions, and **40 numbered predictions** (extended from 38 in v3.4 by adding P39 and P40).

The architecture follows Lakatos: a hard core ($\chi_k \to 1$ triggers dimensional reduction, locally and asynchronously, **in dimension-general form**), a protective belt (CDT geometry, TN entanglement, RVM dynamics, GFT condensate, AQN dark matter, SSM Lagrangian, DA/O cross-check, TGFT-RG rigour, HaPPY QEC), alternative branches (CST, EG, CFS, **eternal-attractor**) that can replace belt components if falsified, resolution paths (Swampland, discrete flavours, Sorkin–HH, overlap Dirac, Diósi–Penrose, Bahr–Steinhaus, conformal bootstrap, quantum simulator, Diósi–Penrose × reducing-volume) that close remaining gaps using existing external tools, and **five cross-framework bridges (CL1–CL5)** providing independent consistency triangulation.

The framework explicitly forbids FTL information transfer, respects thermodynamic time directionality, commits to determining N from observation, and **adopts evaporation as the universe's natural endpoint while preserving the eternal-attractor reading as a swappable branch**. No single failure destroys the programme. Every gap has a resolution path. Every prediction is testable.

v3.5 takes the programme into a five-bracket consistency triangulation regime and adds a **strict architectural separation between present-day predictions and continuation work**. With CL5 added, the framework now has five operationally independent ν extractors: SN-driven, MOND-sequence, time-crystal Q-factor, grain-boundary stratification, and the new (ν_{bulk}, c_W) joint constraint from P39 + P40. All five jointly constrain the bulk RVM dynamics, and the empirical pattern emerging across rounds is one of **density-stratified excesses in multiple independent observables** (P30 κ, CL2 PTA × κ, P3 filament texture, P38 void-wall kurtosis), all in the predicted direction.

The dimension-origin prior is now a first-class part of the interpretation of direct detection — and, in v3.5, has gained a second-class observational handle through P39 (the bulk we evaporate back into is the bulk that birthed us). Higher-dimension-origin species are systematically less likely to be present in our patch; missing heavy peaks are structural, not falsifying.

The programme is ready. The decisive experiments are: tonne-scale direct detection with peak-resolution analysis in the 0.5–3 TeV window (Layer 0 fires); **w(z) reconstruction from DESI DR3 + Euclid Q3 + LSST Y1 (P39 — branch selection)**; **B-mode polarisation at LiteBIRD or CMB-S4 sensitivity (P40 — dimension-general trigger)**. When they fire, the framework will succeed or fail cleanly — under the quantified dimension-origin prior, anchored on five independent ν brackets, and with the universe-endpoint question made empirically distinguishable rather than a metaphysical commitment.

The §22 higher-dimensional continuation programme lays out the work that follows decisive 3+1D-slice testing. Five staged programmes (theoretical generalisation of trigger; bulk RVM dynamics; indirect bulk-geometry constraints; quantum-simulator analogues; anthropic interpretation) with explicit timelines and dependencies. This is a strict architectural separation: present-day work tests the framework in our 3+1D slice; continuation work explores the higher-dimensional structure conditional on positive present-day outcomes.

24. External Support Matrix: CCDR-Adjacent Literature

This section catalogues recent independent literature that strengthens individual CCDR ingredients, supplies held-out test channels, or sharpens existing constraints. **None of these**

papers were written for CCDR, and none should be read as proofs of the framework. The accurate claim is narrower: each entry supports the *plausibility and testability* of one specific CCDR component, while leaving the framework's unifying claims unverified. The matrix is organised by support type rather than by prediction number, making explicit which empirical objects each citation actually constrains.

24.1 Support Type Taxonomy

Four labels are used to classify each entry:

- **Ingredient support** — papers that strengthen a theoretical or methodological building block (CDT, holography/tensor networks, RVM/decaying vacuum, RAR/MOND phenomenology, AQN/composite-DM morphology). They argue that a CCDR sub-component is a live, defensible object in the broader literature.
- **Observable-method support** — papers that establish or refine the measurement machinery on which CCDR predictions depend (Euclid filament extraction, weak-lensing mass-map filament detection, DESI BAO methodology, NANOGrav spectral-running diagnostics). They do not test CCDR but make CCDR's tests executable.
- **Candidate signature** — papers reporting a measurement that, on independent reading, points in a direction consistent with a specific CCDR prediction (P39, P41, P3/P30 density-stratified texture). These are the empirically loaded entries.
- **Constraint / tension** — papers reporting null results or constraints that bound but do not falsify CCDR predictions (XENONnT/LZ direct-detection nulls, P38 lognormal void-null at Round 9, fixed-r_d robustness tests on DESI). They define how close the decisive tests are.

A fifth status is used internally for entries the framework **must not** mis-cite: **convention/correction**, marking instances where the original CCDR text used a sign or convention that diverges from the broader literature's standard reading. These are recorded so that future versions of CCDR can correct without losing the underlying physical claim.

24.2 Candidate Signature — SM-Sector (Layer 3)

LHCb, "A comprehensive analysis of $B^0 \to K^{*0}(\to K^+\pi^-)\mu^+\mu^-$" (arXiv:2512.18053; submitted to Phys. Rev. Lett., Dec 2025). The 8.4 fb^{-1} Run-1 + Run-2 angular analysis with full S-wave treatment and muon-mass effects. The paper explicitly states that CP-averaged observables continue to exhibit the SM-tension pattern of earlier analyses, while CP-asymmetric A_i are consistent with zero. This is the empirical anchor for P41 and CL6.

Convention correction owed (CCDR v7.6 → v7.7). The CCDR-internal text reports $\delta\mathrm{Re}(C_9^{\mathrm{eff}}) \approx +0.9$ in the low-q^2 bins. The dominant convention in the global-fit literature (Altmannshofer & Stangl 2021; Hurth et al.; Alguero et al. and successors, on the H_{eff} convention with $C_9\mathrm{SM} \approx +4.27$) gives a **negative** shift: muon-specific $C_9 \approx -0.7$ to -0.8 relative to the SM. The CCDR sign was therefore inverted relative to the field-standard convention. The physical content — a vector-like, CP-even, real shift in the C_9 sector with no corresponding C_{10} shift — is preserved either way; only the sign convention differs. v7.7 should report $|\Delta C_9| \approx 0.7$–1.0 with explicit sign convention note, and the discriminating CCDR claim then becomes "the shift sits in the C_9 direction, is real, vector-like, lepton-specific, and not accompanied by a C_{10} or C_S/C_T shift" rather than the sign per se.

LHCb, “Angular analysis of $B^0 \to K^{*0}e^+e^-$ decays” (arXiv:2502.10291, Feb 2025). The 9 fb^{-1} electron-mode counterpart in the $q^2 \in [1.1, 6.0]$ GeV^2 bin, including an explicit lepton-flavour-universality test against the muon channel. The angular observables are reported as consistent with both SM expectations and global $b \to s\ell\ell$ fits, with no LFV signal. This is the **control channel** for P41: a CCDR lattice/grain-boundary correction at the TeV scale is in principle lepton-universal, so the electron channel should show the same vector-real shift up to mass-suppressed corrections. The current LHCb electron result is consistent with that picture (no evidence against universality), but is not yet a positive confirmation. CCDR’s predicted discriminator is **same-sign, comparable-magnitude** δC_9 in muons and electrons, with **no** anomalous A_i. The v7.7 text should register this as a held-out P41 sub-test rather than fold the e^+e^- result silently into P41’s confirmation status.

24.3 Candidate Signature — Cosmological Λ-Drift (Layer 4)

DESI Collaboration, DR2 BAO key paper (arXiv:2503.14738) and the broader DR2 dark-energy programme (e.g. Bansal & Huterer arXiv:2502.07185; Wang et al. arXiv:2504.15222 v2 Nov 2025; combined-probe analyses arXiv:2508.10514, arXiv:2604.05849). The DESI DR2 BAO + CMB combination prefers w_0w_a CDM over flat ΛCDM at frequentist significance up to ~3.1σ (BAO+CMB) and ~2.8–4.2σ when SNe Ia are added, depending on the SN sample. The trajectory in (w_0, w_a) space sits in the $w_0 > -1$, w_a < 0 quadrant with a phantom crossing near z ~ 0.5 — i.e. dark energy density was higher in the past than today. **This is the direction P39 predicts** under the evaporation-default endpoint (Λ_eff decreasing in cosmic time as the bulk vacuum cools).

Critical caveats from independent reanalyses. (i) Bayesian model comparison (Bansal & Huterer 2025; Hawkins et al. 2025 arXiv:2511.10631) finds the evidence for w_0w_a CDM is **dataset-combination-driven**: log-Bayes factor for DESI DR2 BAO + Planck alone is *modestly negative* (lnB ≈ −0.6, favouring ΛCDM), and only with DES-Y5 SNe added does the Bayesian preference for evolving DE rise to ~3σ. (ii) Fixed-r_d robustness tests (arXiv:2507.01380) show that the ~3σ frequentist preference does *not* reproduce under controlled sound-horizon anchoring, suggesting the result is partly entangled with early-universe calibration. (iii) Wang et al. (arXiv:2504.15222 v2, Nov 2025) find significant tensions among CMB, BAO and SN datasets within ΛCDM, so the evolving-DE preference may be partly absorbing inter-dataset disagreement.

Net reading for CCDR. DESI DR2 is the strongest external handle on P39 currently available, but it is *suggestive*, not decisive. The framework should continue to flag P39 as awaiting DESI DR3 + Euclid Q3 + LSST Y1 (as stated in §17 / §21 of the restructured CCDR). The fixed-r_d and Bayesian re-readings should be cited explicitly in the P39 status field, not omitted.

Slow-decaying vacuum / RVM context. Turyshev (arXiv:2602.05368), Rezaei & Solà and successors continue to examine running-vacuum/Λ(t) models against DESI DR2; recent analyses (e.g. arXiv:2512.07281) find the late-Universe data consistently favour quintessence-like behaviour while early-Universe fits prefer phantom-like behaviour. These are **ingredient support** rather than direct P39 confirmation: they keep RVM in the live region of theory space.

24.4 Candidate Signature — Density-Stratified Cosmic-Web Texture (Layer 4)

Euclid Collaboration, “Q1 — Galaxy shapes and alignments in the cosmic web**”** (Laigle et al., arXiv:2503.15333, Mar 2025). On 63.1 deg^2 of Euclid Q1 at $z > 0$ with stellar masses $M >$ *10^{10}*

$M_\odot$, the paper reports a detection* of preferential alignment of the major axes of massive early-type galaxies along projected cosmic filaments, plus a clean dependence of morphology on density and on distance to filament spines. **This is observable-method support for P3** — it establishes the alignment signal exists in the public Euclid product and quantifies its dependence on environment. It is **not** a CCDR-specific signature: the standard cosmological reading (tidal torquing, secondary infall) is the null hypothesis Laigle et al. work in. CCDR's reading would require a *sign-coherent excess in density-stratification beyond the standard prediction *in independent catalogues. The Round 7 T17 result ($\Delta r_{texture}(high - low) = +933$ Mpc/h on the Tempel et al. catalogue) is in the direction such an excess would take, but the Round 9 T07 within-field density-shuffle null absorbed the Q1 stratification component at present sensitivity. The Laigle et al. paper makes the predicted signal* measurable* on Euclid; it does not yet test CCDR's specific stratification claim.

DECADE+DES Y3 weak-lensing mass map (Gatti et al., arXiv:2509.03798, Sep 2025; Phys. Rev. D 2026). 270 million galaxies, 13 000 deg², the largest late-Universe weak-lensing mass map to date. The paper reports the **first detection of cosmic filaments directly from a weak-lensing mass map**, validated by SZ-selected clusters from Planck and ACT DR6. **This is observable-method support for P30** (density-correlated κ) and for the §16 (CCDR §22) CL4 cross-stratification: it supplies a public cross-validated mass-map → filament pipeline at the scale CCDR's Round-8/Round-9 tests are now reaching for. A CCDR-specific test on this product would compare κ amplitude inside high-density vs low-density filament/ridge environments; that test has not been executed in the v7.6 audit and should be queued for Round 10.

24.5 Ingredient Support — Microscopic Geometry (Layer 0)

Ambjørn & **Loll, "**Causal Dynamical Triangulations: New Lattice Theory of Quantum Gravity**"** (arXiv:2604.05641, Apr 2026). The current Scholarpedia-style review of CDT covering the C-phase de-Sitter-like geometry, the spectral dimension reduction $d_s : 4 \to 2$ at the Planck scale, and the evidence for a UV fixed point. **This is foundational ingredient support for §1.1 / §3.1** — the dimension-general trigger criterion of CCDR is a phenomenological story laid on top of CDT's dynamical-dimensional-reduction phenomenology. The 2026 review supports the claim that CDT is a viable, actively developed non-perturbative quantum-gravity framework, but does *not* validate CCDR's specific cascade structure or the $\chi_k \to 1$ trigger. v7.7 should cite this review as the canonical Layer-0 ingredient support, replacing or supplementing the older CDT references.

24.6 Ingredient Support — RAR / Local a_0 Anchor (§8.3 / §6.1 of synthesis)

Sands et al., "The radial acceleration relation at the EDGE of galaxy formation**"** (arXiv:2510.06905, Oct 2025). Tests the RAR on twelve nearby low-mass dwarf galaxies ($10^4 < M_{bar}/M_\odot < 10^{7.5}$) using MUSE-Faint and Jeans modelling, comparing to ΛCDM EDGE simulations. The paper finds that most low-mass dwarfs lie *systematically above* the low-mass extrapolation of the RAR, with multi-valued behaviour in some cases. **This is mixed**: it confirms the RAR remains the right empirical object to track, but it complicates the universal-a_0 reading at the very low-mass end. For CCDR's local a_0 → Milgrom anchor (Round-9 T11: $a_0 = 9.44 \times 10^{-11}$ m/s² on n = 175 SPARC galaxies, quadruply anchored), this is a **boundary observation**: the SPARC range used in CCDR's Round-5/8/9 fits sits well above the regime where Sands et al. report the deviation, so the CCDR anchor is intact. The v7.7 text should acknowledge this paper as a constraint on the *extrapolation* of the local a_0 statement to dwarf-spheroidal regimes.

Li & **McGaugh** (A&A 615, A3, 2018; arXiv:1803.00022). The MCMC fit of the mean RAR to 175 individual SPARC galaxies, finding 0.057 dex rms residual scatter and *no evidence* for galaxy-to-galaxy variation in the critical acceleration scale. **Direct ingredient support** for §8.3 and for the CCDR claim of a universal local a_0: the SPARC anchor of CCDR is built on this and earlier RAR work, and that statistical foundation has not been undermined by post-2018 results in the SPARC range.

24.7 Ingredient Support — Holographic / Tensor-Network Tools (Layer 2)

The HaPPY-style holographic-QEC literature (Pastawski et al. 2015; Almheiri–Dong–Harlow 2015; Hayden et al. 2016) carried forward in the CCDR References list remains the canonical ingredient support for the §5.3 / §17 holographic-encoding picture. No 2025-2026 paper directly displaces it, but the broader review literature on holographic tensor networks for boundary error-correcting codes continues to expand. CCDR uses these tools at the level of *language for boundary encoding*, not as a derivation; the framework is therefore robust to refinements in this literature but does not gain direct empirical leverage from them either.

24.8 Ingredient Support — AQN / DCN_k Morphology (CCDR §13, Synth §8)

The AQN literature (Zhitnitsky 2003 onwards) supplies the worked QCD-stage example for the DCN_k generalisation introduced in v7.6. Recent “AQN glow” analyses (e.g. arXiv:2406.12122, arXiv:2504.15382) examine electromagnetic and CMB-side signatures of axion/quark-nugget composite dark matter. **These are ingredient support, not candidate signatures**: they argue that compact, surface-stabilised, charge-asymmetric DM residues are a testable morphology with diverse channels (macro-impact, microlensing, gravitational granularity, thermal-anomaly), exactly the channels CCDR §13 (DCN_k) lists for non-recoil constraints. AQN should remain in the CCDR core at Ω_4 / baryogenesis as it is in v7.6; DCN_{k>4} should remain a protective-belt extension as labelled.

24.9 Constraint / Tension — Direct-Detection Nulls

XENONnT, “WIMP Dark Matter Search using a 3.1 Tonne-Year Exposure**”** (arXiv:2502.18005; Phys. Rev. Lett. 135 221003, Nov 2025). 3.1 t·yr exposure, no significant nuclear-recoil excess above 3.8 keV_NR. Spin-independent WIMP-nucleon cross-section limit at 1.7×10^{-47} cm² for a 30 GeV/c² WIMP at 90% CL; sensitivity improvement up to 1.8× over the first XENONnT result. **This is a constraint, not a falsification** of CCDR’s hard-core peak-counting prediction (P10/P25/P27): the mass window is partially covered, but the experiment does not yet have peak-resolution sensitivity in the 0.5–3 TeV range that CCDR’s dimension-origin prior (§4.5) flags as the relevant test region. The Round 8 T11 / Round 9 T18 readiness assessment in CCDR remains correct: window overlap = true, peak-count readiness = false. v7.7 should cite this paper as the most recent direct-detection bound and update P10/P25 status accordingly.

LZ 4.2 t·yr (arXiv:2410.17036) and PandaX-4T continue to push the WIMP-nucleon cross-section frontier in the same way. None of these results are CCDR-specific; all are constraint-defining for when the hard-core test fires.

24.10 Constraint / Tension — Round-9 Internal Nulls

The Round-9 audit itself (consolidated in §18.4 of the restructured CCDR) registered three structural nulls that are part of this matrix's "constraint" column: (i) T02 DR2-only BAO null on P39's mean-vector trend; (ii) T07 within-field density-shuffle null absorbing the Q1 P3 stratification component; (iii) T09 lognormal-matched null on the VAST P38 k_4 result. Each of these is a *protective-belt tightening* — the CCDR framework retains the prediction but the empirical content shifts to other catalogues / estimators (Tempel et al. for P3 bulk-texture amplitude; DESI Y3 / Euclid DR1 for P38 confirmation; DESI DR3 + Euclid Q3 + LSST Y1 for P39). They define what it would take for the framework's empirical content to remain non-trivial.

24.11 What This Matrix Is *Not*

This matrix does **not** claim that CCDR is supported as a unified framework. The accurate statement is the converse: each ingredient has independent literature-level standing, several CCDR observables are now measurable on public products, and at least two predictions (P39, P41) have current public-data results sitting in the direction CCDR predicts — but with caveats (sign convention for P41, dataset-combination dependence for P39) that any honest reading must register. CCDR's unifying content — that all of these effects derive from the same local, asynchronous, dimension-general cascade — is *not* tested by any single paper in this matrix. It is testable, in principle, only via the joint sign-coherence of P39 + P40 + P41 + the cascade-residue spectroscopy from tonne-scale direct detection (§17). Until those joint constraints are available, the framework is at the same epistemic status the v7.6 conclusion places it in: structurally complete, partially anchored, with explicit branch-selection observables ahead.

24.12 Future Updates to This Matrix

The matrix is intended as a living document. Entries to be added in v7.7+ under each label:

- **Candidate signature**: any forthcoming Belle II / LHCb-upgrade angular analysis covering the C_9 sign and CP-parity discriminator across $b \to s\ell\ell$ channels (P41 cross-channel coherence); first DESI DR3 w(z) reconstruction (P39 first decisive read); LiteBIRD or CMB-S4 first B-mode bandpower release (P40 first decisive read).
- **Observable-method**: Euclid DR1 filament/ridge/void catalogue products as they appear; DESI Y3 dedicated void catalogue with full 3D galaxy density profiles (P38 collaboration-grade replication target); Rubin/LSST Y1 weak-lensing maps (filament + density-stratification at unprecedented S/N).
- **Constraint / tension**: continued XENONnT / LZ / PandaX-4T / DARWIN-XLZD direct-detection updates; any ringdown analyses targeting the χ_4 → 1 signature (P32) on LIGO-VK O5 or LISA.
- **Convention / correction**: any further convention drift between CCDR-internal language and field-standard global-fit conventions in flavour, weak-lensing, or BAO sectors.

Round 10 (queued in §21 of the restructured CCDR) will execute the first cross-channel P41 test and the first joint w(z) reconstruction with DESI DR3 hooks where available; that round will populate the matrix's first true "candidate-signature confirmed" or "candidate-signature failed" entries.

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v3.4 additional references (Round-7 and Round-8 public catalogues):

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v3.5 additional references (evaporation default, dimension-general trigger, P39, P40, §22):

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Appendix A. Proposed Laboratory Experiment: Flying-Focus Vacuum Tomography (VFT-1) — *out of main core*

This appendix proposes a near-term laboratory experiment — **Flying-Focus Vacuum Tomography (VFT-1)**, also referred to as the **Vacuum-Phonon Pump** — designed as the smallest-scale, highest-leverage test of CCDR's projection mechanism. It is presented **out of main core**: VFT-1 is an *engineering boost* on top of v7.6, generating a new prediction **SE11 (Sensors & Metrology)** and a **Stage-4 Quantum-Simulator Extension** to the §22 Higher-Dimensional Physics Exploration Programme. The hard-core architecture of CCDR (P1–P42, the falsifiability layers, the cross-links CL1–CL7, the open problems OP1–OP18, the alternative branches) is **independent of VFT-1's outcome**: if VFT-1 fails to detect the predicted deviation, only SE11 is retracted; if it succeeds, only SE11 and the Stage-4 calibration are added.

A.1 Motivation: the explanatory gap addressed

Standard QED describes vacuum fluctuations via an infinite sum of Feynman diagrams whose probabilities are extracted statistically. It supplies no underlying continuous physical mechanism for *how* those virtual processes actually occur in a non-perfect vacuum. CCDR / Synthesis supplies the missing layer:

- The laboratory vacuum is the **projection** of the bulk vacuum onto our 3+1D acoustic-phonon sector.
- There are orthogonal optical-phonon (dark-sector) modes with a small but non-zero coupling set by the bulk RVM coefficient ν_bulk (§2.5).
- The dimension-general trigger χ_k (§1.1, §3.1) admits an explicit Weyl-squared term W_k at $k \geq 5$.

Energising the vacuum should reveal deviations from pure perturbative QED that are signatures of (i) real phonon-mode orthogonality, (ii) dimension-general trigger saturation as $\chi\_k \to 1$, and (iii) bulk-vacuum projection fluctuations. **VFT-1 is the cleanest near-term lab-scale probe of this mechanism**: actively pump the vacuum with extreme but controllable energy density, and perform a full tomographic scan of interaction probabilities across multiple Feynman channels.

A.2 Experimental concept

Use a **flying-focus laser beam (FFB)** — a space-time-structured pulse whose focal spot co-propagates with a counter-propagating ultra-relativistic electron or gamma-ray beam at velocity ≈ c. This dramatically extends the effective interaction time (from femtoseconds to picoseconds or longer) while keeping peak intensity high, allowing measurable strong-field QED effects at orders-of-magnitude lower laser power than fixed-focus setups.

- **Pump.** Multi-petawatt flying-focus laser (baseline: 10–25 PW class, e.g. NSF OPAL design or ELI-NP upgrade). Focus velocity tuned to −c (counter-propagating geometry) to maximise interaction length.
- **Probe / seed.** Ultra-relativistic electron beam (GeV–TeV, laser-wakefield accelerated in the same facility) or high-energy gamma-ray beam (produced via Compton backscattering or bremsstrahlung) counter-propagating into the flying focus.
- **Chamber.** Ultra-high vacuum ($\leq 10^{-10}$ Torr) interaction region with polarisation-preserving X-ray/gamma optics and particle detectors.

Key innovation. The flying focus turns the vacuum into a tunable “nonlinear medium” whose response (pair production, birefringence, higher-order scattering, radiation reaction) can be scanned as a function of local energy density.

A.3 Measurables — full Feynman-diagram probability tomography

Record simultaneously:

- **Vacuum birefringence.** Δn induced by the pump; conversion of circular → linear polarisation in the probe; expected Stokes parameter $S_1 \sim 0.01$–0.02 in optimised geometry.
- **Real e^+e^- pair production yield** and angular/momentum spectra (Schwinger-like tunnelling + multiphoton channels).
- **Higher-order QED scattering** (photon emission/absorption rates, radiation reaction force on electrons).
- **Polarisation-dependent cross-sections** and correlations across dozens of effective Feynman channels.
- **Anomalous noise floor or secular drift** consistent with ν_bulk projection (predicted fractional fluctuation $\sim 10^{-8}$–10^{-15} depending on integration time and volume).

All observables are integrated probabilities from large numbers of virtual diagrams; any systematic deviation from pure QED predictions — especially non-perturbative scaling or new angular structures — would signal underlying phonon-sector physics.

A.4 CCDR-specific predictions (the programme boost)

Under CCDR v7.6 + the compactification principle of §3.6:

- Vacuum response is not pure probabilistic QED but **acoustic-phonon propagation** through a locally energised crystal boundary. The optical-phonon sector remains orthogonal until χ_k approaches 1, producing a **sharper-than-QED threshold** for pair production and a calculable suppression of certain higher-order diagrams.
- **v_bulk contributes** a small, measurable background fluctuation floor in all channels ($\delta \sim v_{bulk} \times (t_{int} \times H_\infty)^{-1/2}$), uncorrelated with classical noise — directly testable against the bulk-vacuum projection prediction (FR9, QC11, SE2).
- The **dimension-general trigger term W_k** appears as a tiny but detectable correction to vacuum polarisation; the flying-focus geometry naturally mimics a small-N laboratory analogue of the cosmological $\chi_k \to 1$ saturation.
- **Expected signature.** A ~1–10 % deviation in selected channels (especially vacuum birefringence and pair spectra) once energy density exceeds the current QED-validated regime, scaling with the same acoustic–optical orthogonality that protects dark-sector qubits and ball-lightning analogues.

New prediction SE11 (v7.6+ engineering boost). Flying-focus vacuum tomography at 10–25 PW will reveal v_bulk-coupled deviations from standard QED at the **10^{-3}–10^{-2}** level in polarisation and pair-yield observables, providing the **first laboratory-grade calibration of the bulk-RVM coefficient v_bulk** independent of cosmological channels (P39 / P40).

A.5 Feasibility and timeline (2026–2030)

- **Near-term (1–3 years).** Use existing or funded facilities (NSF OPAL design, ELI-NP 10 PW beams, European XFEL HIBEF upgrades) with flying-focus optics already demonstrated in theory and in early experiments.
- **Mid-term (3–5 years).** Dedicated VFT-1 beamline at a multi-PW facility — cost comparable to current strong-field QED campaigns.
- **Diagnostics are mature.** High-resolution gamma polarimetry, pair spectrometers, and Stokes-parameter analysis are already proposed in 2025–2026 literature.

This experiment is **not speculative hardware** — it leverages the exact flying-focus technology now being actively developed for strong-field QED (Di Piazza et al.; Formanek et al.; recent PPPL/SLAC colloquia 2026).

A.6 Why this is a genuine “programme boost” for CCDR / Synthesis

- Directly addresses the explanatory gap of pure QED: replaces statistical Feynman probabilities with measurable **phonon-mode dynamics** in an energised vacuum.
- Provides a **laboratory anchor for v_bulk**, the dimension-general trigger, and optical-sector orthogonality — independent of cosmology.
- Fits naturally into **Phase 4 (Bulk-Vacuum Projection Exploitation)** and the **Stage-4 quantum-simulator analogue** of the §22 engineering programme.

- Generates **cross-domain calibration**: the same setup tests fusion-edge noise floors (FR9), qubit decoherence limits (QC11), and metrology floors (SE2–SE10).

A.7 Status

VFT-1 / SE11 is registered as an **out-of-main-core engineering boost**. It does not alter the v7.6 hard-core architecture, the empirical scorecard, or any P1–P42 status. It supplies a *near-term laboratory channel* that — if executed — would give CCDR its first non-cosmological, non-astrophysical empirical handle on the bulk-vacuum projection layer. Round-10 / Round-11 status of VFT-1 readiness is to be reviewed when the next public flying-focus campaign at OPAL / ELI-NP reports its first vacuum-birefringence bandpower.

Appendix B. Double-Slit Experiment as Pedagogical Analogue — *out of main core*

The dependence of double-slit interference patterns on slit geometry and shape is used in this appendix as a **conceptual / illustrative parallel** for the crystallographic-projection physics of CCDR (§3.4 projection map, §8 cosmic web as crystal grain structure, §12 dimensional projection and quantum statistics, P41 / CL6 lattice grain-boundary correction). It is presented **out of main core** because:

- It is **not direct empirical confirmation** of any CCDR prediction (P1–P42).
- It does **not close** any open problem (OP1–OP18).
- It is a **strong pedagogical parallel** that makes several of the framework's least-intuitive claims more intuitive and internally consistent.

B.1 The core CCDR claim that the analogy supports

The universe is a **spacetime crystal** formed by local, asynchronous dimensional reductions (§8 "Cosmic Web as Crystal Grain Structure", §10). The BAO scale $r_* \approx 147$ Mpc is the lattice constant; the galaxy two-point correlation shows Bragg-like peaks at $k_n = n\pi/r_*$; filaments and walls are grain boundaries; voids are grains (§8.1–8.2). The observed cosmic-web texture (P3 filament-orientation excess, P38 void-wall kurtosis $k_4 > 4$, P30 density-correlated κ) is **density-stratified** because grain-boundary geometry and local reducing-volume shape modulate the projected pattern (§1.2, §4.5, §22 CL4). Quantum statistics and interference itself **emerge from dimensional projection** of a higher-D classical state (§12): the 3+1D wavefunction is the geometric projection of an N-D phase-space vector, and the Born rule appears as a large-N limit.

B.2 Direct laboratory parallel from double-slit physics

In the double-slit (or any multi-aperture) setup:

- **Fringe spacing** $\Delta x = \lambda D/d$ depends only on the **center-to-center separation** d (the "lattice constant").
- **Envelope shape and visibility** (how many fringes are bright, how sharply they decay, whether minima reach zero) are set by the **single-slit diffraction pattern**, which is the squared Fourier transform of the individual slit geometry (width a, shape, edge profile, orientation, curvature). Rectangular slits → sinc² envelope. Circular pinholes → Airy

rings. Non-identical or tilted slits → reduced contrast, asymmetry. Finite length or apodised edges → suppressed side-lobes.

This is exactly the **separation vs. shape** distinction that CCDR invokes for the cosmic lattice:

Double-slit optics	CCDR cosmic-web analogue (§8, §3.4, §4.5)	CCDR prediction / observable
Center-to-center separation d	BAO lattice constant $r_\star$ (Bragg peaks in P(k))	Harmonic peaks at $k_n = n\pi/r_\star$
Single-slit width / shape	Local reducing-volume geometry + grain-boundary shape	Density-stratified modulation of texture (P3, P30, P38)
Envelope = FT[aperture function]	Projected pattern = geometric projection of higher-D defects	Filament orientation excess $\Delta r_{texture}$, void-wall kurtosis $k_4 > 4$, $\kappa(\rho)$
Non-identical slits → lower contrast	Reducing-volume caveat + patch-to-patch variation (§4.5, OP13)	Patch spread > baseline $\Delta\kappa$ (Round-8 T2)

The observed density-stratified cosmic-web signatures (Round 7/8/9 T17, T08, T06) are precisely what one would expect if the “apertures” (reducing volumes / grain boundaries) have geometry-dependent Fourier content that modulates the far-field pattern — exactly as slit shape modulates the interference envelope.

B.3 Deeper resonance with CCDR architecture

- **Projection map (§3.4).** Bulk point defects / strings / walls project to 3D monopoles / filaments / surfaces. The shape of the pre-reduction 5D (or higher-k) region sets the Fourier content of the projected 3+1D metric perturbations — identical to how slit shape sets the diffraction envelope.
- **Layer-3 lattice correction (P41 / CL6).** The CDT simplicial lattice plus residual grain-boundary defects from 5→4 nucleation leave a small vector-like, CP-even shift in the effective Dirac operator. The $b \to s\mu\mu$ angular anomaly (LHCb arXiv:2512.18053) is interpreted as the low-energy footprint of this lattice “slit geometry”. The double-slit experiment is the textbook demonstration that lattice / aperture geometry imprints a measurable far-field pattern.
- **Emergent QM from projection (§12).** The double-slit interference *itself* is the canonical laboratory demonstration of the wavefunction whose probability distribution arises from higher-D geometry. CCDR’s claim that the Born rule is geometric (large-N projection) is directly illustrated by the fact that changing the physical slit geometry changes the observed interference pattern without changing the underlying wave equation.
- **Boundary-deformation principle (§3.6).** The boundary-deformation spectrum $\varepsilon_{\ell m}$ introduced in §3.6 has its direct double-slit analogue in the aperture-function expansion: changing the slit shape redistributes spectral content; changing the reducing-volume shape redistributes the cosmic-web texture spectrum across P3, P30, P38. P42 (boundary-deformation residuals) is the cosmological-scale version of “single-slit envelope analysis”.

B.4 Status

The double-slit analogue is added as a **consistent-with-data illustrative parallel**, not a falsifiable prediction. It is *not falsifiable on its own*, and it does *not* add new free parameters or metaphysical commitments. It strengthens the crystallographic intuition without changing the v7.6 / §3.6 architecture. It dovetails cleanly with the existing Bragg-peak language in §8.1 and provides a textbook handle for §12.

B.5 Bottom line

The double-slit experiment is one of the cleanest everyday demonstrations of the exact mechanism CCDR invokes at cosmological scales: **geometry of the "**aperture**" (slit / reducing volume / grain boundary) controls the far-field / projected pattern**. It makes the otherwise abstract projection + crystal picture immediately graspable and reinforces the internal coherence of Layers 0–3.